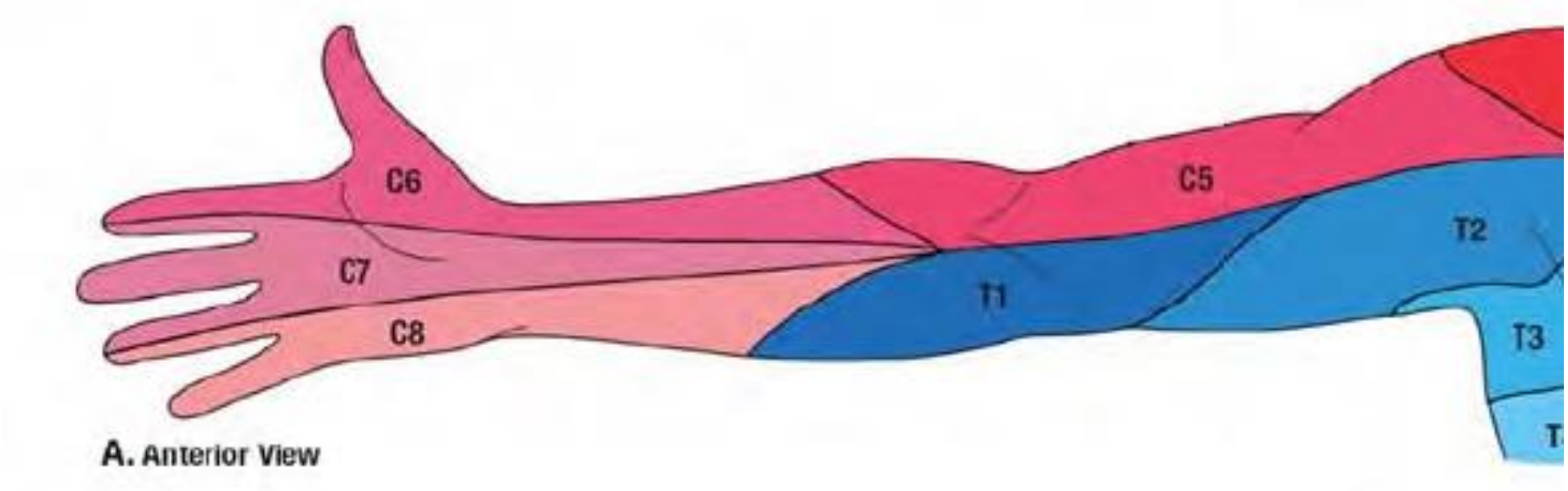
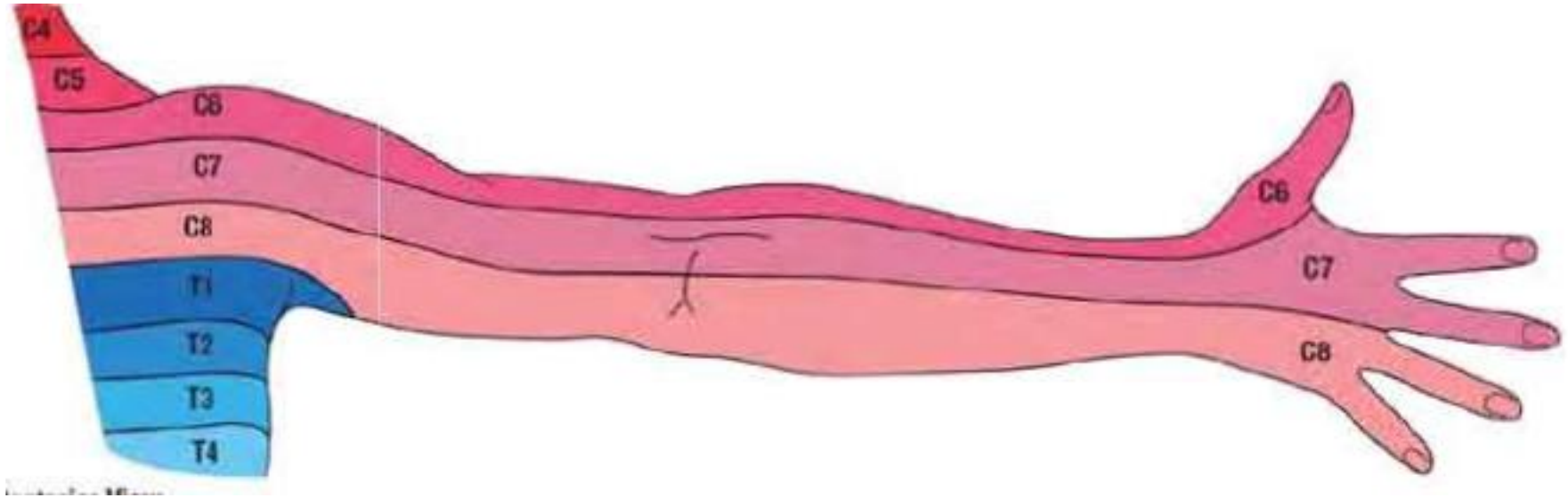
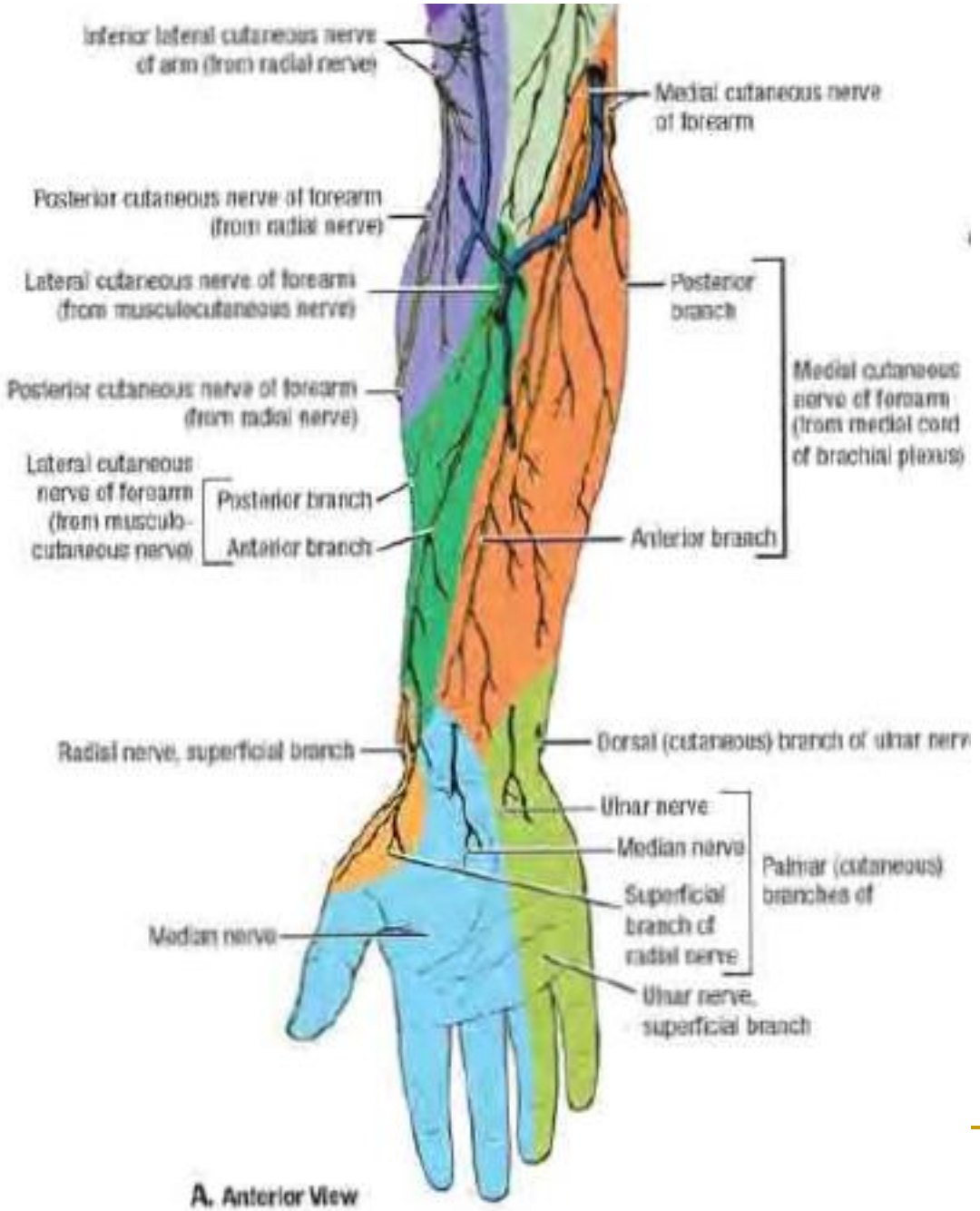
FOREARM

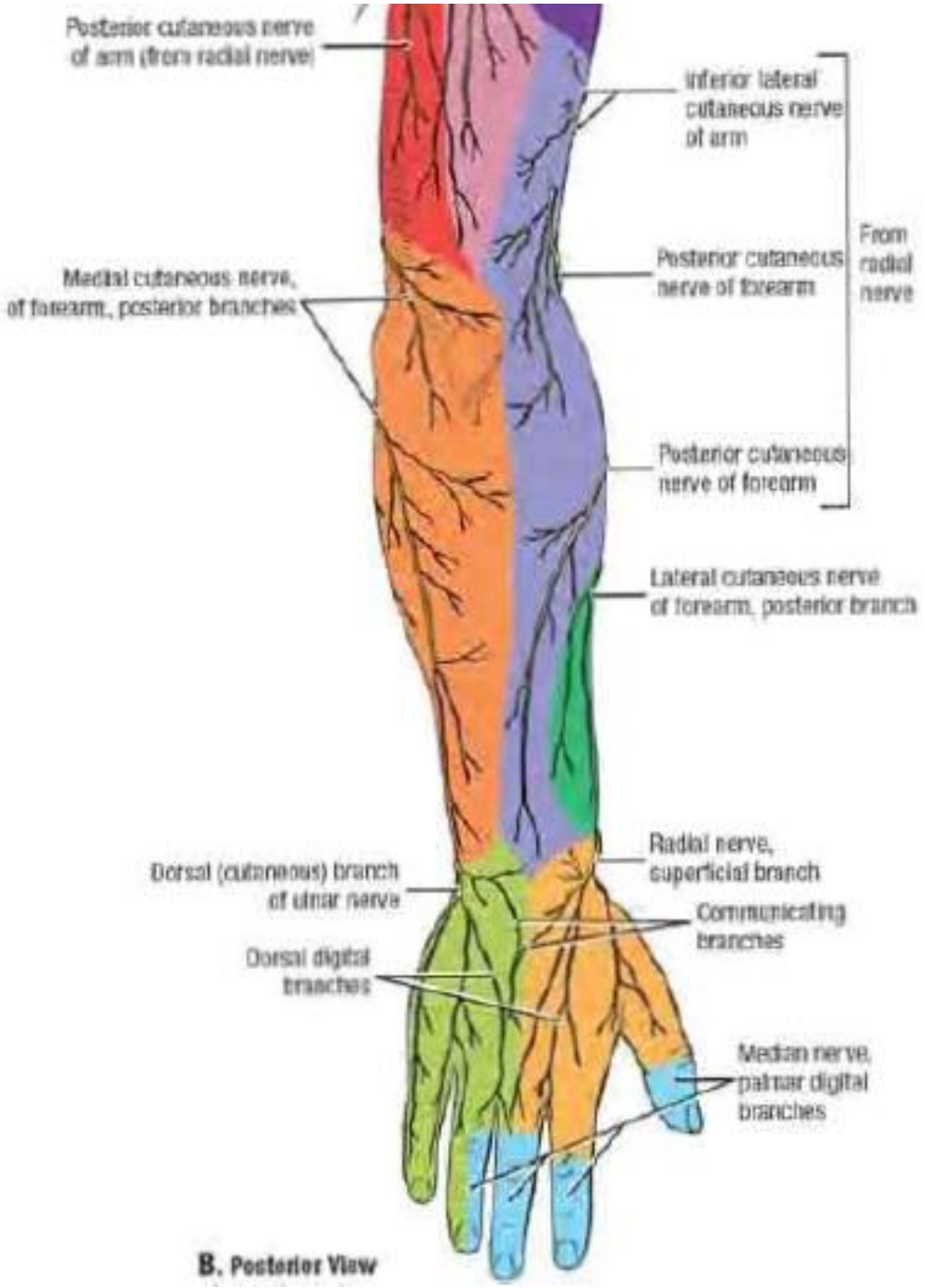
OYESHIKU

Dermatome

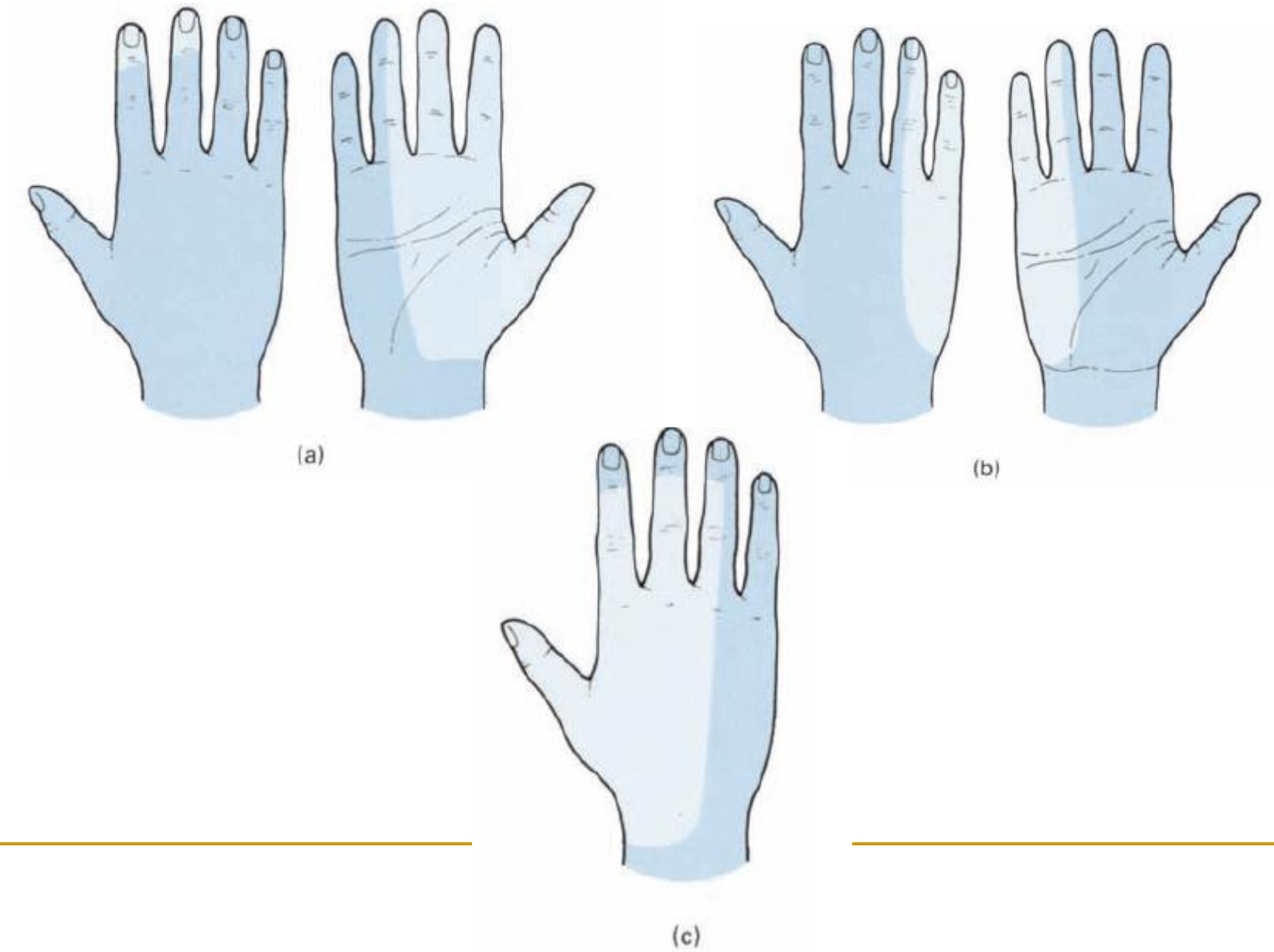


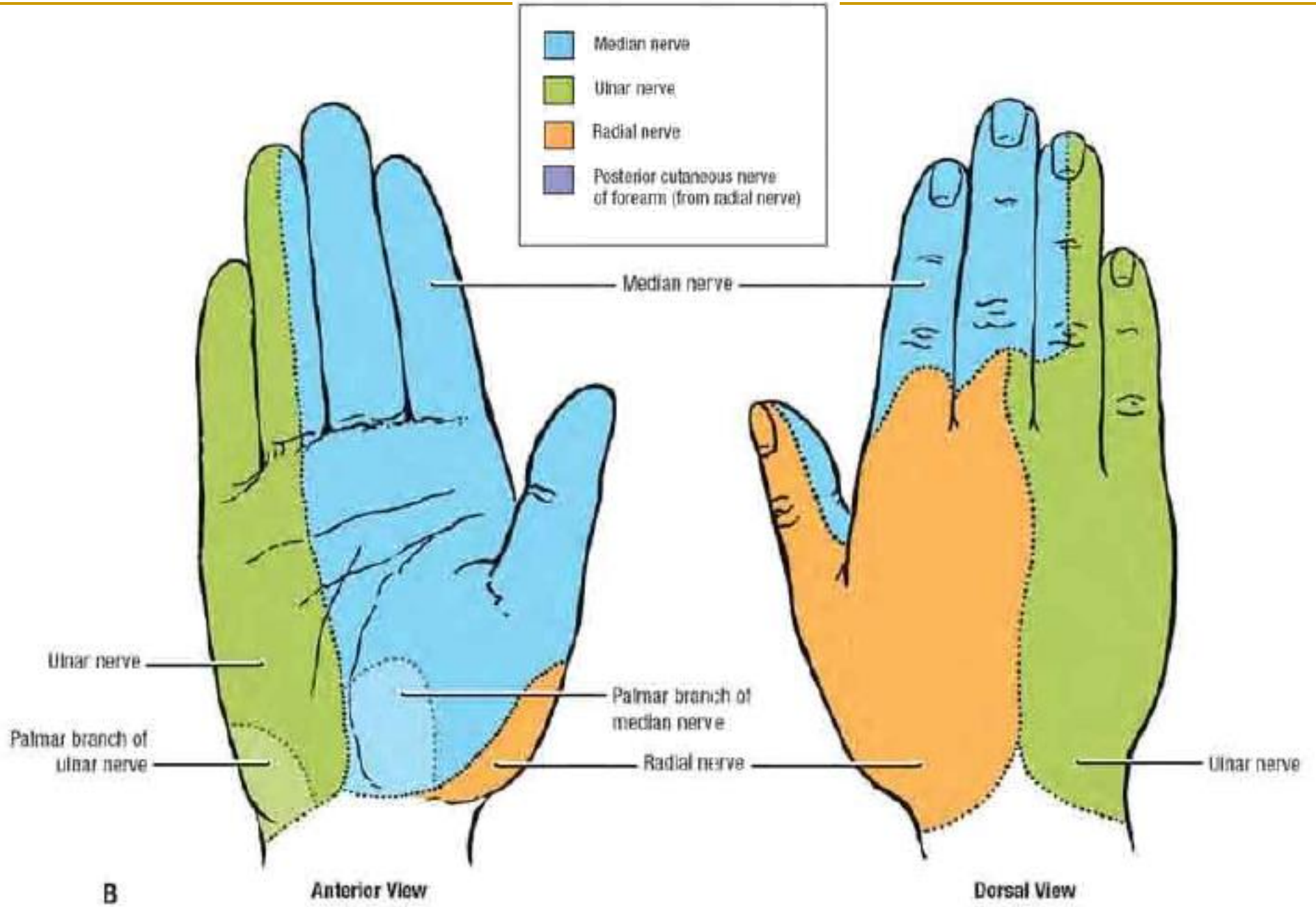

C4
C5
C6
C7
C8
T1
T2
T3
T4
C6
C7
C8

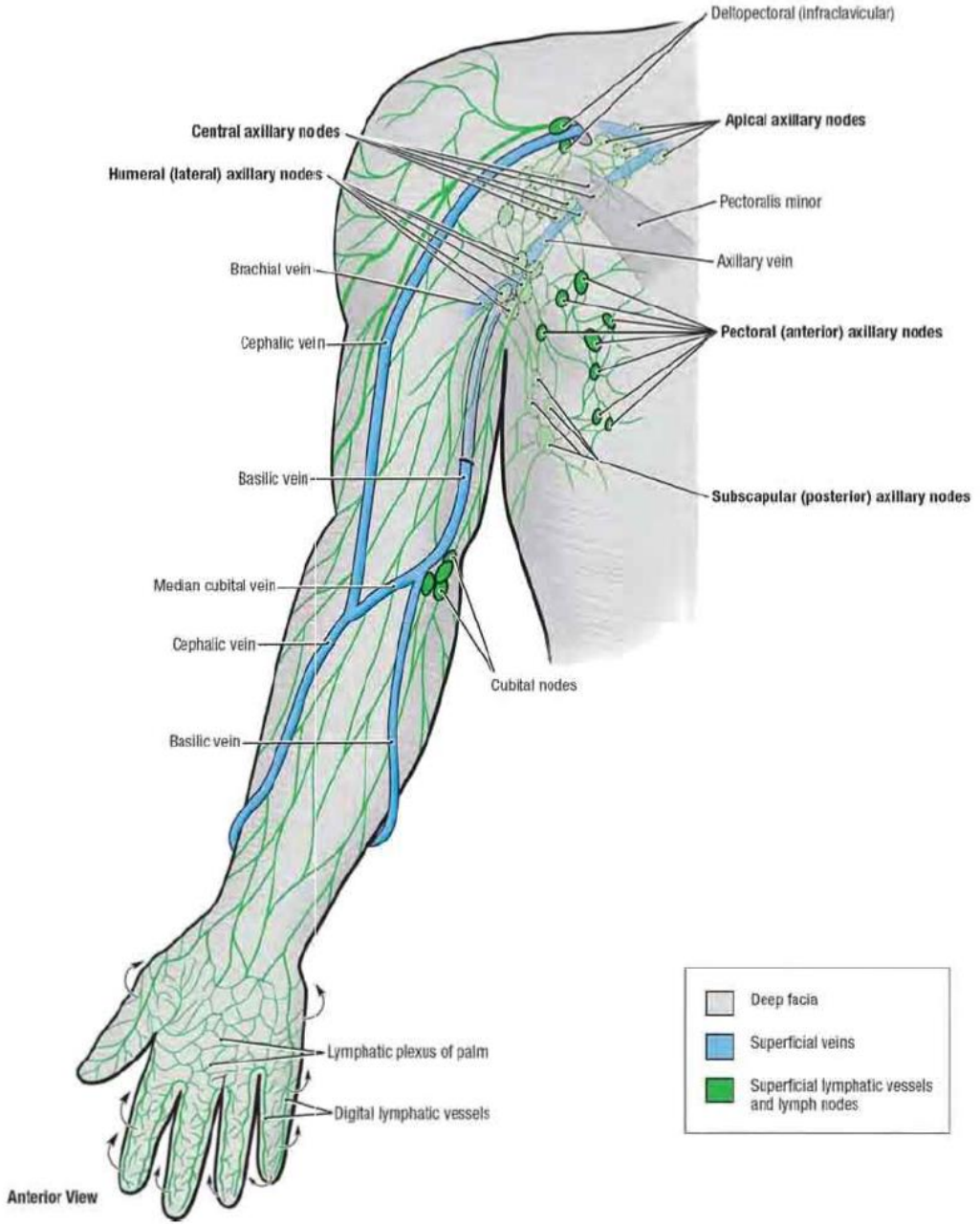

Inferior lateral cutaneous nerve of arm (from radial nerve)
Medial cutaneous nerve of forearm
Posterior cutaneous nerve of forearm (from radial nerve)
Lateral cutaneous nerve of forearm (from musculocutaneous nerve)
Posterior branch
Medial cutaneous nerve of forearm (from medial cord of brachial plexus)
Posterior cutaneous nerve of forearm (from radial nerve)
Lateral cutaneous nerve of forearm (from musculo-cutaneous nerve)
Posterior branch
Anterior branch
Anterior branch
Radial nerve, superficial branch
Dorsal (cutaneous) branch of ulnar nerve
Ulnar nerve
Median nerve
Palmar (cutaneous) branches of
Superficial branch of radial nerve
Median nerve
Ulnar nerve, superficial branch
A. Anterior View

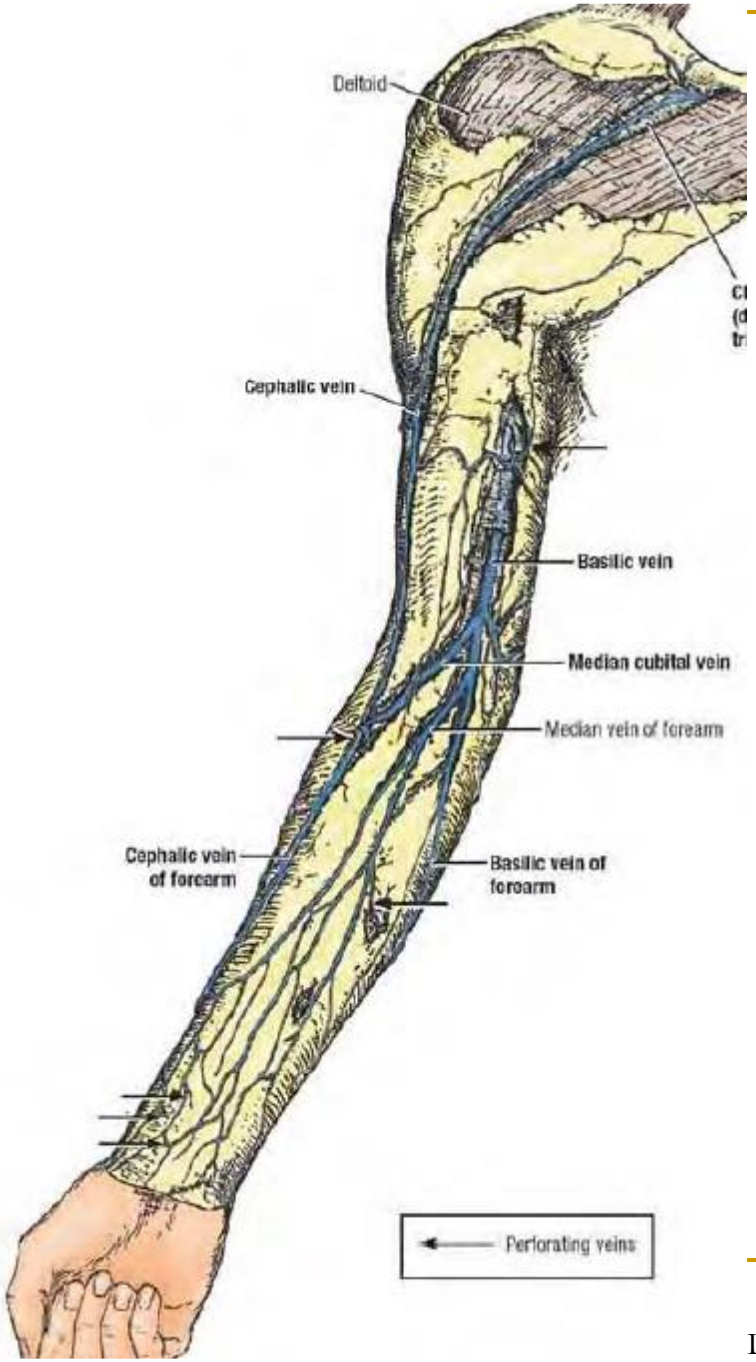

Posterior cutaneous nerve of arm (from radial nerve)
Inferior lateral cutaneous nerve of arm
From radial nerve
Posterior cutaneous nerve of forearm
Medial cutaneous nerve of forearm, posterior branches
Posterior cutaneous nerve of forearm
Lateral cutaneous nerve of forearm, posterior branch
Radial nerve, superficial branch
Dorsal (cutaneous) branch of ulnar nerve
Communicating branches
Dorsal digital branches
Median nerve, palmar digital branches
B. Posterior View

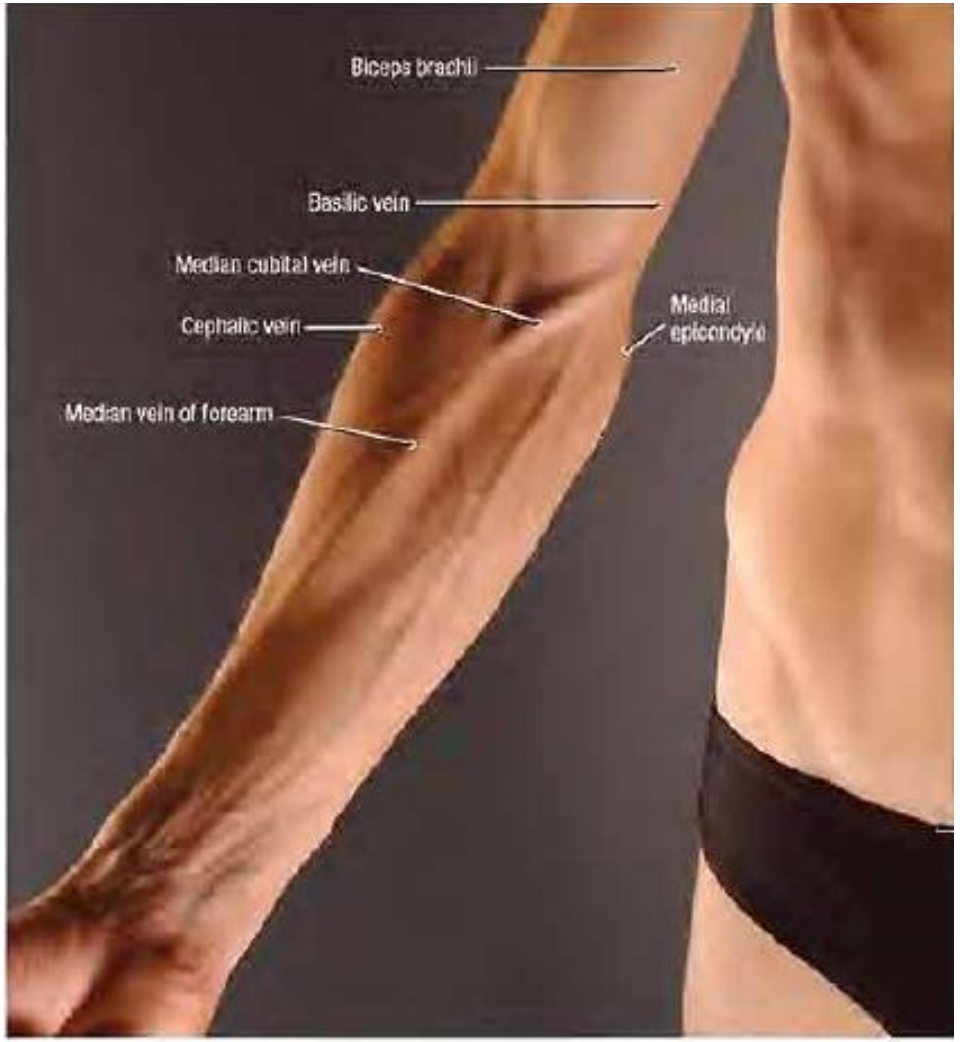
The usual cutaneous distribution (shown in pale blue) of the (a) median, (b) ulnar and (c) radial nerves in the hand (considerable variations and overlap occur).



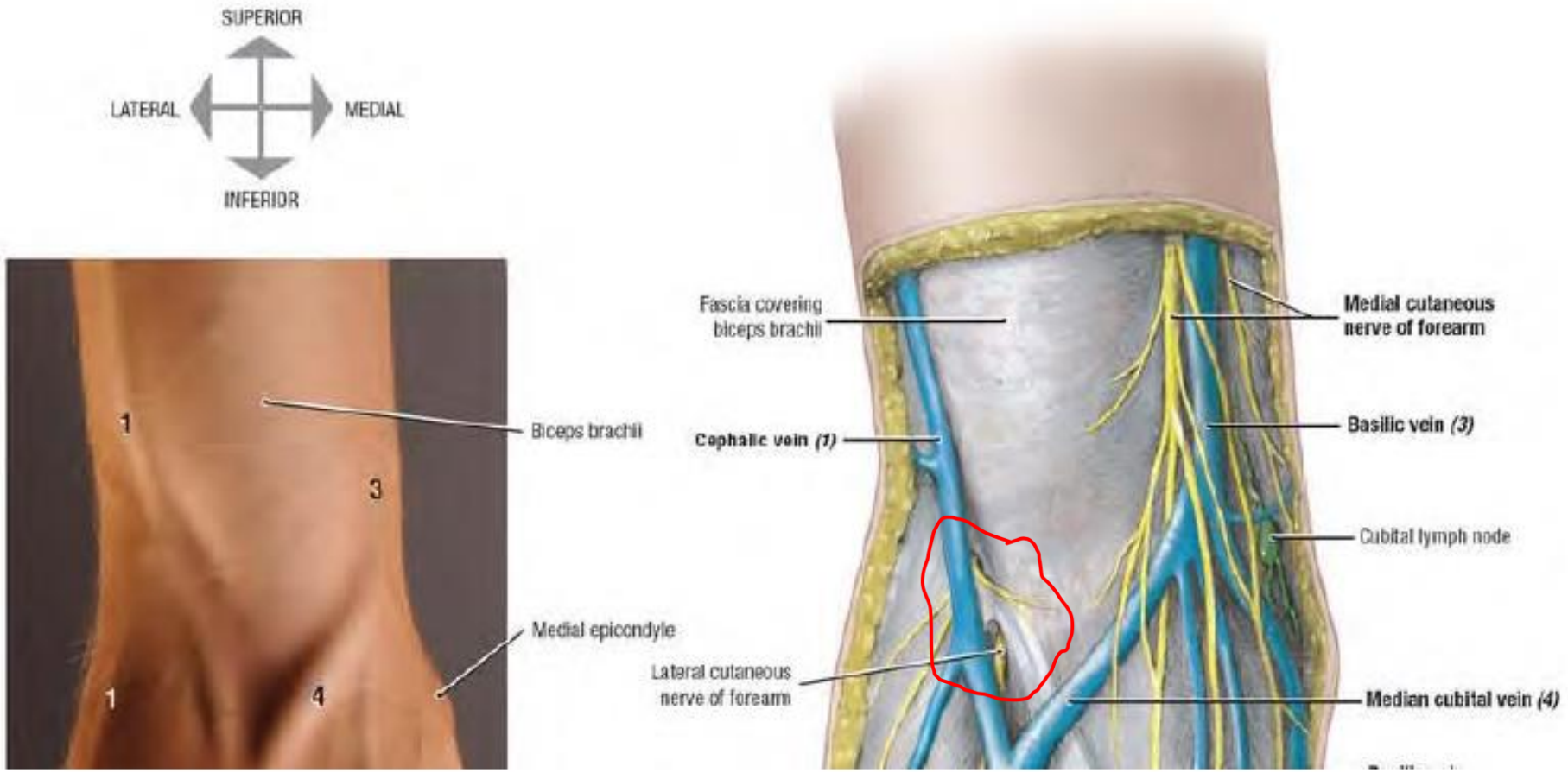

Median nerve
Ulnar nerve
Radial nerve
Posterior cutaneous nerve of forearm (from radial nerve)
Median nerve
Ulnar nerve
Palmar branch of ulnar nerve
Palmar branch of median nerve
Radial nerve
Ulnar nerve
B
Anterior View
Dorsal View

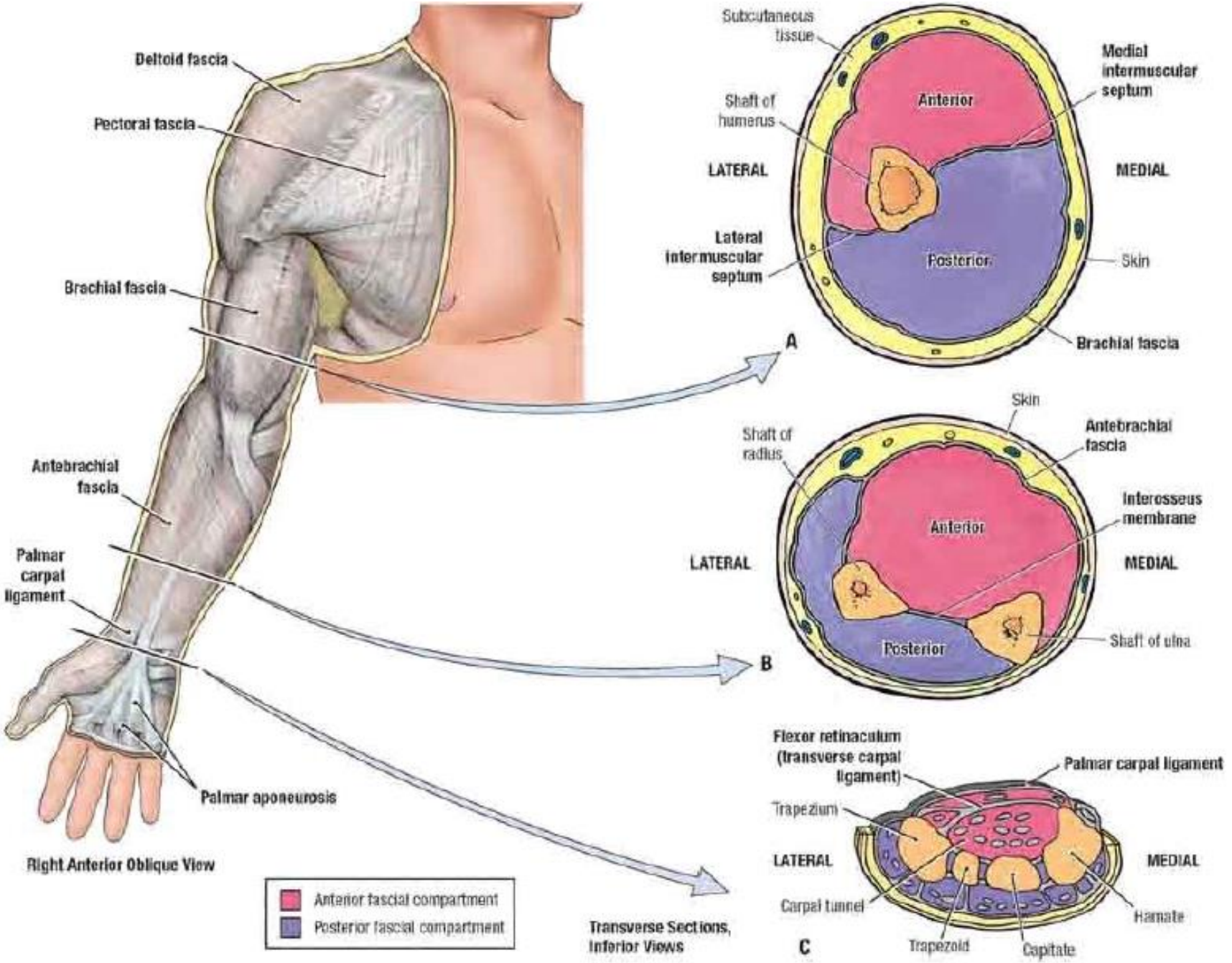

Deltopectoral (infraclavicular)
Central axillary nodes
Apical axillary nodes
Humeral (lateral) axillary nodes
Pectoralis minor
Axillary vein
Brachial vein
Cephalic vein
Pectoral (anterior) axillary nodes
Basilic vein
Subscapular (posterior) axillary nodes
Median cubital vein
Cephalic vein
Cubital nodes
Basilic vein
Lymphatic plexus of palm
Digital lymphatic vessels
Anterior View
Deep facia
Superficial veins
Superficial lymphatic vessels and lymph nodes

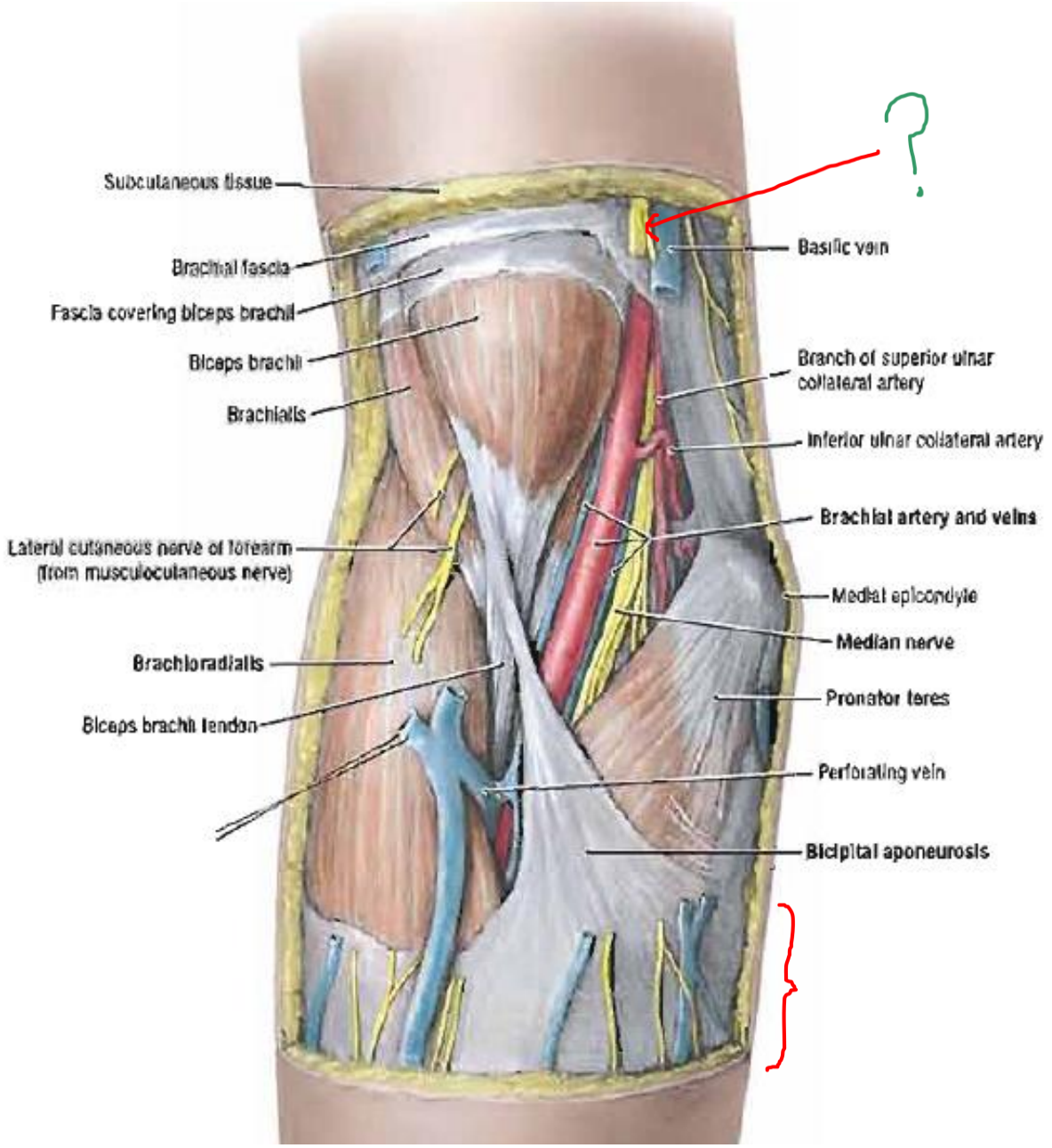


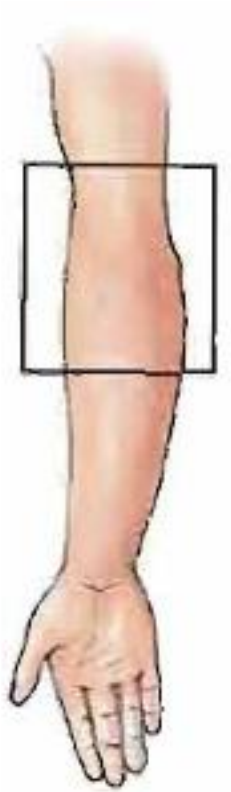


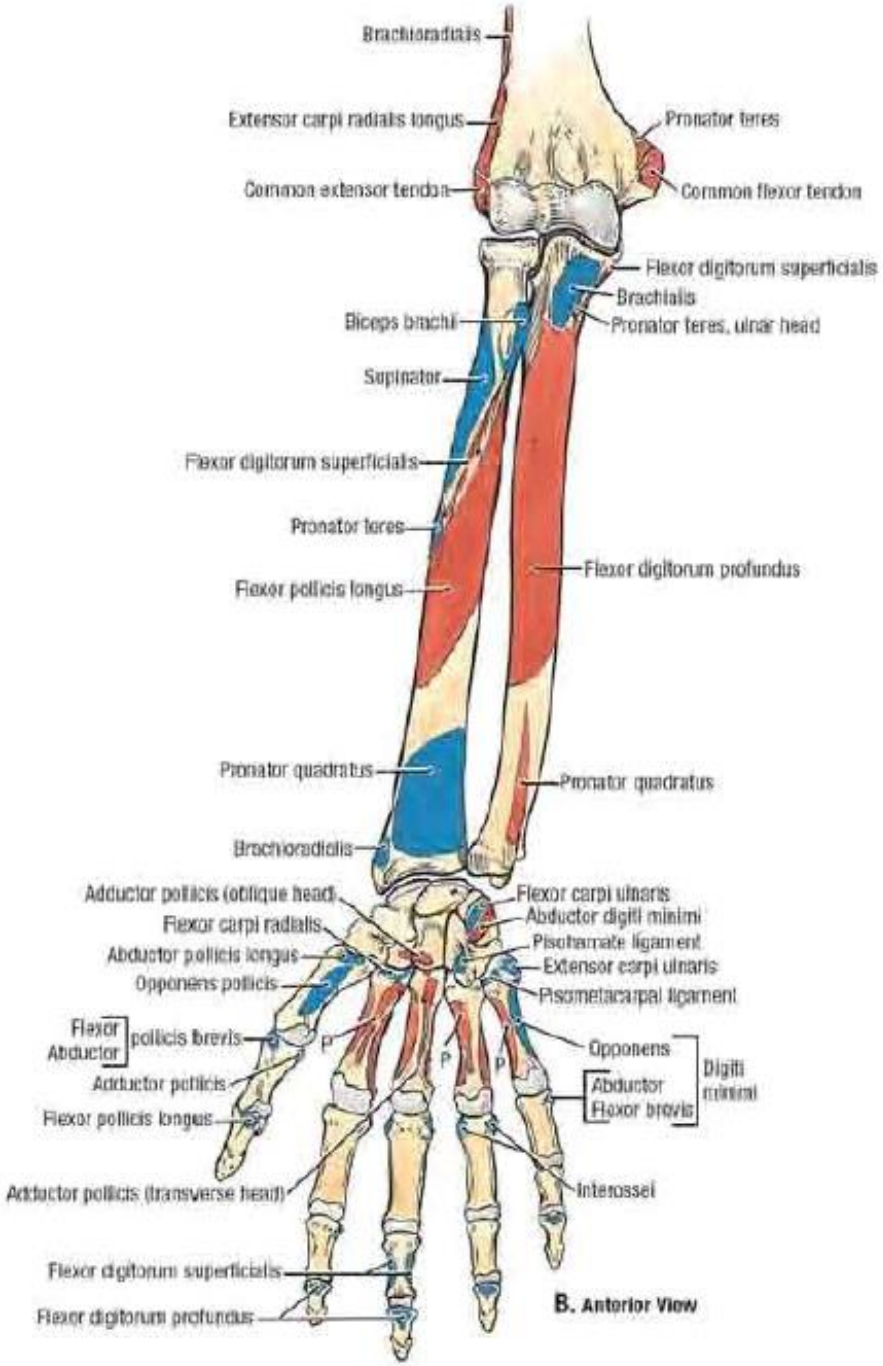
Anterior View

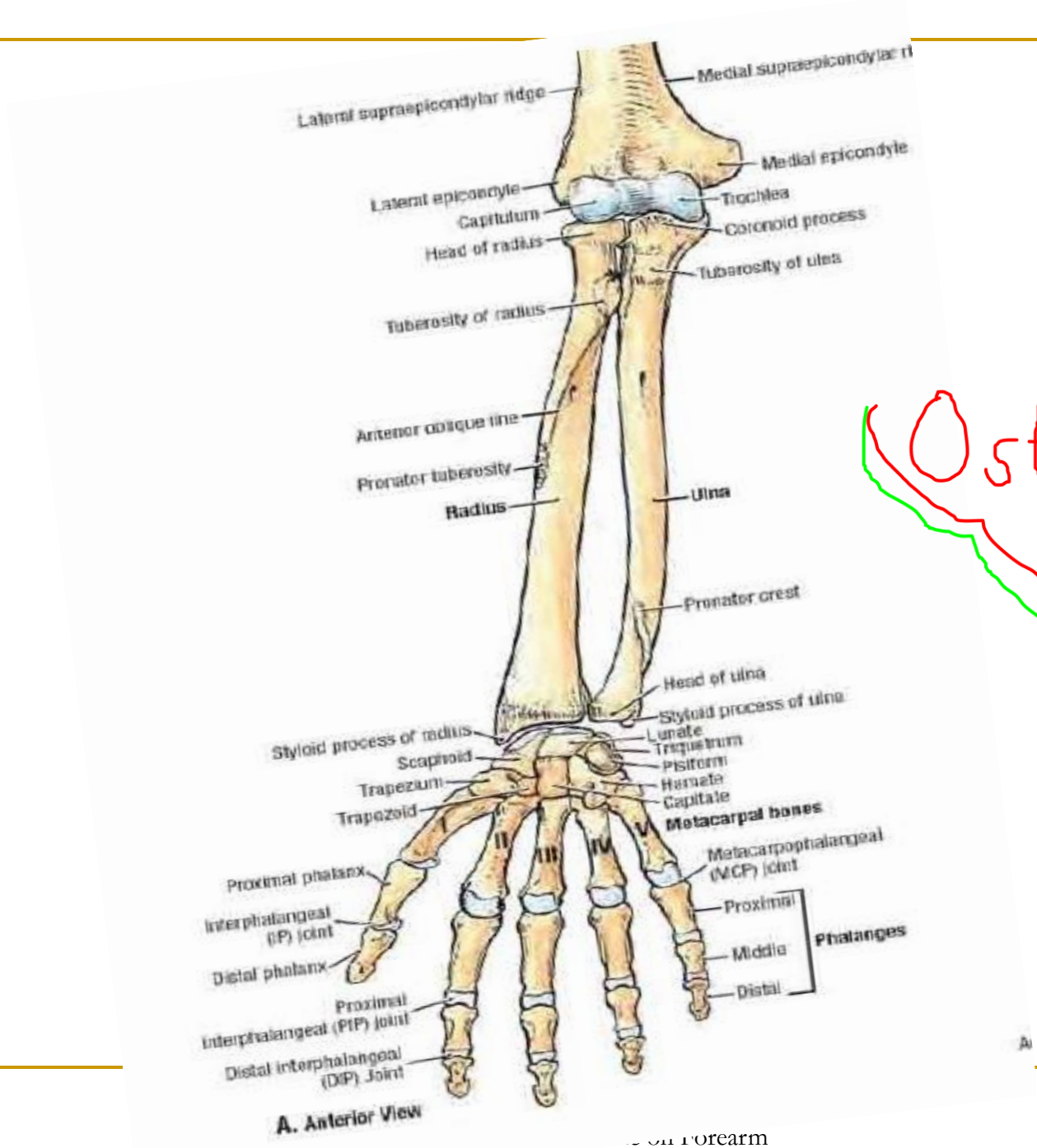

SUPERIOR
LATERAL
MEDIAL
INFERIOR
1
3
1
4
Biceps brachii
Medial epicondyle
Fascia covering biceps brachii
Cephalic vein (1)
Lateral cutaneous nerve of forearm
Medial cutaneous nerve of forearm
Basilic vein (3)
Cubital lymph node
Median cubital vein (4)


Deltoid fascia
Pectoral fascia
Brachial fascia
Antebrachial fascia
Palmar carpal ligament
Palmar aponeurosis
Right Anterior Oblique View
Anterior fascial compartment
Posterior fascial compartment
Transverse Sections, Inferior Views
Subcutaneous tissue
Shaft of humerus
LATERAL
Lateral intermuscular septum
Medial intermuscular septum
Anterior
Posterior
MEDIAL
Skin
Brachial fascia
A
Skin
Shaft of radius
Antebrachial fascia
Interosseus membrane
Anterior
LATERAL
MEDIAL
Posterior
Shaft of ulna
B
Flexor retinaculum (transverse carpal ligament)
Palmar carpal ligament
Trapezium
LATERAL
MEDIAL
Carpal tunnel
Hamate
Trapezoid
Capitate
C

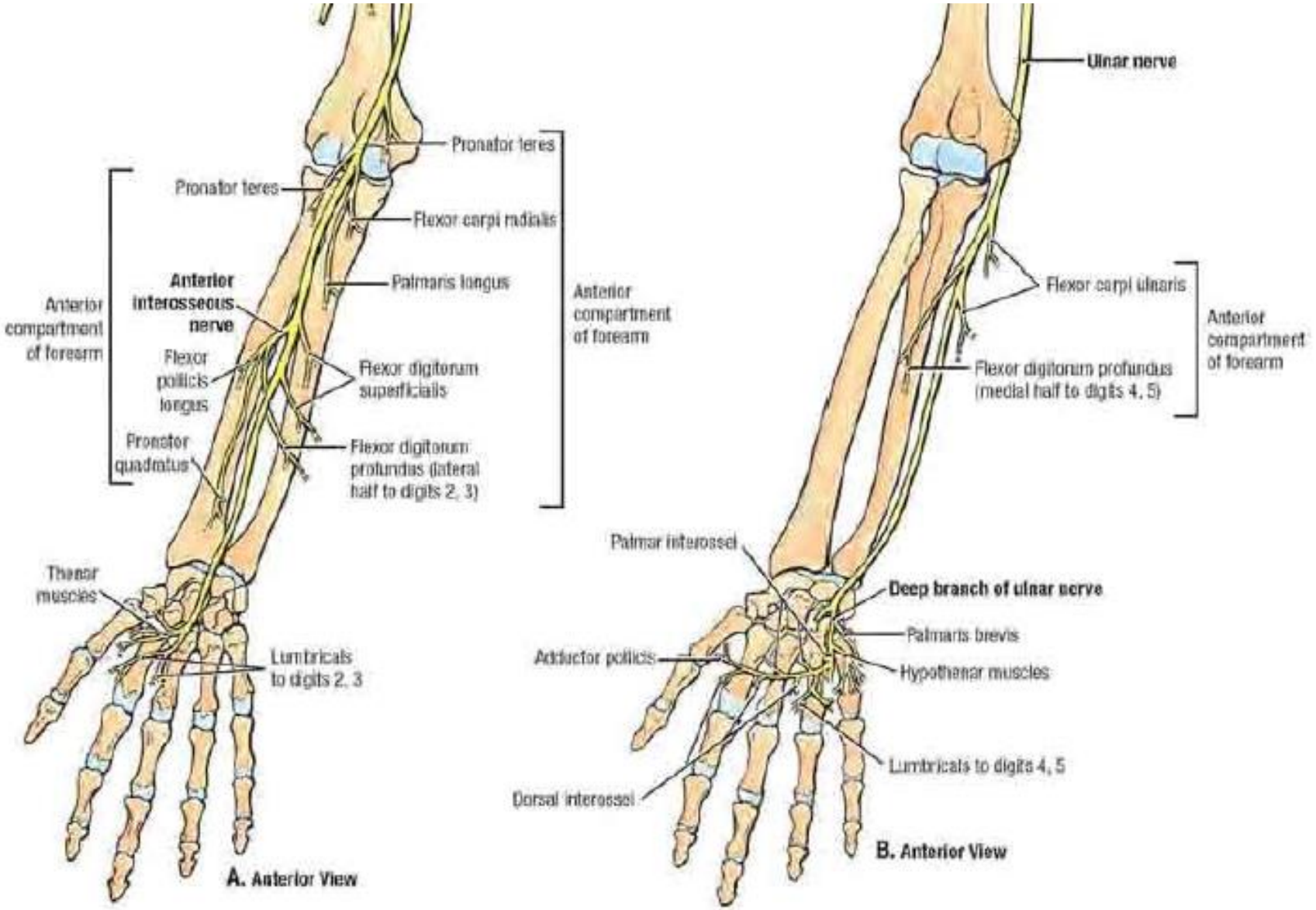

Subcutaneous tissue
Brachial fascia
Fascia covering biceps brachii
Biceps brachii
Brachialis
Lateral cutaneous nerve of forearm (from musculocutaneous nerve)
Brachioradialis
Biceps brachii tendon
Basilic vein
Branch of superior ulnar collateral artery
Inferior ulnar collateral artery
Brachial artery and veins
Medial epicondyle
Median nerve
Pronator teres
Perforating vein
Bicipital aponeurosis

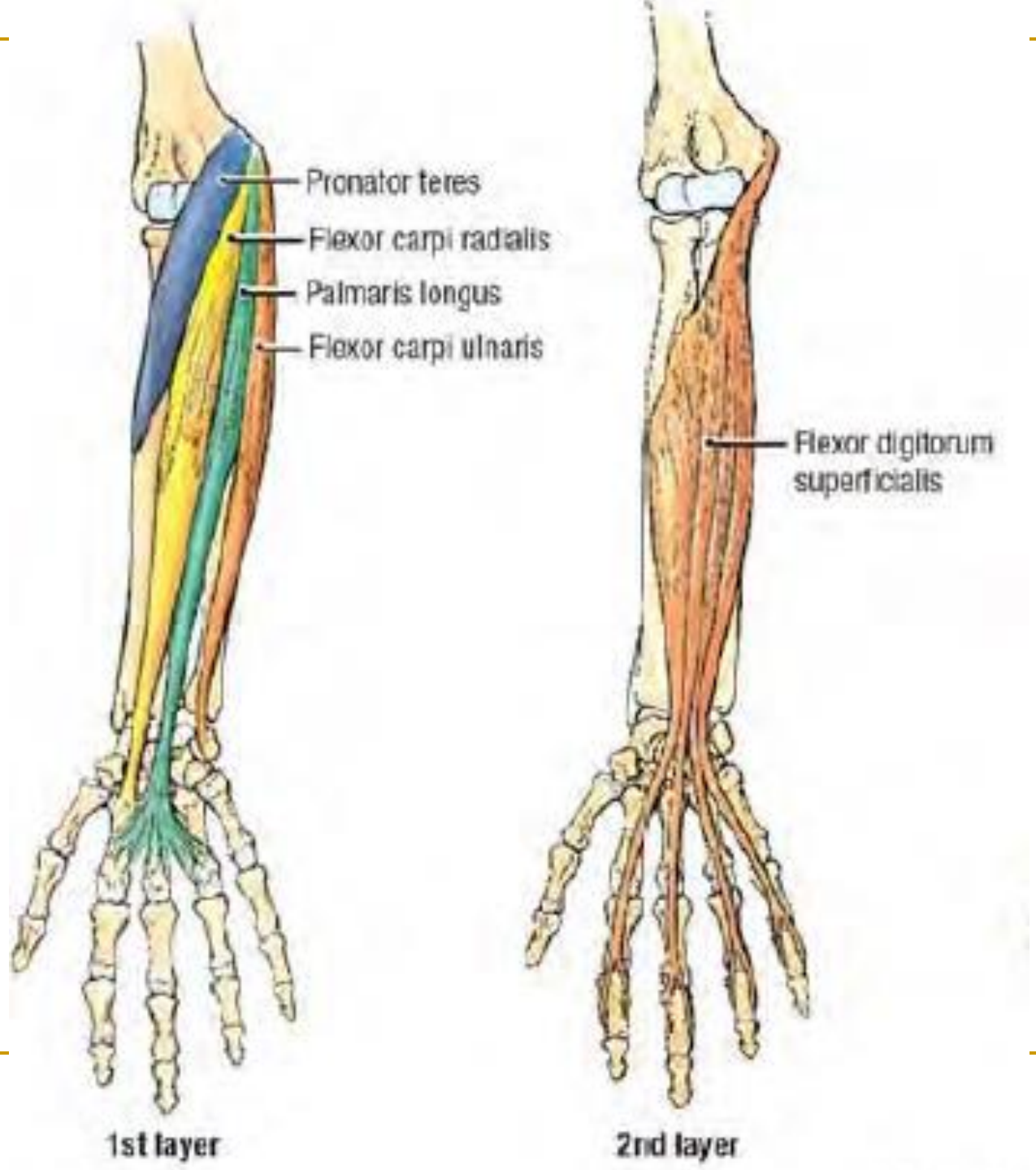


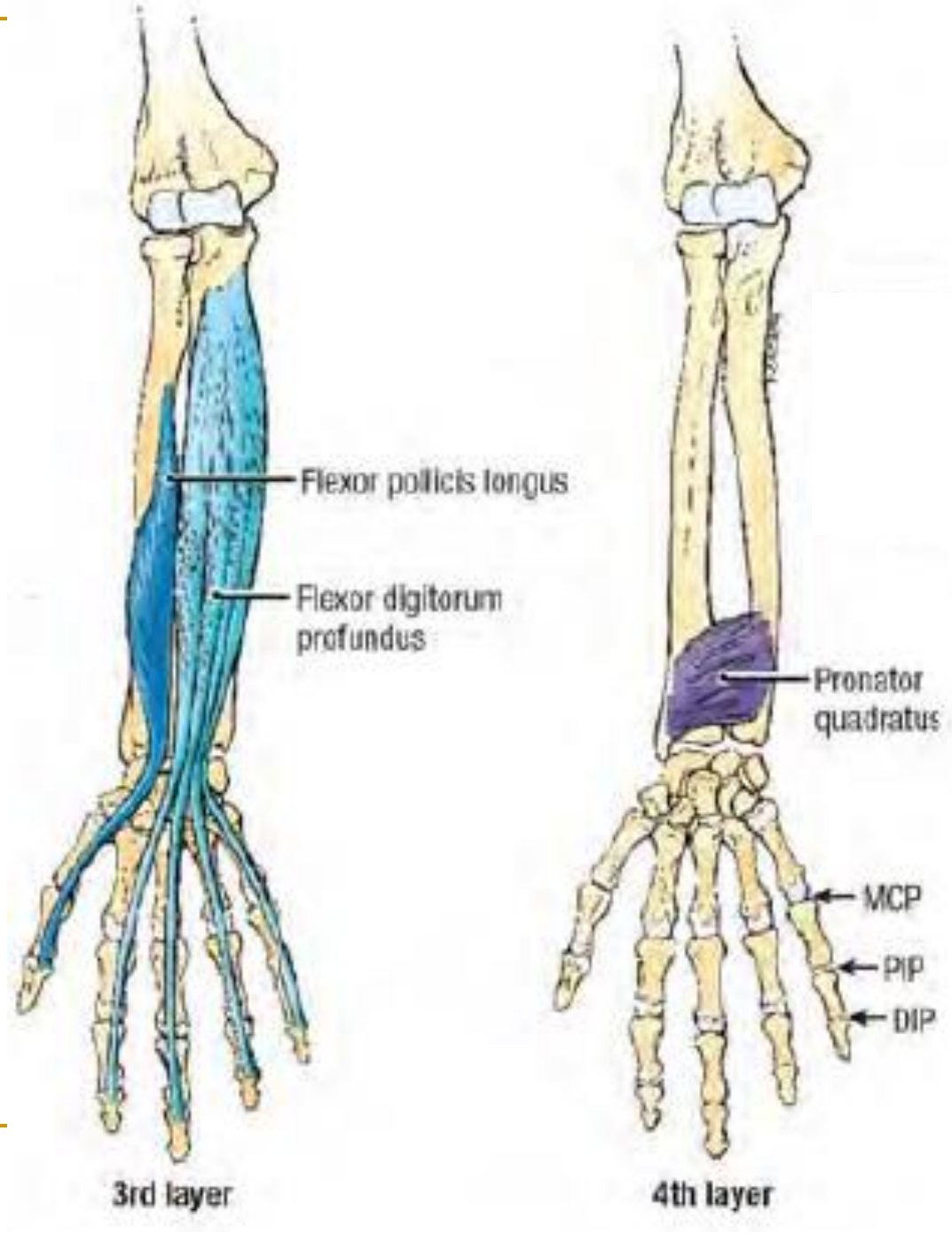

Brachioradialis
Extensor carpi radialis longus
Common extensor tendon
Pronator teres
Common flexor tendon
Flexor digitorum superficialis
Brachialis
Pronator teres, ulnar head
Biceps brachii
Supinator
Flexor digitorum superficialis
Pronator teres
Flexor pollicis longus
Flexor digitorum profundus
Pronator quadratus
Pronator quadratus
Brachioradialis
Adductor pollicis (oblique head)
Flexor carpi radialis
Abductor pollicis longus
Opponens pollicis
Flexor Abductor pollicis brevis
Adductor pollicis
Flexor pollicis longus
Adductor pollicis (transverse head)
Flexor carpi ulnaris
Abductor digiti minimi
Pisohamate ligament
Extensor carpi ulnaris
Pisometacarpal ligament
Opponens
Abductor Flexor brevis
Digiti minimi
Interossei
Flexor digitorum superficialis
Flexor digitorum profundus
B. Anterior View


Lateral supraepicondylar ridge
Medial supraepicondylar ri
Medial epicondyle
Lateral epicondyle
Capitulum
Trochlea
Head of radius
Coronoid process
Tuberosity of ulna
Tuberosity of radius
Anterior oblique line
Pronator tuberosity
Radius
Ulna
Pronator crest
Head of ulna
Styloid process of ulna
Styloid process of radius
Lunate
Triquetrum
Scaphoid
Pisiform
Trapezium
Hamate
Capitate
Trapezoid
Metacarpal bones
Metacarpophalangeal (MCP) joint
Proximal phalanx
Interphalangeal (IP) joint
Distal phalanx
Proximal
Middle
Distal
Phalanges
Proximal interphalangeal (PIP) joint
Distal interphalangeal (DIP) joint
A. Anterior View

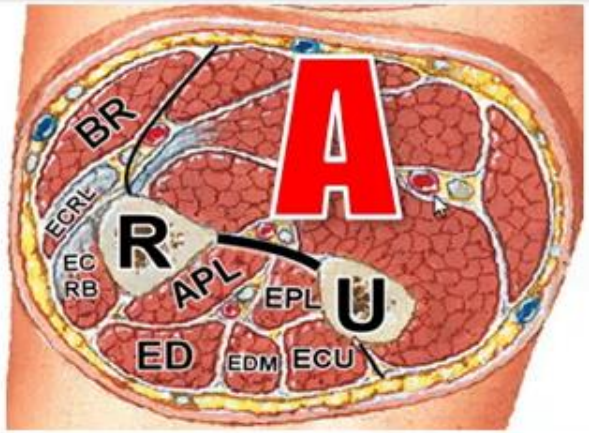
Osteo

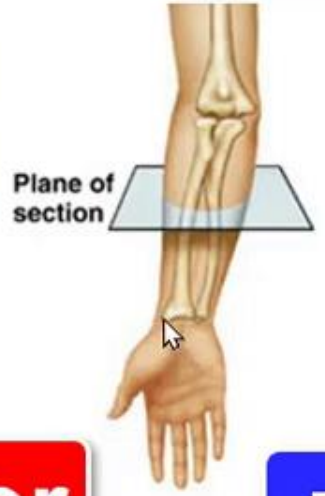

Pronator teres
Pronator teres
Flexor carpi radialis
Palmaris longus
Anterior interosseous nerve
Anterior compartment of forearm
Flexor pollicis longus
Flexor digitorum superficialis
Pronator quadratus
Flexor digitorum profundus (lateral half to digits 2, 3)
Anterior compartment of forearm
Thenar muscles
Lumbricals to digits 2, 3
A. Anterior View
Ulnar nerve
Flexor carpi ulnaris
Anterior compartment of forearm
Flexor digitorum profundus (medial half to digits 4, 5)
Palmar interossei
Deep branch of ulnar nerve
Palmaris brevis
Adductor pollicis
Hypothenar muscles
Lumbricals to digits 4, 5
Dorsal interossei
B. Anterior View

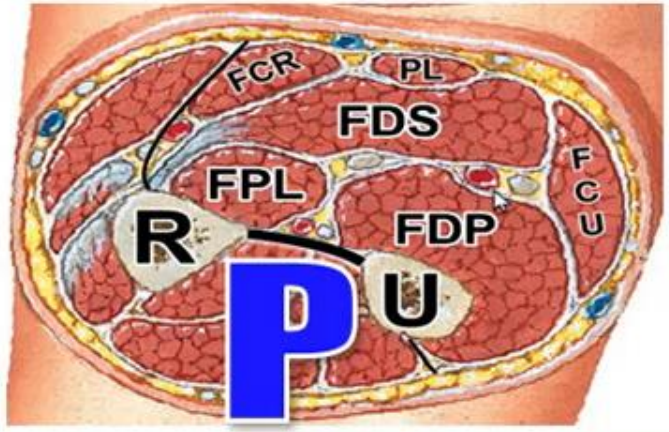

Pronator teres
Flexor carpi radialis
Palmaris longus
Flexor carpi ulnaris
Flexor digitorum superficialis
1st layer
2nd layer


Flexor pollicis longus
Flexor digitorum profundus
3rd layer
Pronator quadratus
MCP
PIP
DIP
4th layer

Forearm Muscles: 2 Compartments

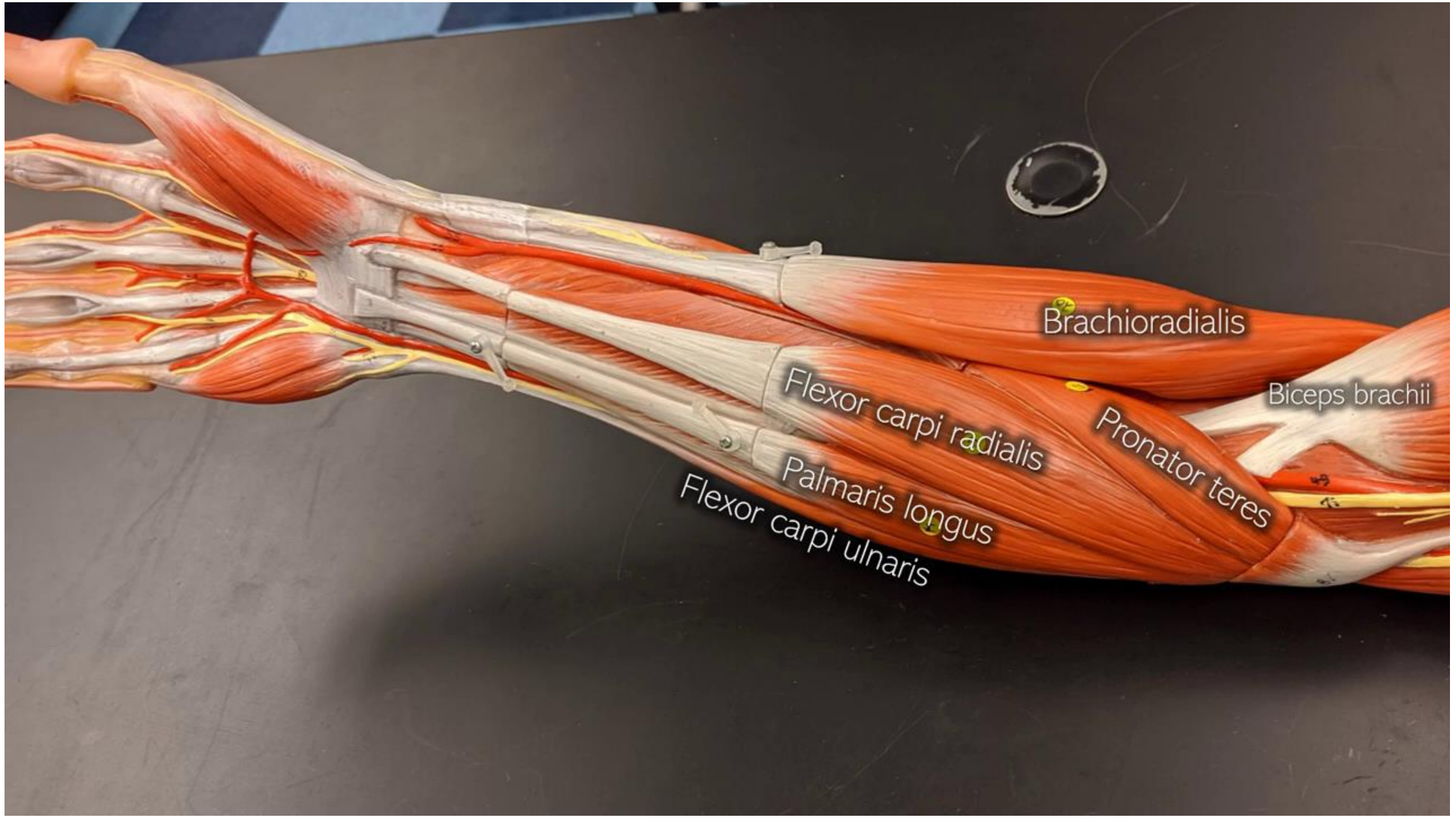


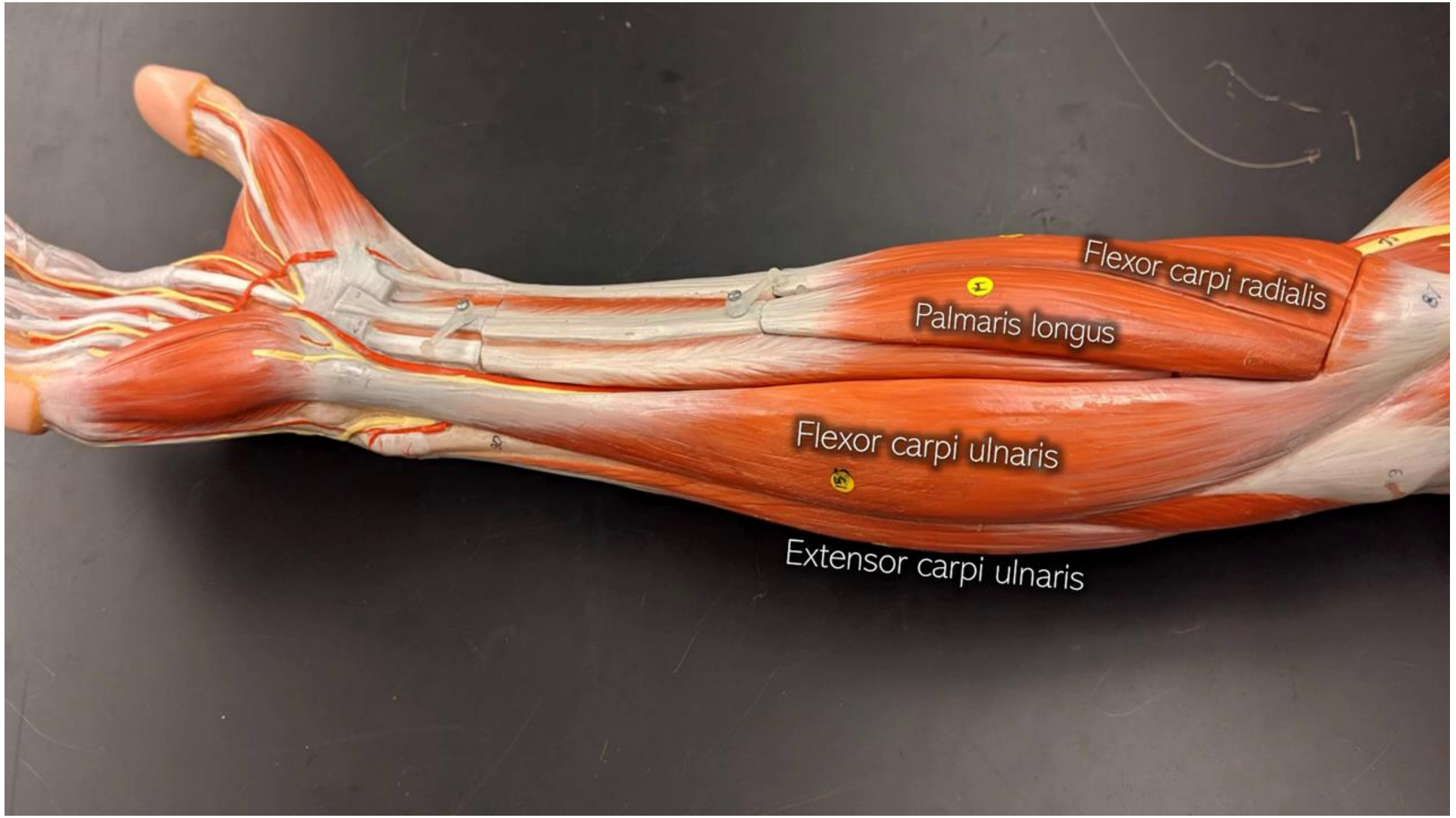




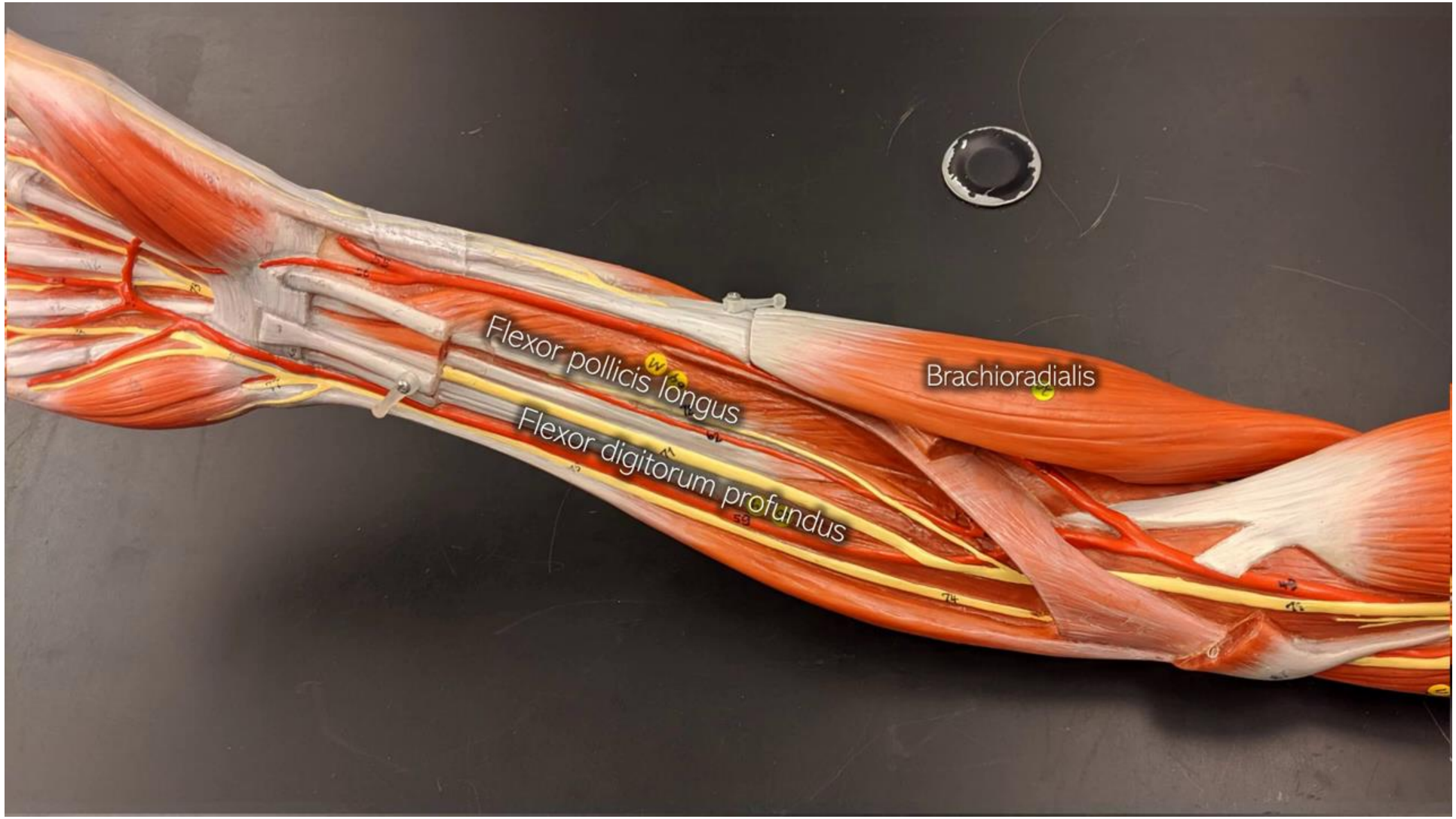
Anterior Flexor

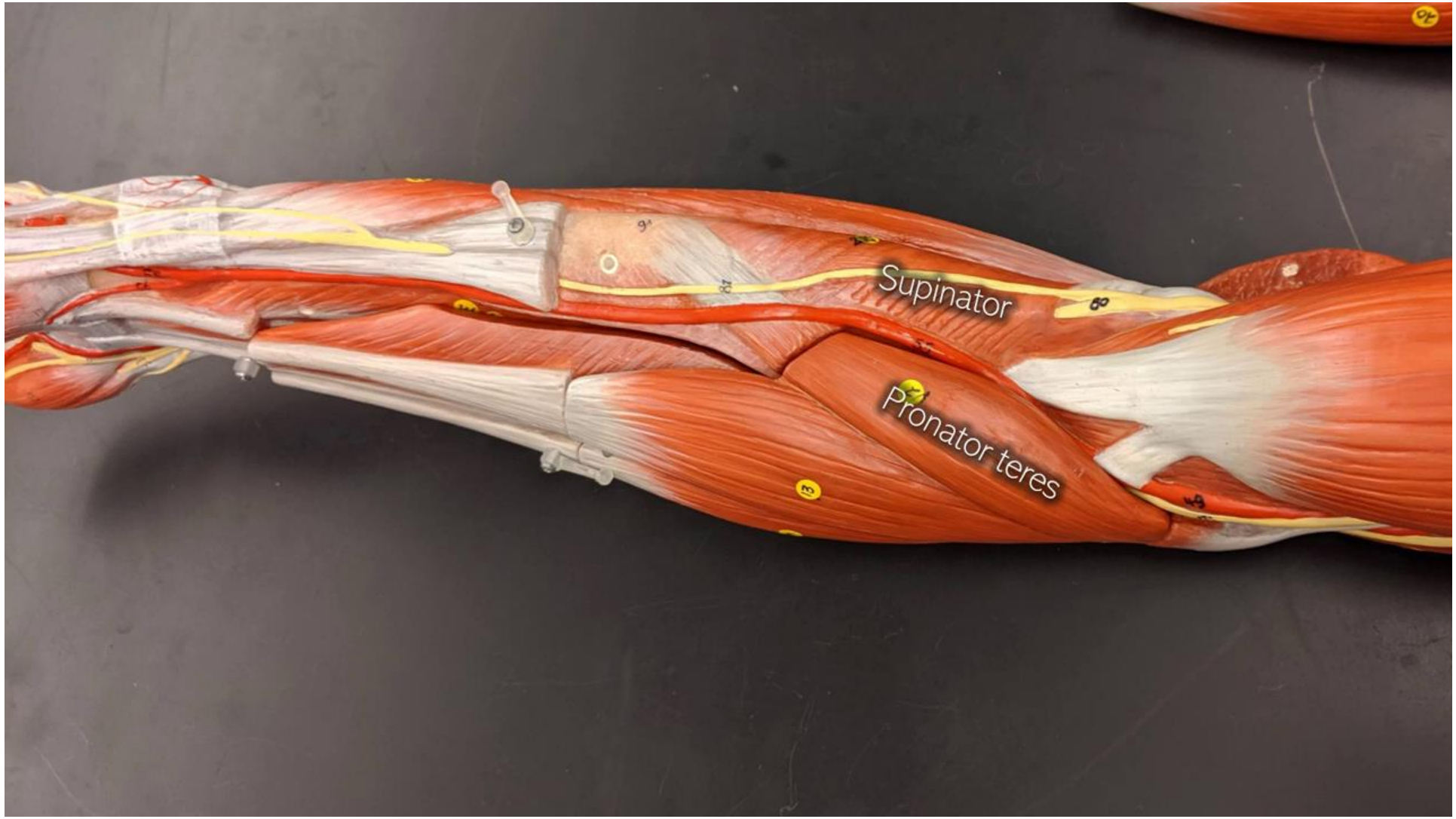
Posterior Extensor

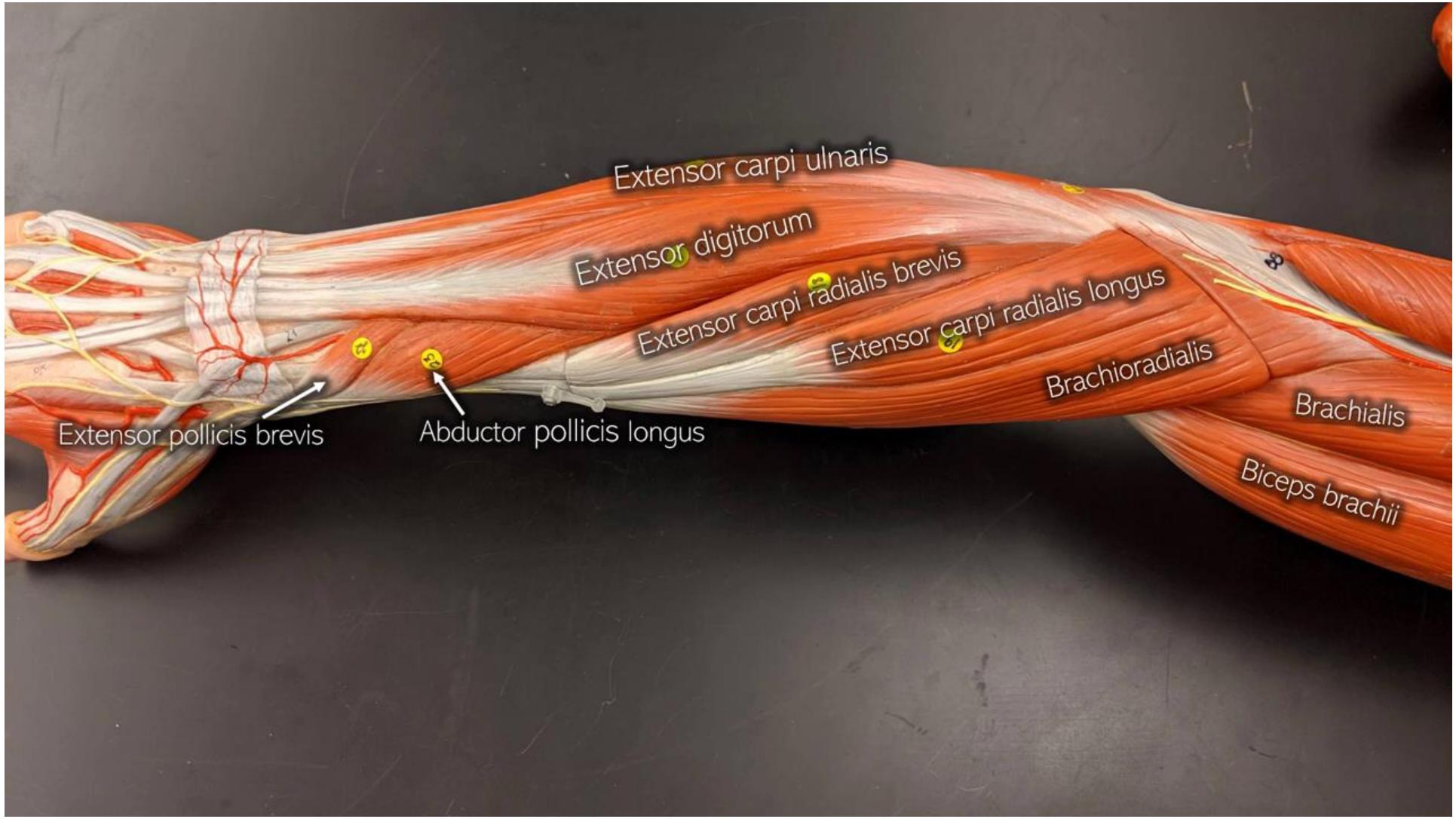

Brachioradialis
Biceps brachii
Pronator teres
Flexor carpi radialis
Palmaris longus
Flexor carpi ulnaris

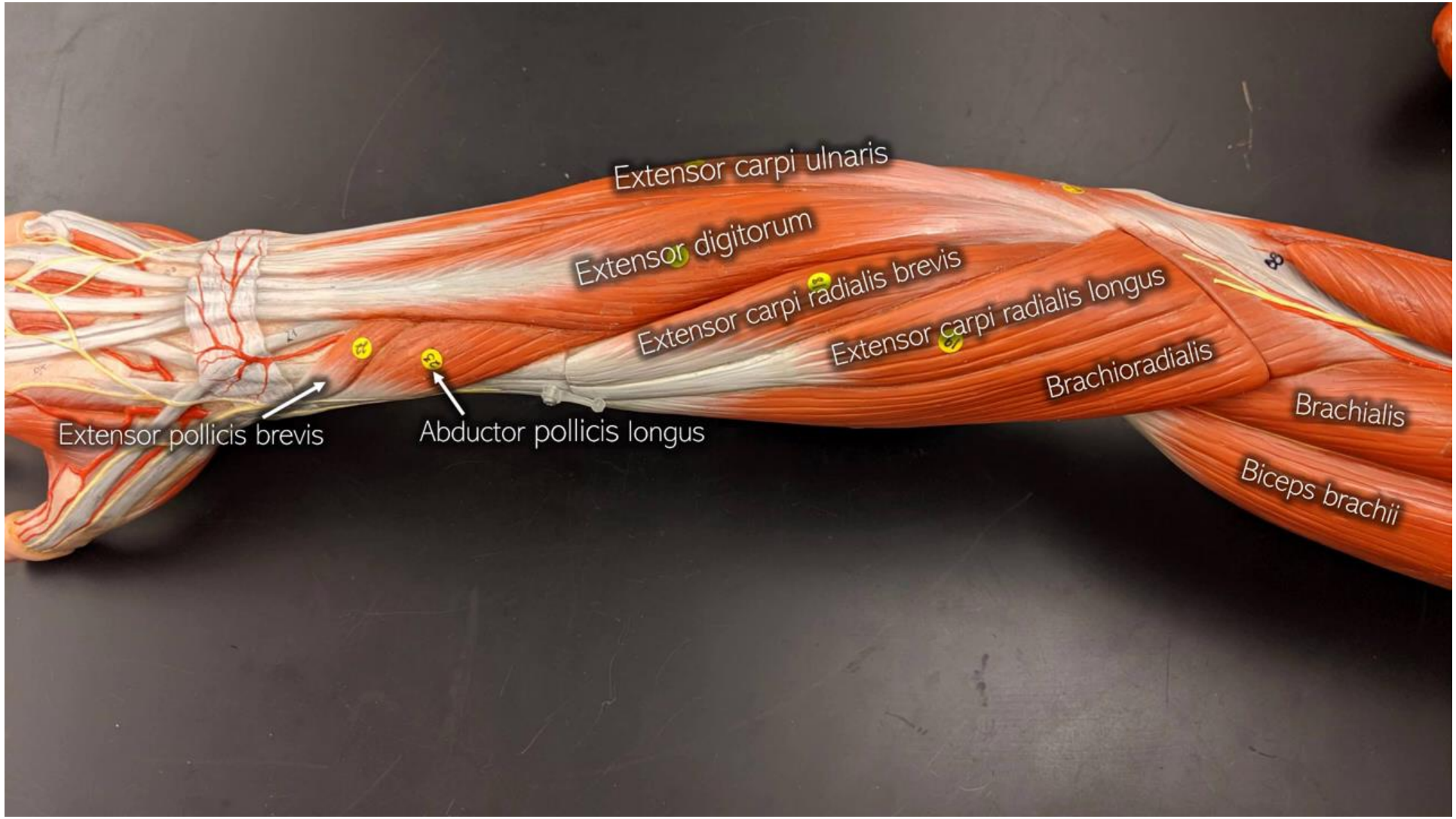

Flexor carpi radialis
Palmaris longus
Flexor carpi ulnaris
Extensor carpi ulnaris

Brachioradialis
Flexor digitorum superficialis

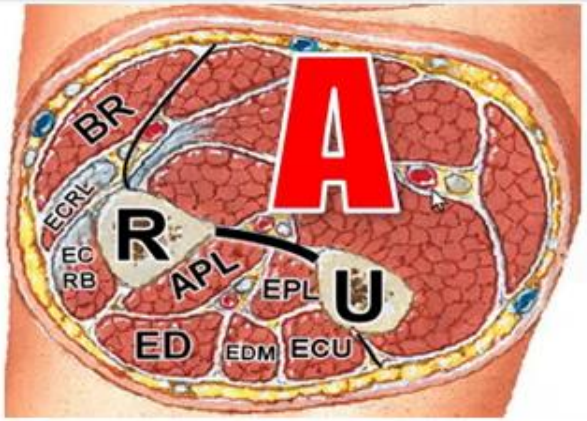

Flexor pollicis longus
Flexor digitorum profundus
Brachioradialis

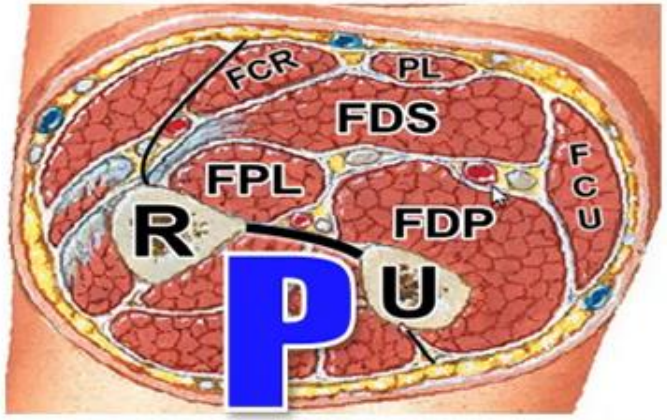

Supinator
Pronator teres


Extensor carpi ulnaris
Extensor digitorum
Extensor carpi radialis brevis
Extensor carpi radialis longus
Brachioradialis
Brachialis
Biceps brachii
Extensor pollicis brevis
Abductor pollicis longus


Extensor carpi ulnaris
Extensor digitorum
Extensor carpi radialis brevis
Extensor carpi radialis longus
Brachioradialis
Brachialis
Biceps brachii
Extensor pollicis brevis
Abductor pollicis longus

Forearm Muscles: 2 Compartments





Anterior Flexor

1. Flexion of the wrist
2. Flexion of the fingers
3. Pronation

Supplied by median nerve except FCU & medial ½ of FDP by ulnar nerve

Posterior Extensor

1. Extension of the wrist
2. Extension of the fingers
3. Supination

Supplied by radial nerve and its posterior interosseous branch

Anterior Compartment: 8 Muscles

8 Muscles:

Superficial: 5

Deep: 3

2 Nerves:

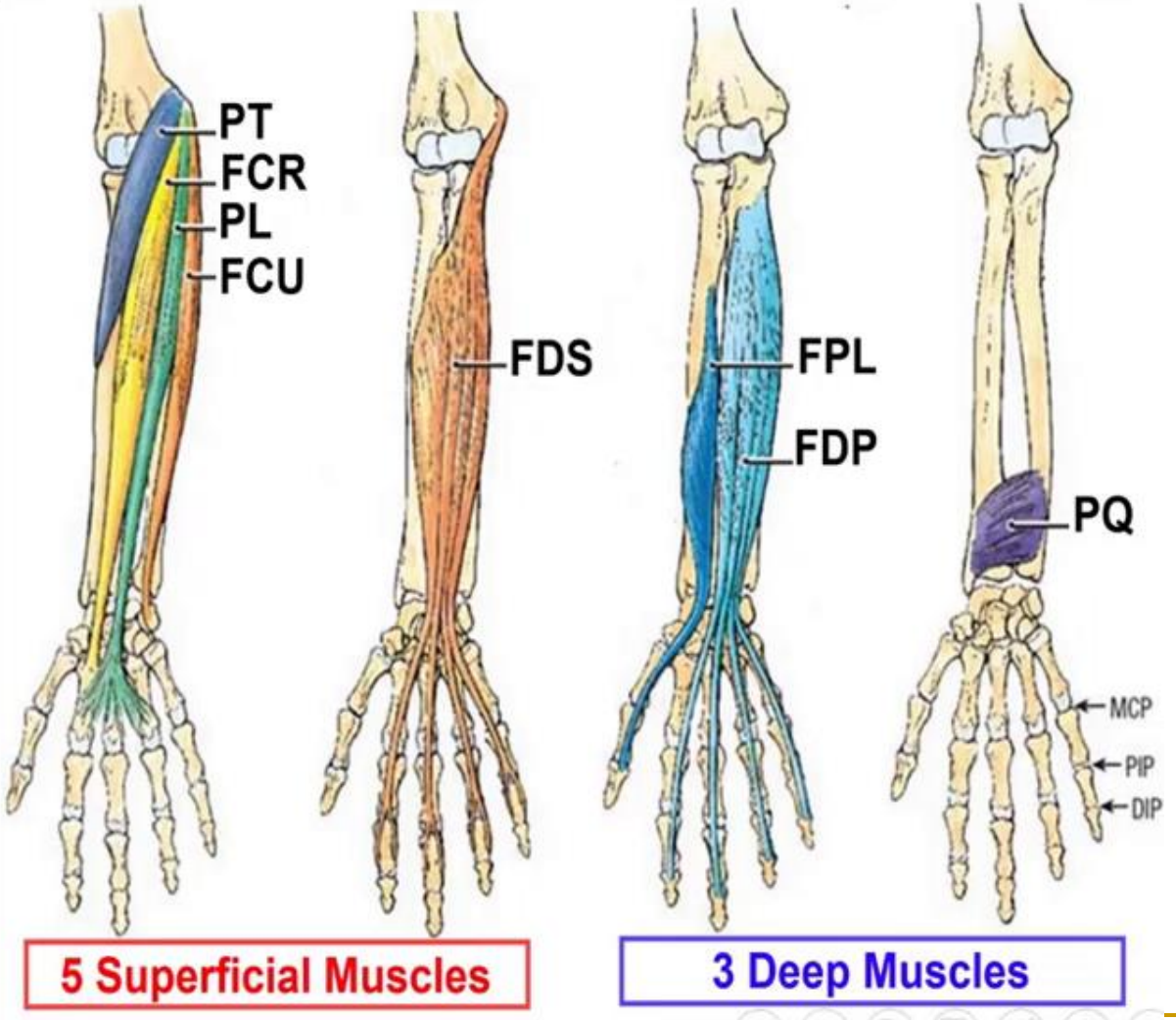
Median nerve

Ulnar nerve

2 Arteries:

Radial artery

Ulnar artery



5 Superficial Muscles

All arise from **common flexor origin**

All are supplied by the **median nerve**
Except FCU by the **ulnar nerve**

All flex the hand (wrist) **Except PT**

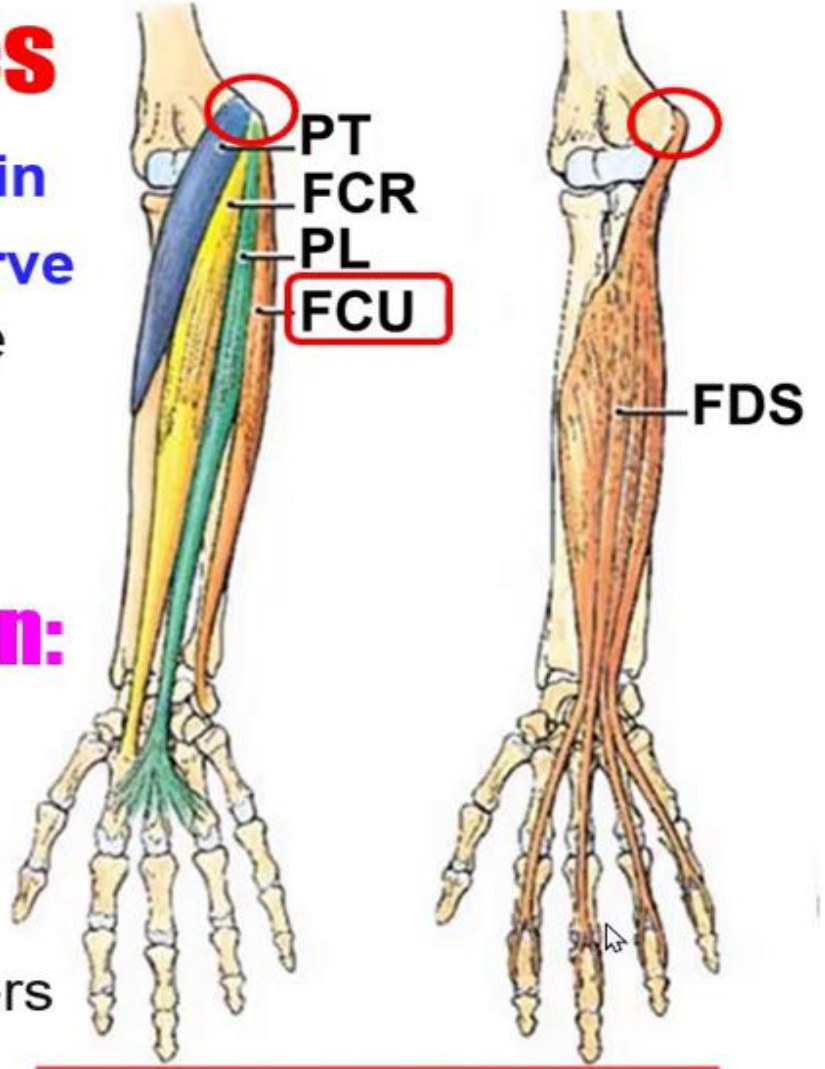
Pronator teres → Pronation

Each has Additional Action:

FCR → Abduction of the hand

FCU → Adduction of the hand

FDS → Flexion of the MPJ &
proximal IPJ of the medial 4 fingers



5 Superficial Muscles

Pronator Teres: PT

Origin by 2 heads:

1. **Humeral head:**
   common flexor origin (front of the medial epicondyle)
2. **Ulnar head:** coronoid process

The median nerve passes between the 2 heads

Insertion

Middle of the lateral surface of the radius

Nerve Supply: Median nerve

Action:

Pronation of the forearm

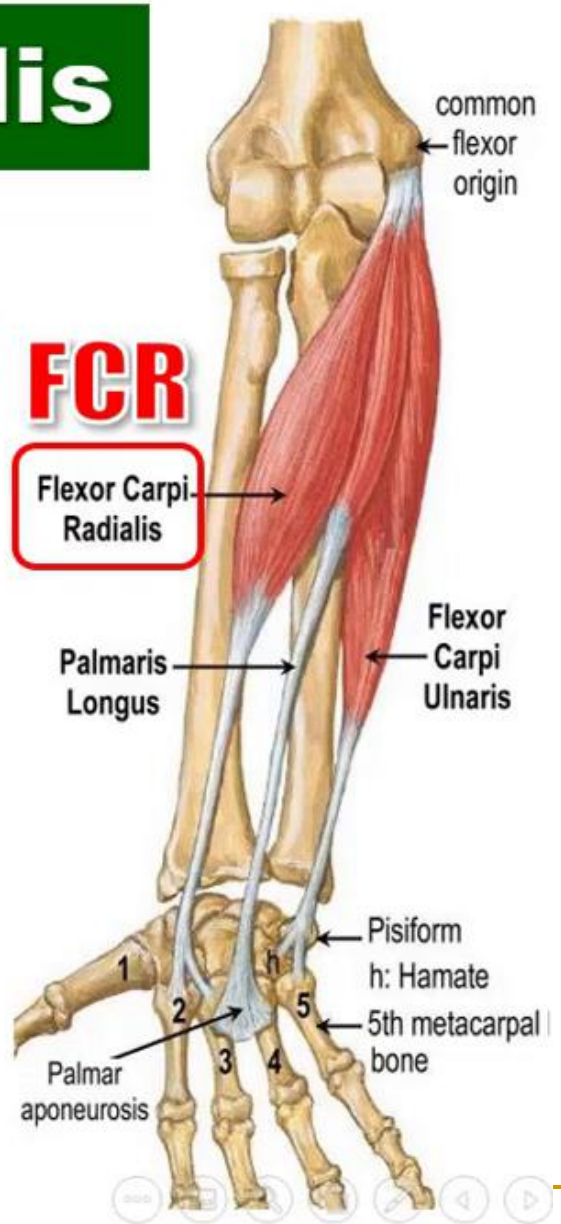
Flexor Carpi Radialis

Origin: common flexor origin from front of the medial epicondyle

Insertion:
2nd & 3rd metacarpal bones

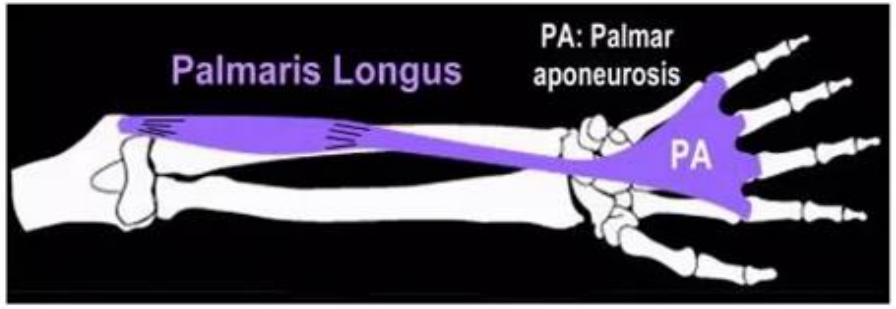
Nerve Supply:
Median nerve

Action:
Flexion & **Abduction** of the wrist joint (hand)



Palmaris Longus: PL

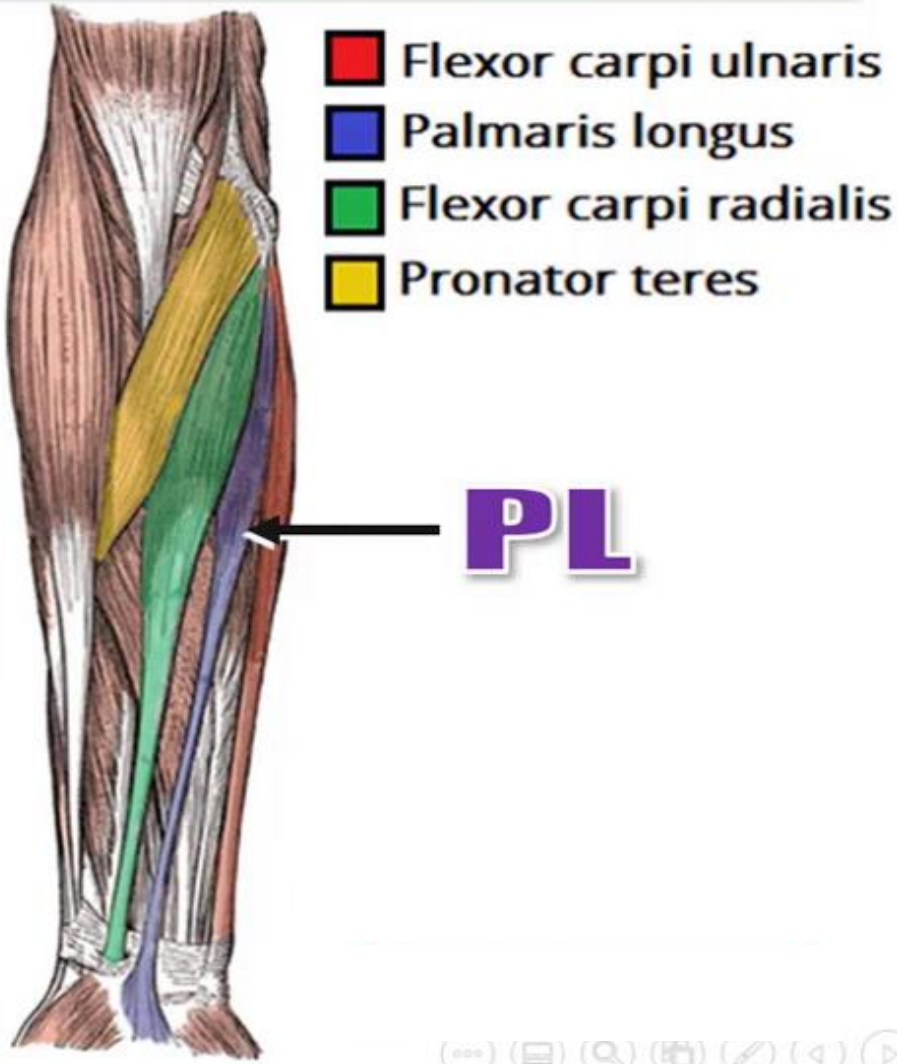
Origin: common flexor origin: front of the medial epicondyle



Insertion:
Palmar aponeurosis

Nerve Supply:
Median nerve

Action:
Flexion of the wrist joint



Flexor Digitorum Superficialis: FDS

Origin by 2 heads:

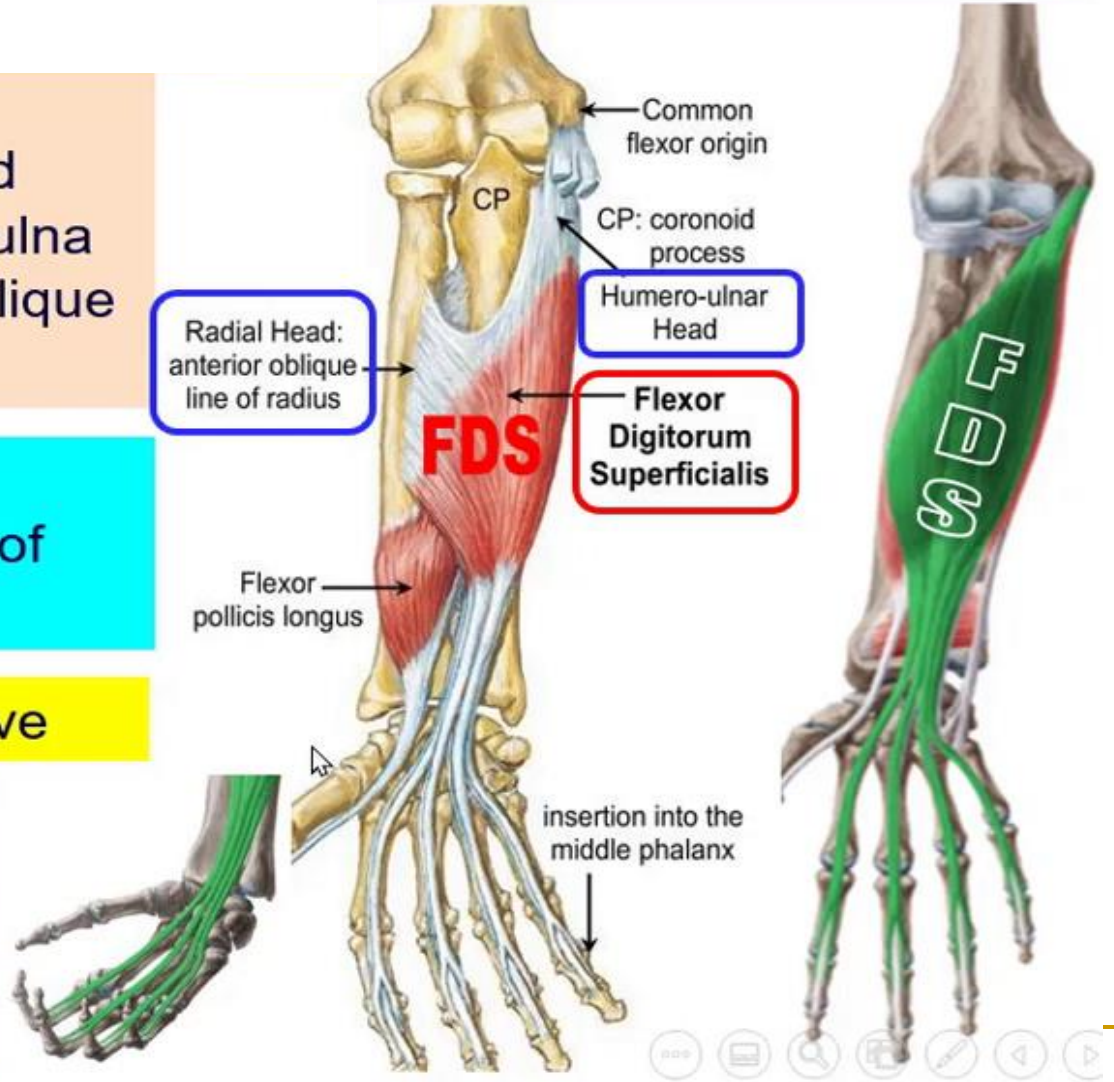
1. **Humero-ulnar head:** common flexor origin and coronoid process of the ulna
2. **Radial head:** anterior oblique line of the radius

Insertion: 4 tendons into the middle phalanx of the medial 4 fingers

Nerve Supply: Median nerve

Action:

Flexion of the **proximal IPJ** and the **MPJ** of the medial 4 fingers.



Flexor Carpi Ulnaris

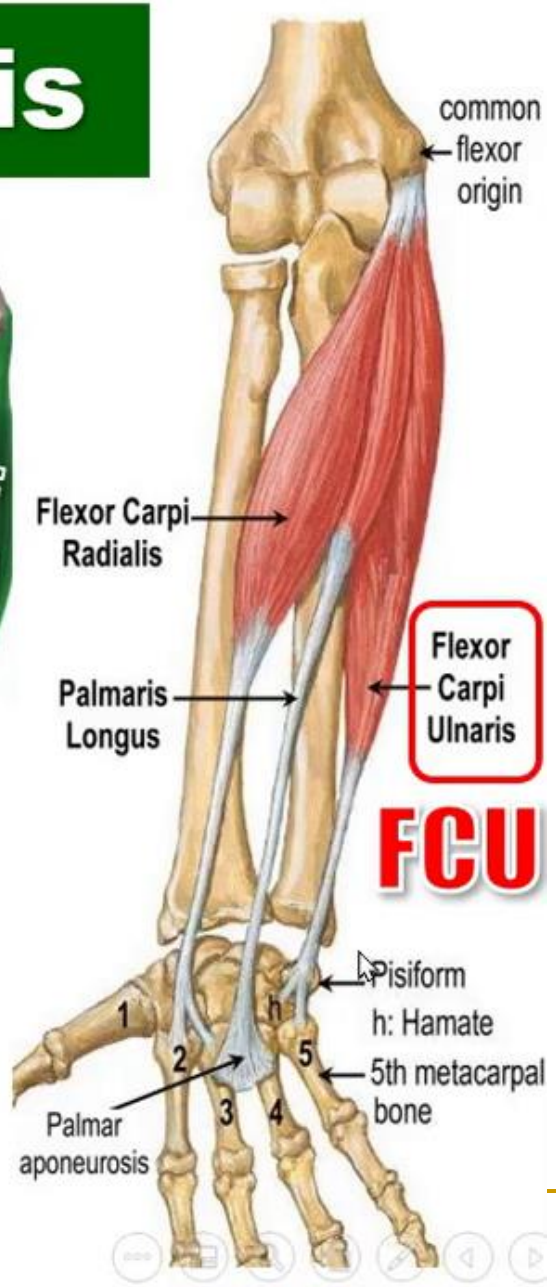
Origin by 2 heads:

1. **Humeral head:** common flexor origin front of the medial epicondyle
2. **Ulnar head:** posterior border of the una

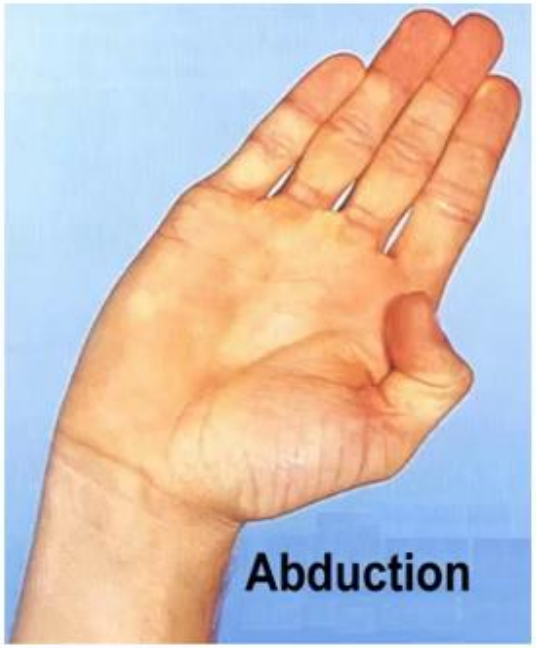
Insertion:
Mainly into the **pisiform bone**

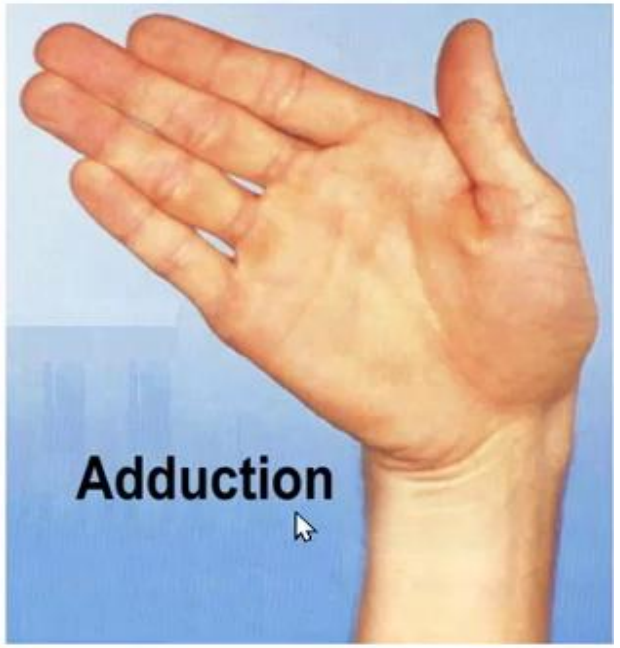
Nerve Supply:
Ulnar nerve

Action:
Flexion & **Adduction** of the wrist joint (hand)



Action of Superficial Muscles on the Wrist





Flexion of the Wrist:

Mainly by the carpi muscles and PL

FCR – FCU - PL

Wrist Abduction:

By the flexor carpi radialis FCR

Wrist Adduction:

By the flexor carpi ulnaris FCU

3 Deep Muscles

1. **Flexor Pollicis Longus:**
   from radius & Inteross membrane
2. **Flexor Digitorum Profundus:**
   from ulna & Inteross membrane
3. **Pronator Quadratus:**
   from the ulna to the radius

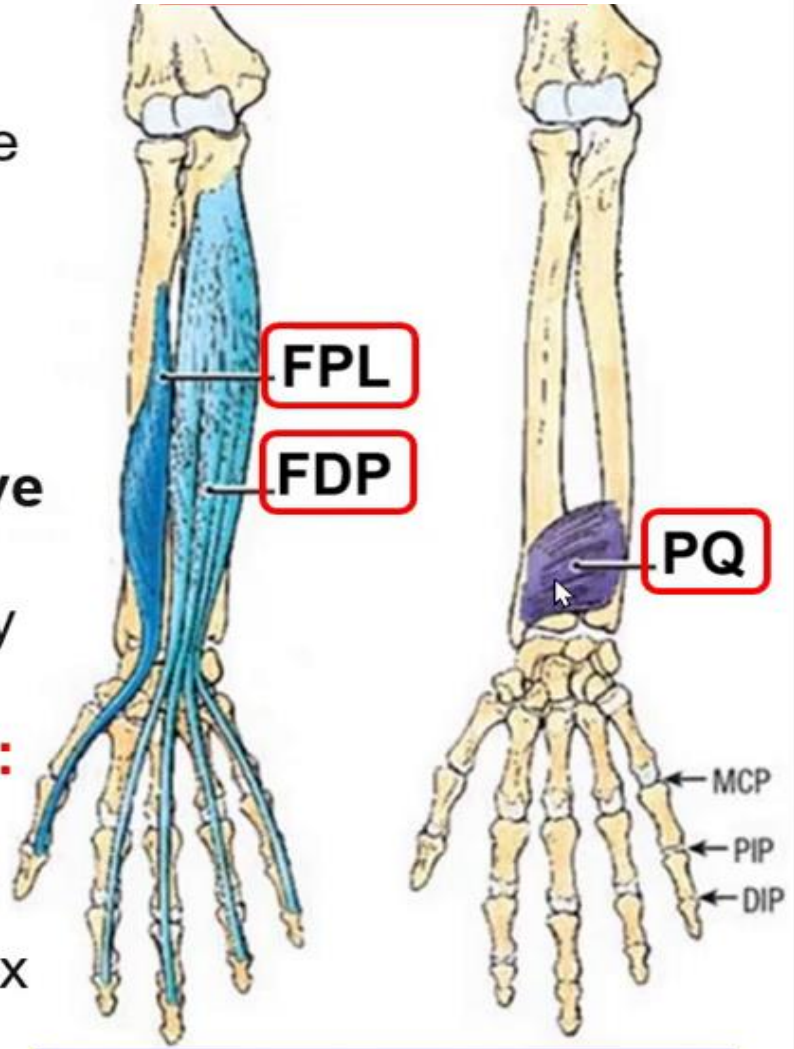
All are supplied by the **median nerve** (**anterior interosseous nerve**)
Except the medial ½ of the **FDP** by the ulnar nerve

Each muscle has a specific action:

Flexor Pollicis Longus ➔ flex all joints of the thumb

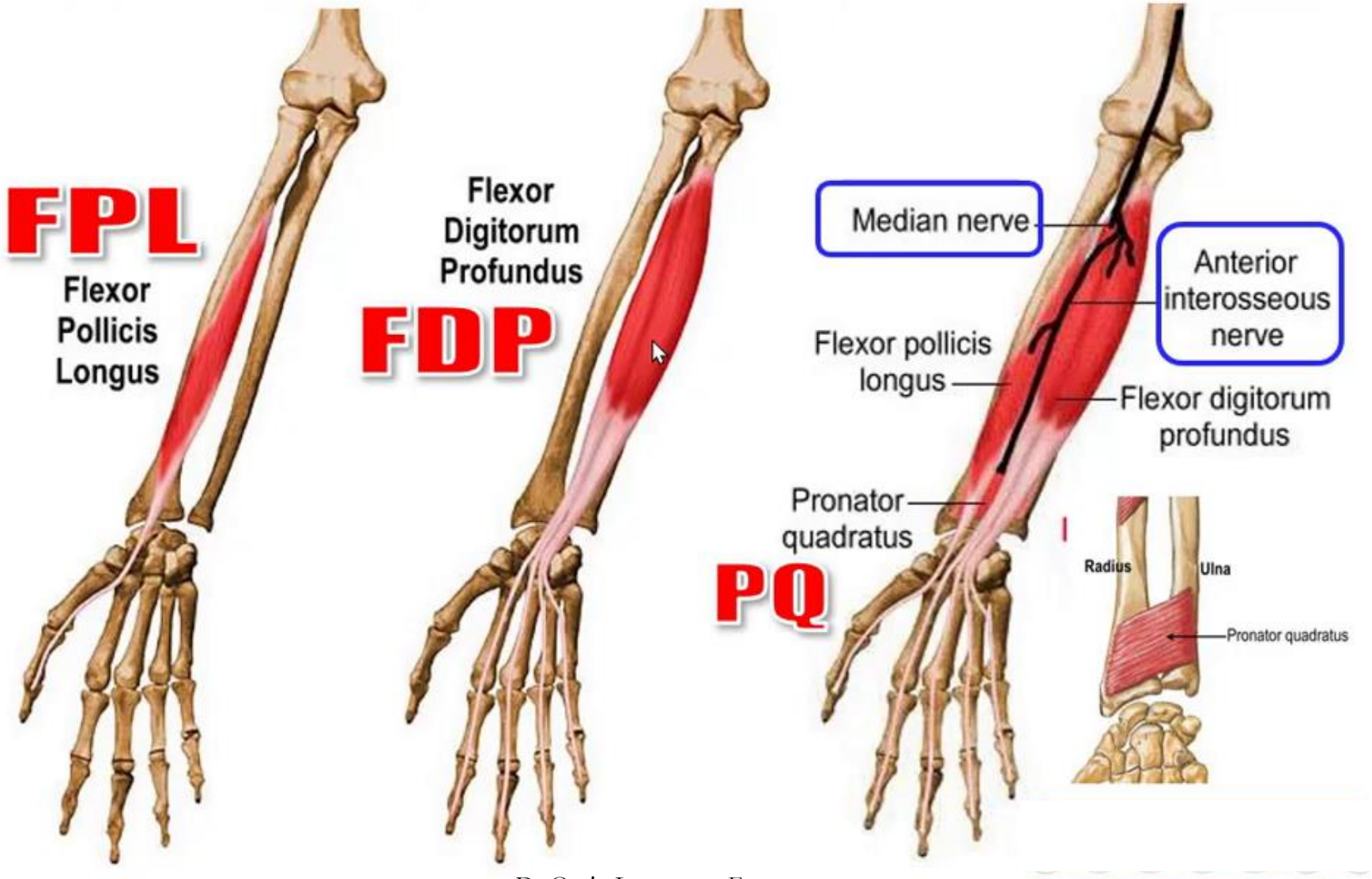
Flexor Digitorum Profundus ➔ flex all joints of the medial 4 fingers

Pronator Quadratus ➔ pronation



3 Deep Muscles

DEEP MUSCLES OF FOREARM



Flexor Digitorum Profundus

Origin:

1. Anterior & medial surfaces of the ulna.
2. Interosseous membrane

Insertion: 4 tendons

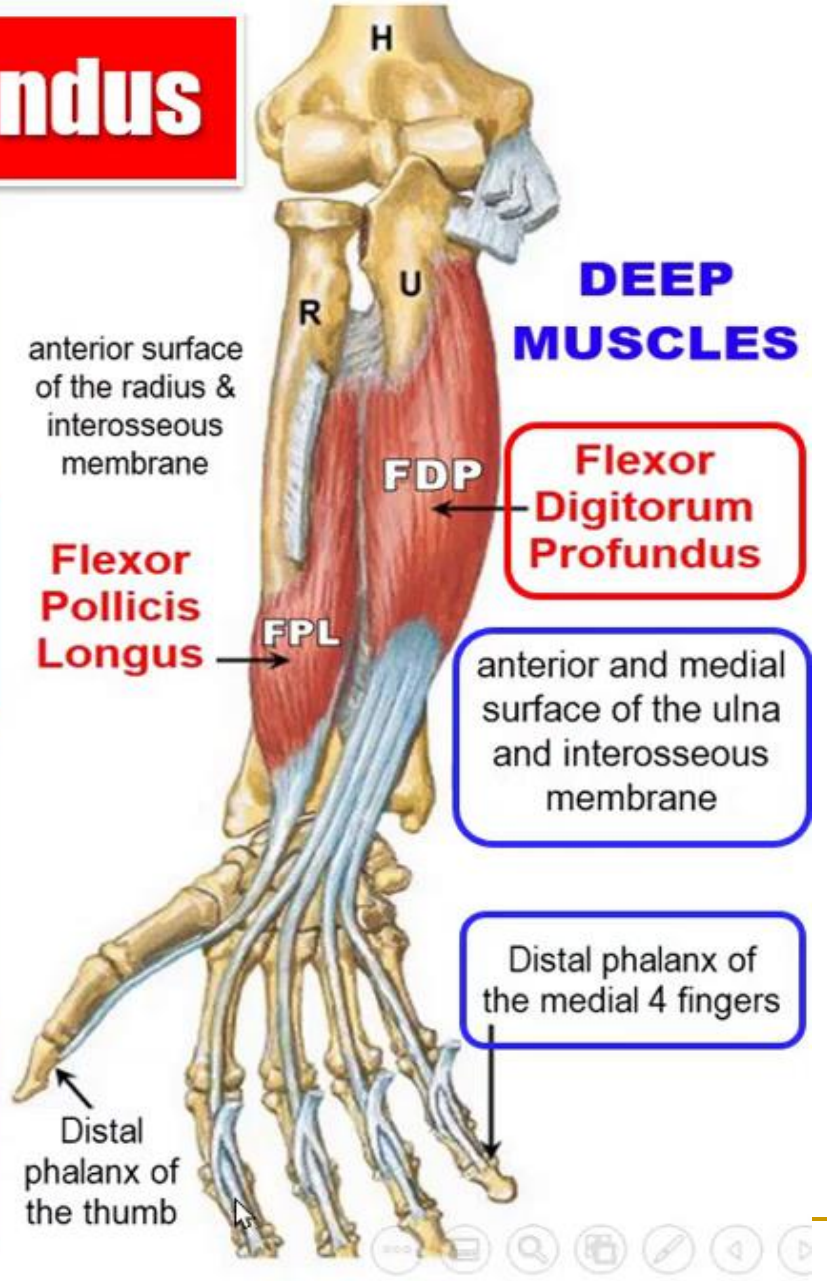
into the distal phalanx of the medial 4 fingers

Nerve Supply:

1. **Medial ½:** ulnar nerve
2. **Lateral ½:** anterior interosseous nerve from the median nerve

Action:

Flexion of **all joints** of the medial 4 fingers.



Flexor Pollicis Longus: FPL

Origin:

1. Anterior surface of the radius.
2. Interosseous membrane

Insertion:

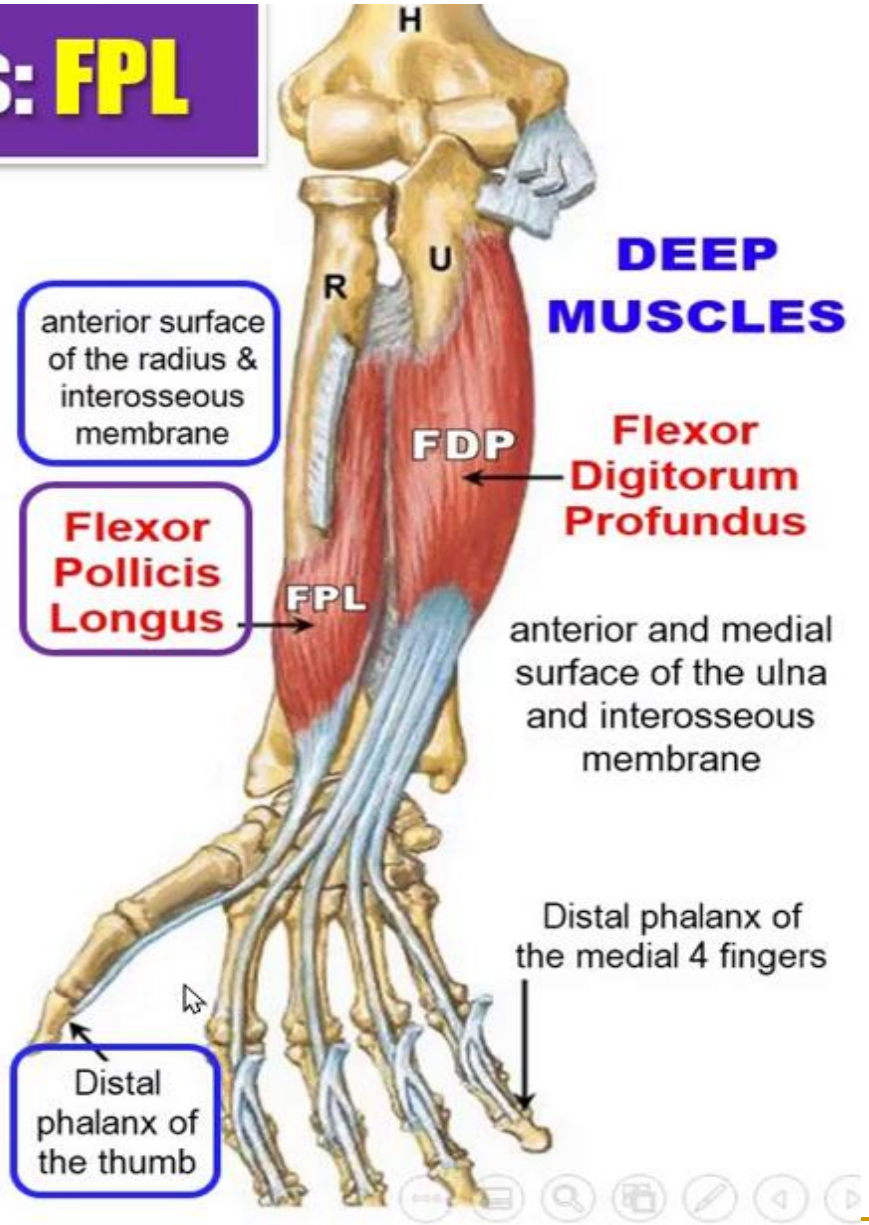
into the distal phalanx of the thumb

Nerve Supply:

anterior interosseous nerve from the median nerve

Action:

Flexion of **all joints** of the thumb



Pronator Quadratus: PQ

Origin:

Lower ¼ of the anterior surface of the **ulna**

Insertion:

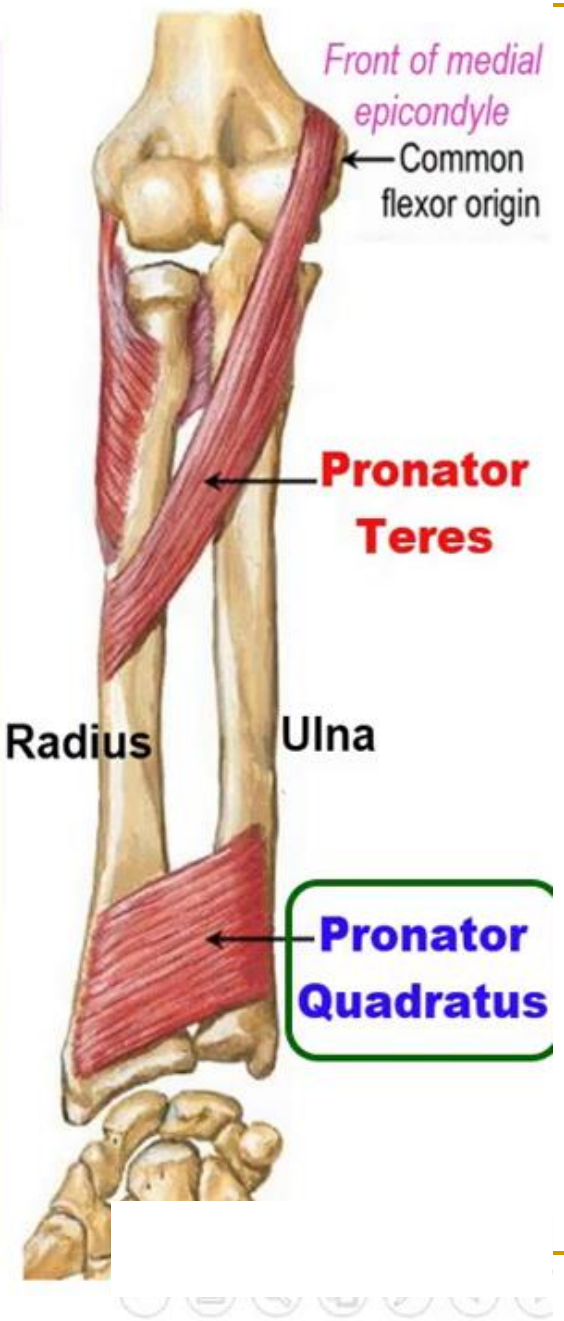
Lower ¼ of the anterior surface of the **radius**

Nerve Supply:

anterior interosseous nerve from the median nerve

Action:

Main pronator of the forearm



Pronator teres

- Medial epicondyle of humerus and coronoid process of ulna
- Middle of lateral surface of radius (pronator tuberosity)
- Median nerve (C6, **C7**)
- Pronates forearm and flexes elbow

Flexor carpi radialis

- Medial epicondyle of humerus
- Base of 2nd metacarpal
- Flexes wrist and abducts hand

Palmaris longus

- Distal half of flexor retinaculum and palmar aponeurosis
- Median nerve (C7,**C8**)
- Flexes wrist and tightens palmar aponeurosis

Flexor carpi ulnaris

- *Humeral head:* medial epicondyle of humerus; *Ulnar head:* olecranon and posterior border of ulna
- Pisiform, hook of hamate, and 5th metacarpal
- Ulnar nerve (C7,**C8**)
- Flexes wrist and adducts hand

Flexor digitorum superficialis

- *Humeroulnar head:* medial epicondyle of humerus, ulnar collateral ligament, and coronoid process of ulna.
- *Radial head:* superior half of anterior border of radius
- Bodies of middle phalanges of medial four digits
- Median nerve (C7, **C8** , and T1)
- Flexes PIPs of medial four digits; acting more strongly, it flexes MCPs and hand

Flexor digitorum profundus

- Proximal three quarters of medial and anterior surfaces of ulna and interosseous membrane
- Bases of distal phalanges of medial four digits
- *Medial part:* ulnar nerve (**C8,**T1)
- *Lateral part:* median nerve (**C8,**T1)
- Flexes DIPs of medial four digits; assists with flexion of wrist

Flexor pollicis longus

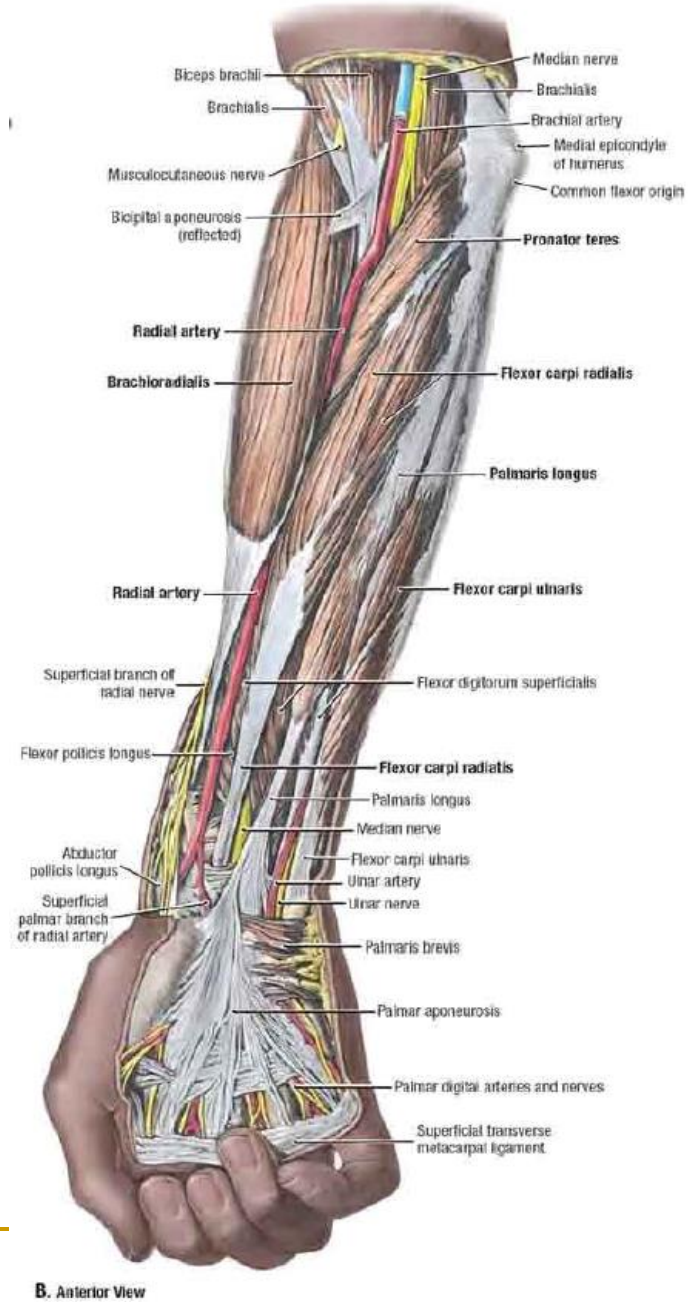
- Anterior surface of radius and adjacent interosseous membrane
- Base of distal phalanx of thumb
- Anterior interosseous nerve from median (**C8,**T1)
- Flexes phalanges of 1st digit (thumb)

Pronator quadratus

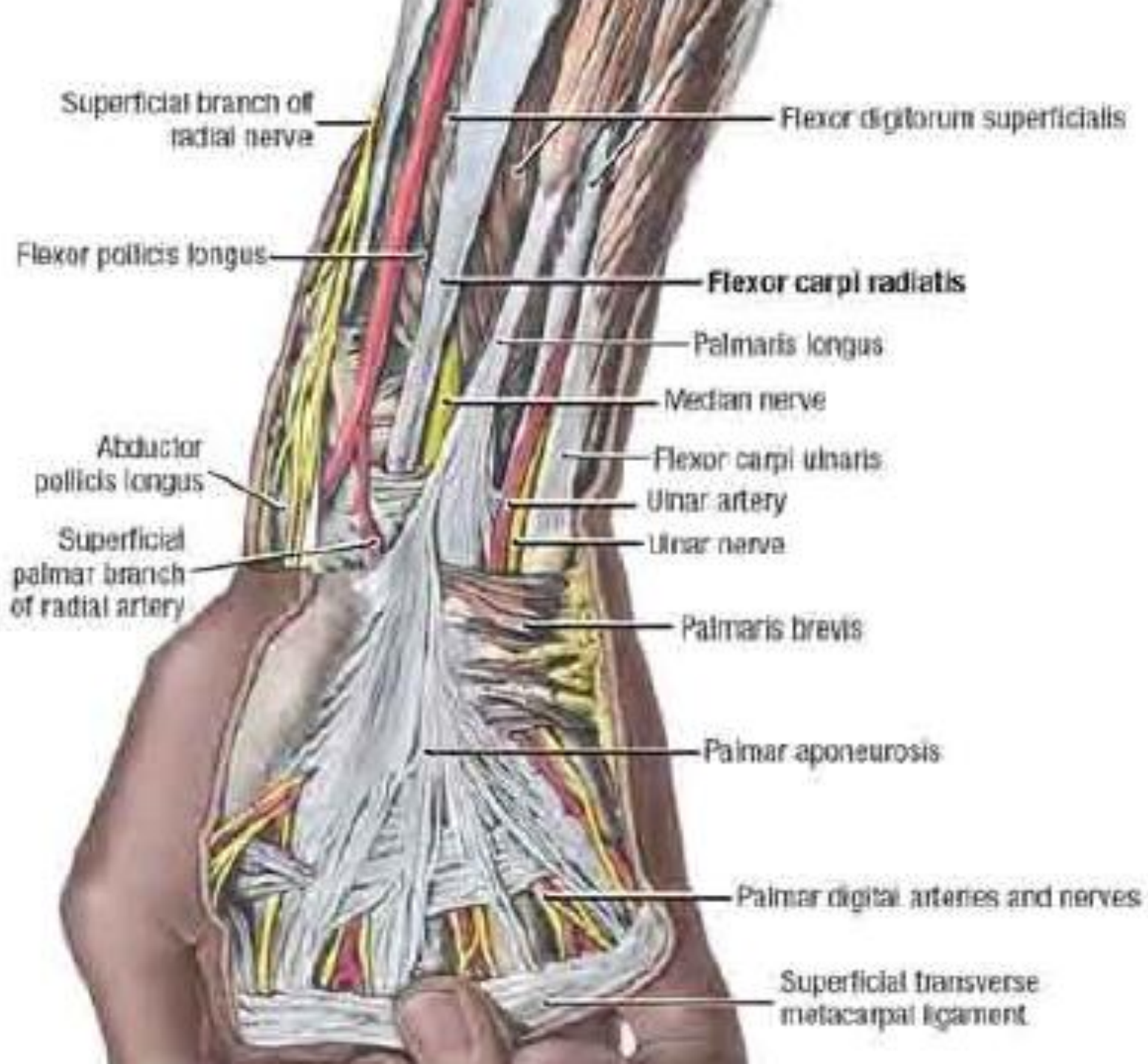
- Distal fourth of anterior surface of ulna
- Distal fourth of anterior surface of radius
- Pronates forearm; deep fibers bind radius and ulna together

Muscles of Anterior Surface of Forearm and relations

Common flexor origin
Pronator teres
Brachioradialis
Palmaris longus
Flexor carpi radialis
Flexor carpi ulnaris
Flexor retinaculum
A. Anterior View



B. Anterior View


Superficial branch of radial nerve
Flexor digitorum superficialis
Flexor pollicis longus
Flexor carpi radialis
Palmaris longus
Median nerve
Abductor pollicis longus
Flexor carpi ulnaris
Ulnar artery
Superficial palmar branch of radial artery
Ulnar nerve
Palmaris brevis
Palmar aponeurosis
Palmar digital arteries and nerves
Superficial transverse metacarpal ligament

At the elbow,

- the brachial artery lies between the biceps tendon and median nerve.
- It then bifurcates into the radial and ulnar arteries.

At the wrist,

- The radial artery is **lateral** to the flexor carpi radialis tendon,
- The ulnar artery is **lateral** to flexor carpi ulnaris tendon.

- In the forearm, the radial artery lies between the flexor and extensor compartments.
- The muscles lateral to the artery are supplied by the radial nerve
- Those medial to it by the median and ulnar nerves; thus, no motor nerve crosses the radial artery.
- The brachioradialis muscle slightly overlaps the radial artery, which is otherwise superficial.

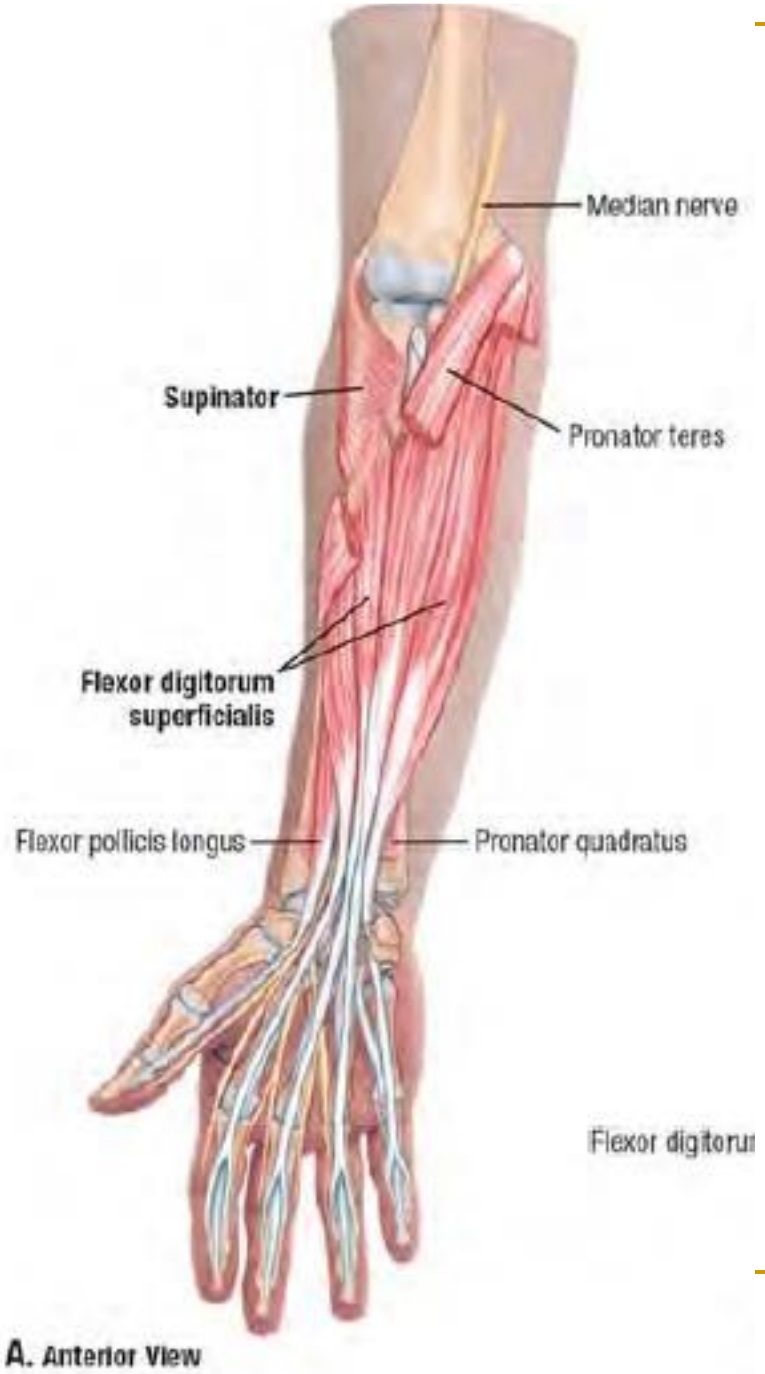
The **four superficial muscles**:

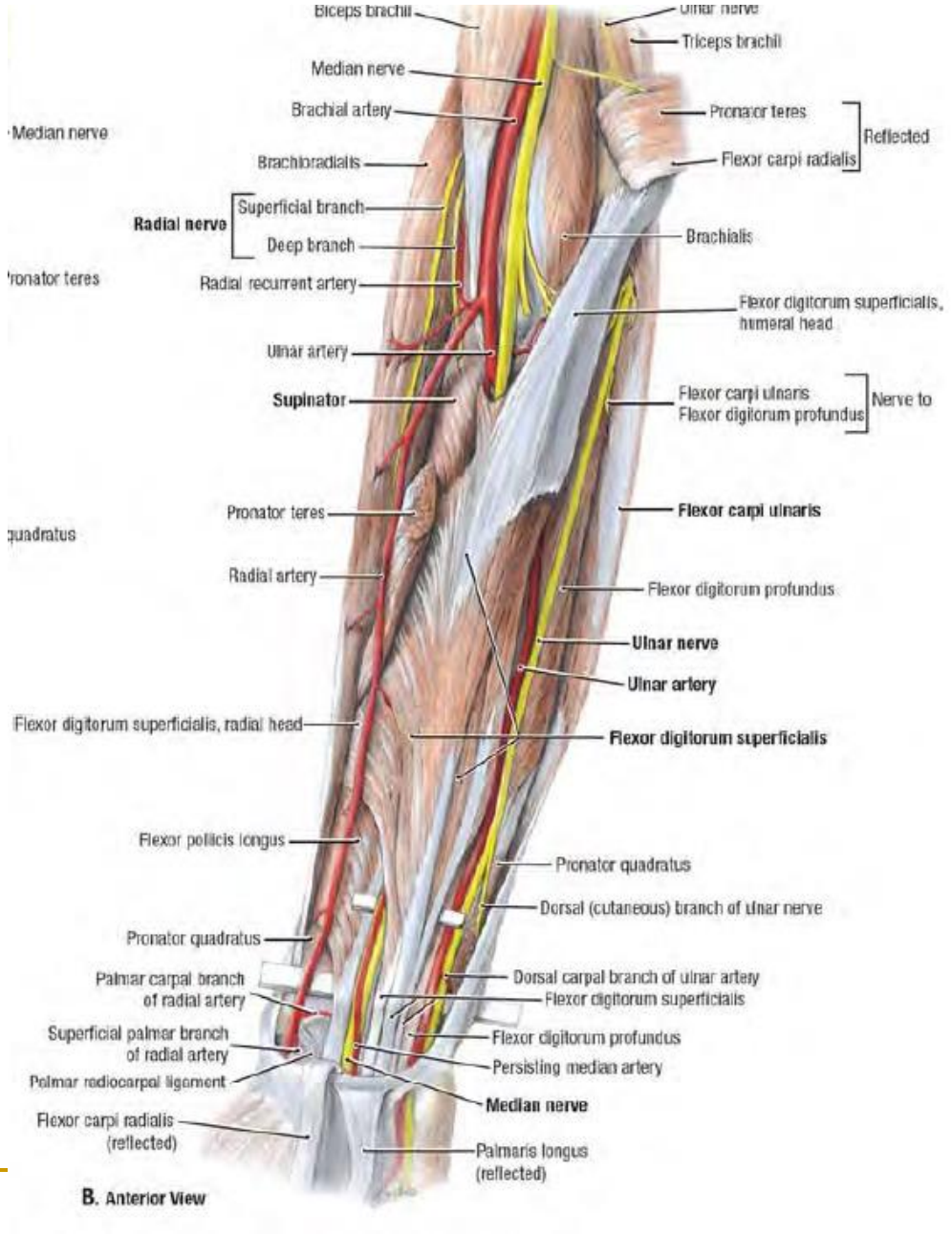
- Pronator teres,
- Flexor carpi radialis,
- Palmaris longus,
- Flexor carpi ulnaris

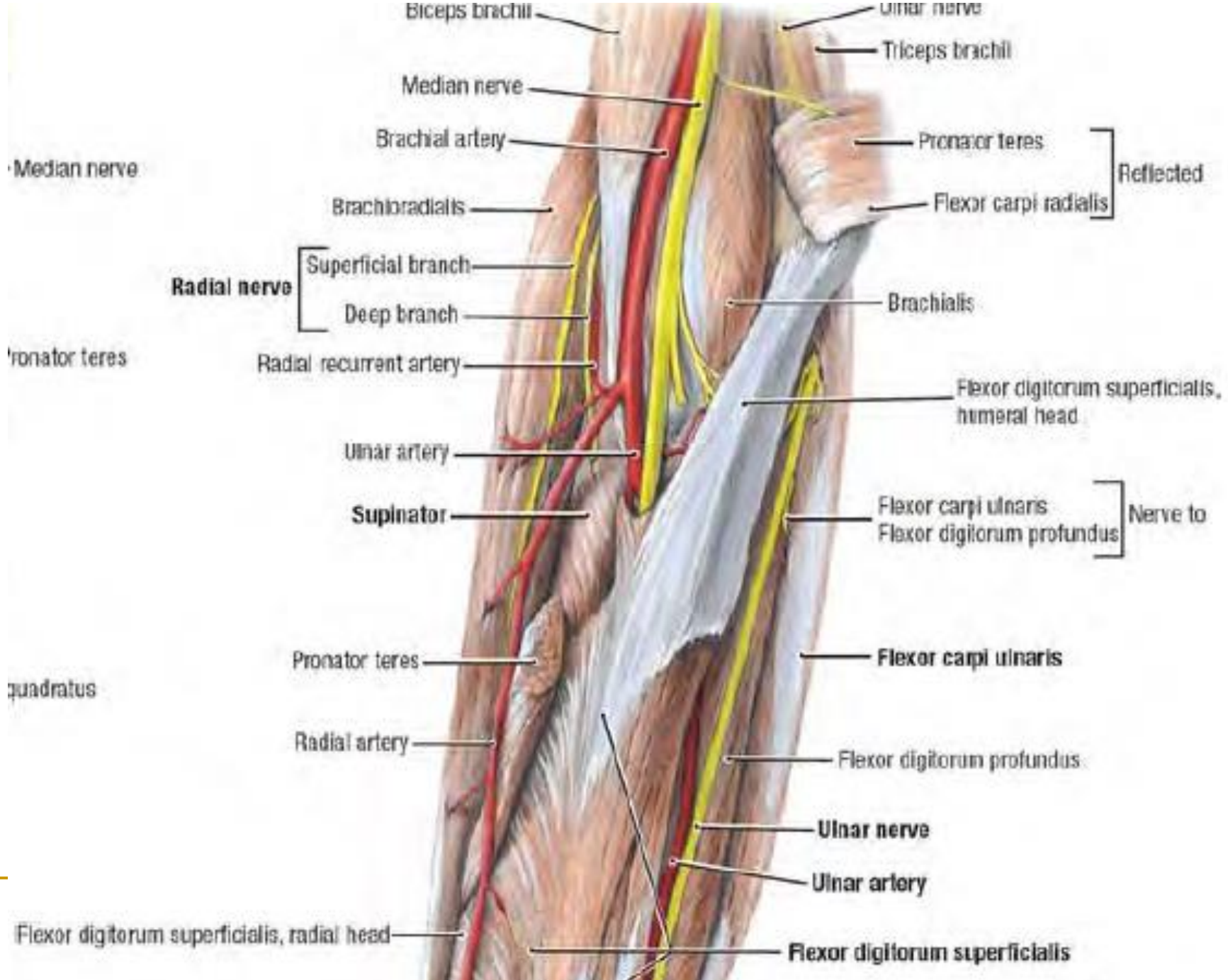
all attach proximally to the medial epicondyle of the humerus (common flexor origin)

The palmaris longus muscle,

- This muscle usually has a **small belly at the common flexor origin**
- A long tendon that is continued into the palm as the palmar aponeurosis but it could have an anomalous distal belly
- The palmaris longus is absent in approximately 14% of limbs


Median nerve
Supinator
Pronator teres
Flexor digitorum superficialis
Flexor pollicis longus
Pronator quadratus
Flexor digitoru
A. Anterior View

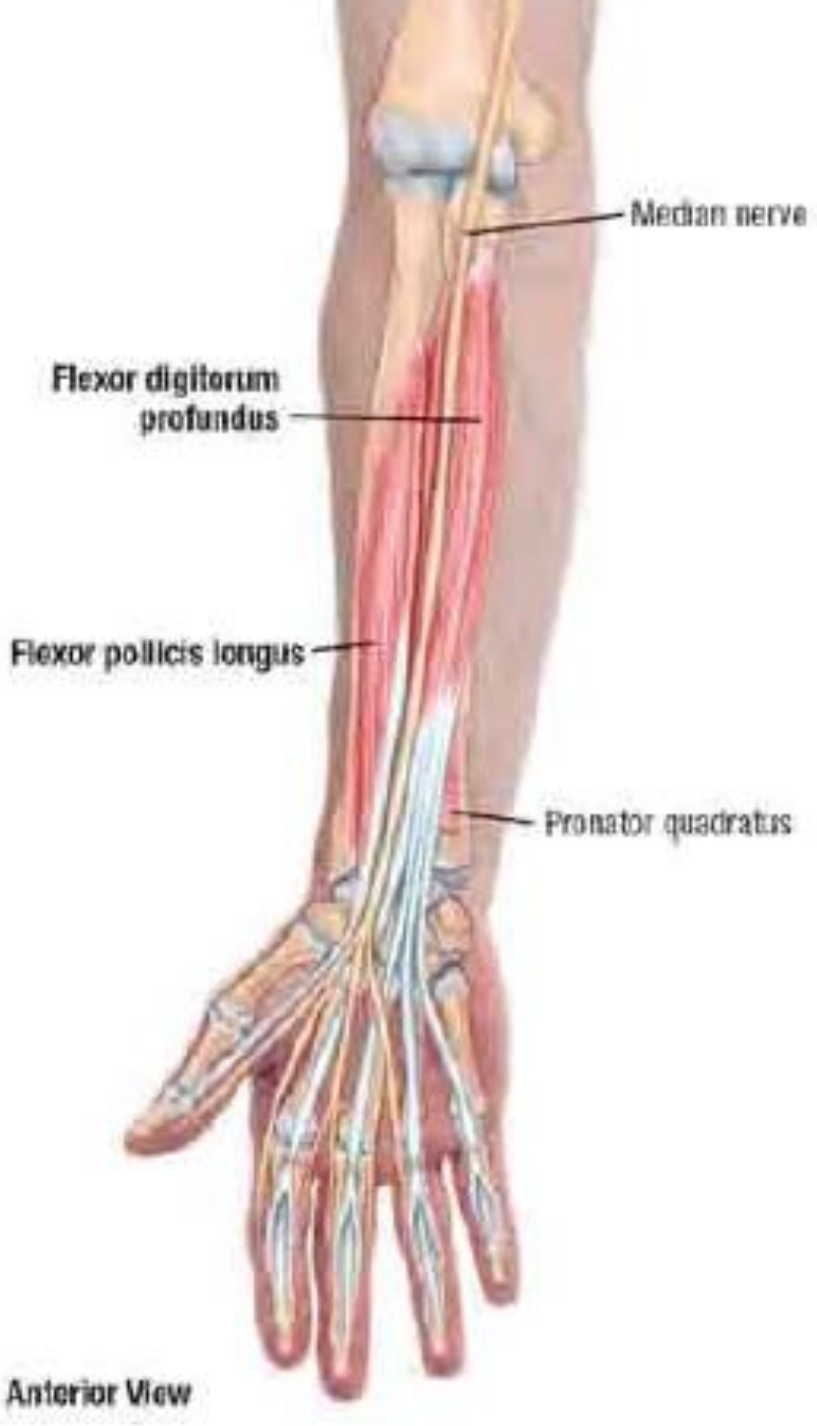

Biceps brachii
Ulnar nerve
Triceps brachii
Median nerve
Brachial artery
Median nerve
Pronator teres
Reflected
Flexor carpi radialis
Brachioradialis
Radial nerve
Superficial branch
Deep branch
Brachialis
Pronator teres
Radial recurrent artery
Flexor digitorum superficialis, humeral head
Ulnar artery
Supinator
Flexor carpi ulnaris
Flexor digitorum profundus
Nerve to
Pronator teres
Flexor carpi ulnaris
quadratus
Radial artery
Flexor digitorum profundus
Ulnar nerve
Ulnar artery
Flexor digitorum superficialis, radial head
Flexor digitorum superficialis
Flexor pollicis longus
Pronator quadratus
Dorsal (cutaneous) branch of ulnar nerve
Pronator quadratus
Palmar carpal branch of radial artery
Dorsal carpal branch of ulnar artery
Flexor digitorum superficialis
Superficial palmar branch of radial artery
Flexor digitorum profundus
Persisting median artery
Palmar radiocarpal ligament
Median nerve
Flexor carpi radialis (reflected)
Palmaris longus (reflected)
B. Anterior View

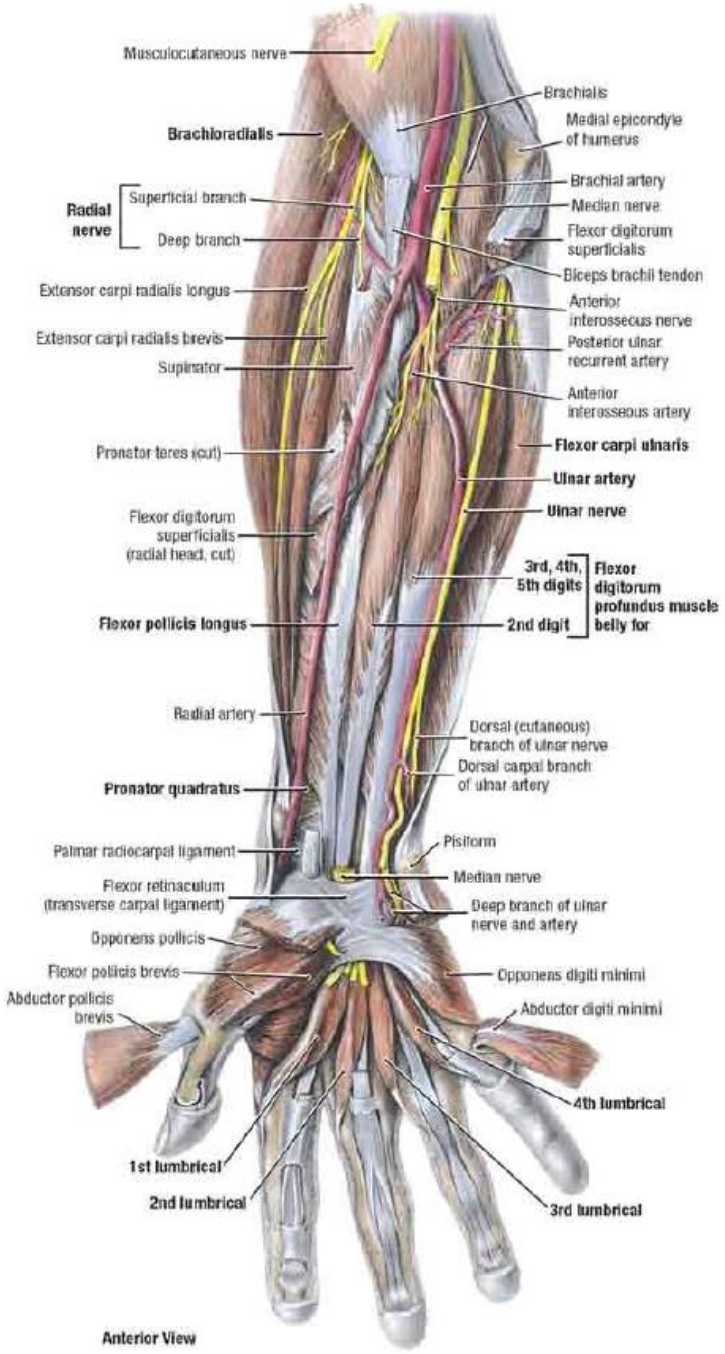

Biceps brachii
Ulnar nerve
Triceps brachii
Median nerve
Brachial artery
Median nerve
Pronator teres
Reflected
Flexor carpi radialis
Brachioradialis
Radial nerve
Superficial branch
Deep branch
Brachialis
Pronator teres
Radial recurrent artery
Flexor digitorum superficialis, humeral head
Ulnar artery
Supinator
Flexor carpi ulnaris
Flexor digitorum profundus
Nerve to
Pronator teres
Flexor carpi ulnaris
quadratus
Radial artery
Flexor digitorum profundus
Ulnar nerve
Ulnar artery
Flexor digitorum superficialis, radial head
Flexor digitorum superficialis

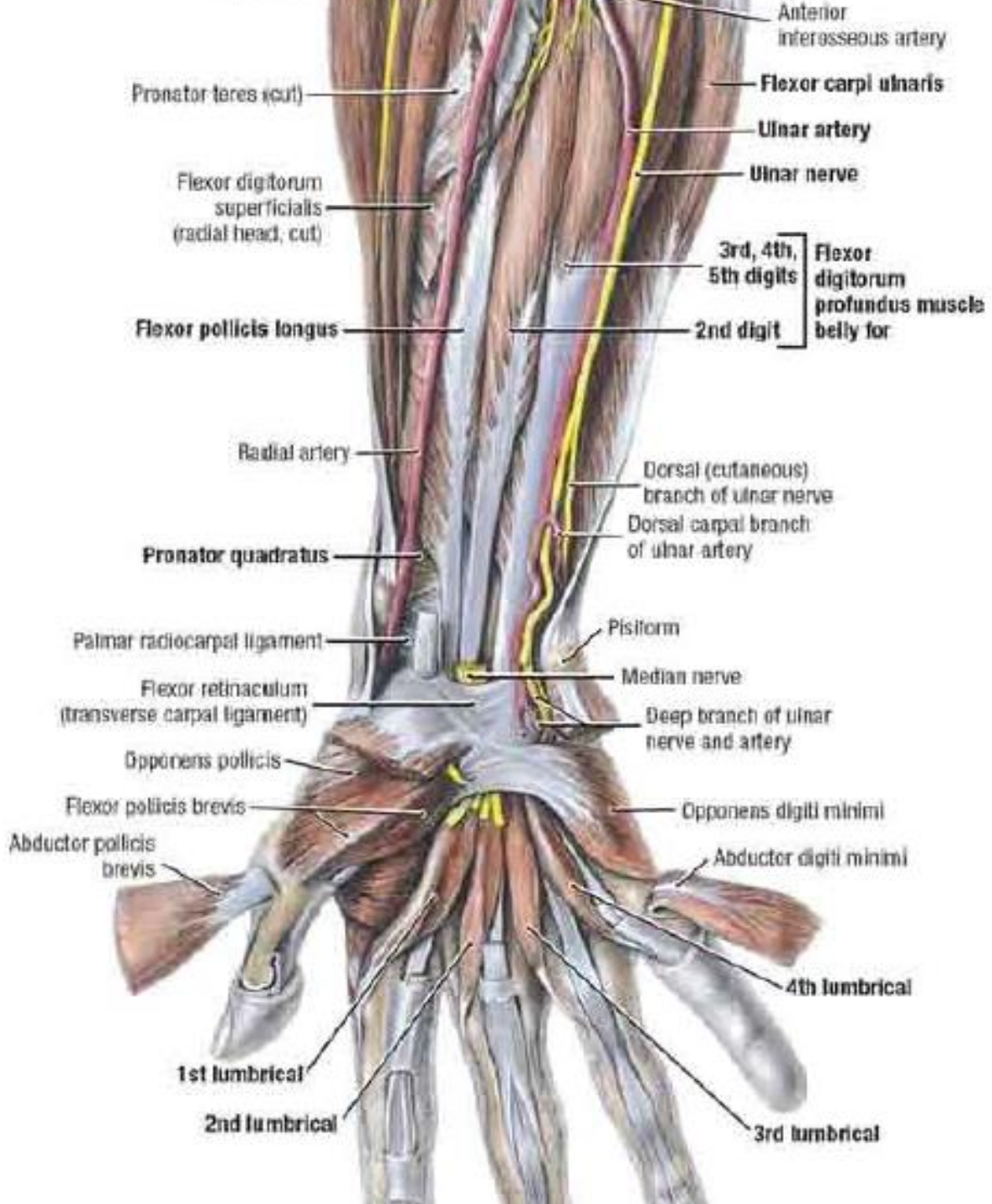
Tracing the path of Ulnar Artery and Nerve

- The flexor digitorum superficialis muscle is attached proximally to the humerus, ulna, and radius.
- The ulnar artery passes obliquely posterior to the flexor digitorum superficialis; at the medial border of the muscle, the ulnar artery joins the ulnar nerve.
- The ulnar nerve lies between the flexor digitorum profundus and flexor carpi ulnaris.

- The median nerve descends vertically posterior to the flexor digitorum superficialis and appears distally at its lateral border.


Median nerve
Flexor digitorum profundus
Flexor pollicis longus
Pronator quadratus
Anterior View

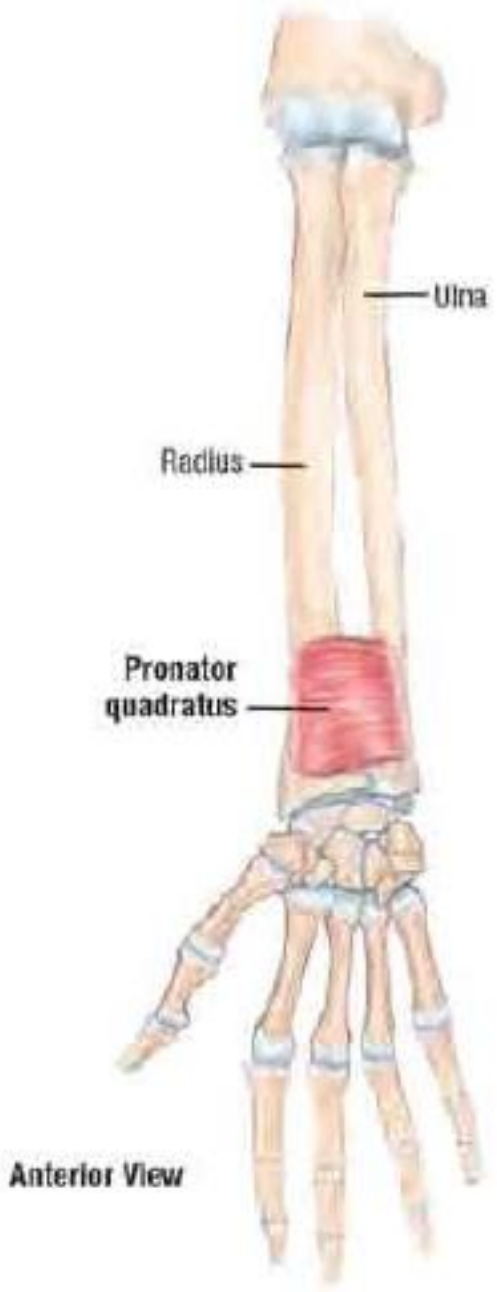

Musculocutaneous nerve
Brachialis
Brachioradialis
Medial epicondyle of humerus
Radial nerve
Superficial branch
Deep branch
Brachial artery
Median nerve
Flexor digitorum superficialis
Biceps brachii tendon
Extensor carpi radialis longus
Anterior interosseous nerve
Extensor carpi radialis brevis
Posterior ulnar recurrent artery
Supinator
Anterior interosseous artery
Pronator teres (cut)
Flexor carpi ulnaris
Ulnar artery
Ulnar nerve
Flexor digitorum superficialis (radial head, cut)
3rd, 4th, 5th digits
Flexor digitorum profundus muscle belly for
Flexor pollicis longus
2nd digit
Radial artery
Dorsal (cutaneous) branch of ulnar nerve
Dorsal carpal branch of ulnar artery
Pronator quadratus
Pisiform
Palmar radiocarpal ligament
Median nerve
Flexor retinaculum (transverse carpal ligament)
Deep branch of ulnar nerve and artery
Opponens pollicis
Flexor pollicis brevis
Opponens digiti minimi
Abductor pollicis brevis
Abductor digiti minimi
4th lumbrical
1st lumbrical
2nd lumbrical
3rd lumbrical
Anterior View

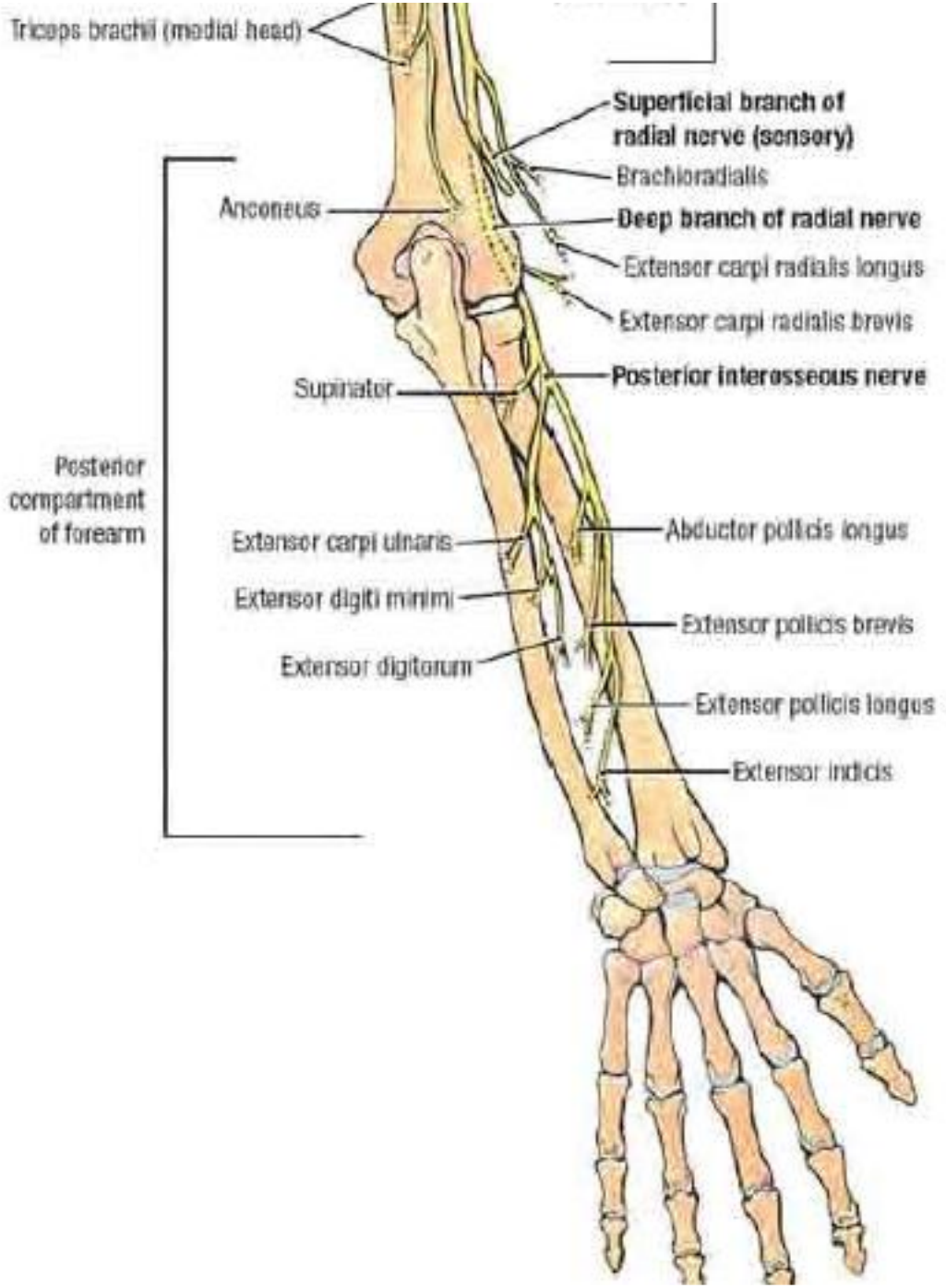

Anterior interosseous artery
Pronator teres (cut)
Flexor carpi ulnaris
Ulnar artery
Ulnar nerve
Flexor digitorum superficialis (radial head, cut)
3rd, 4th, 5th digits
Flexor digitorum profundus muscle belly for
Flexor pollicis longus
2nd digit
Radial artery
Dorsal (cutaneous) branch of ulnar nerve
Dorsal carpal branch of ulnar artery
Pronator quadratus
Pisiform
Palmar radiocarpal ligament
Median nerve
Flexor retinaculum (transverse carpal ligament)
Deep branch of ulnar nerve and artery
Opponens pollicis
Flexor pollicis brevis
Opponens digiti minimi
Abductor pollicis brevis
Abductor digiti minimi
4th lumbrical
1st lumbrical
2nd lumbrical
3rd lumbrical

- The two deep digital flexor muscles, flexor pollicis longus and flexor digitorum profundus, arise from the flexor aspects of the radius, interosseous membrane, and ulna between the origin of flexor digitorum superficialis proximally and pronator quadratus distally.

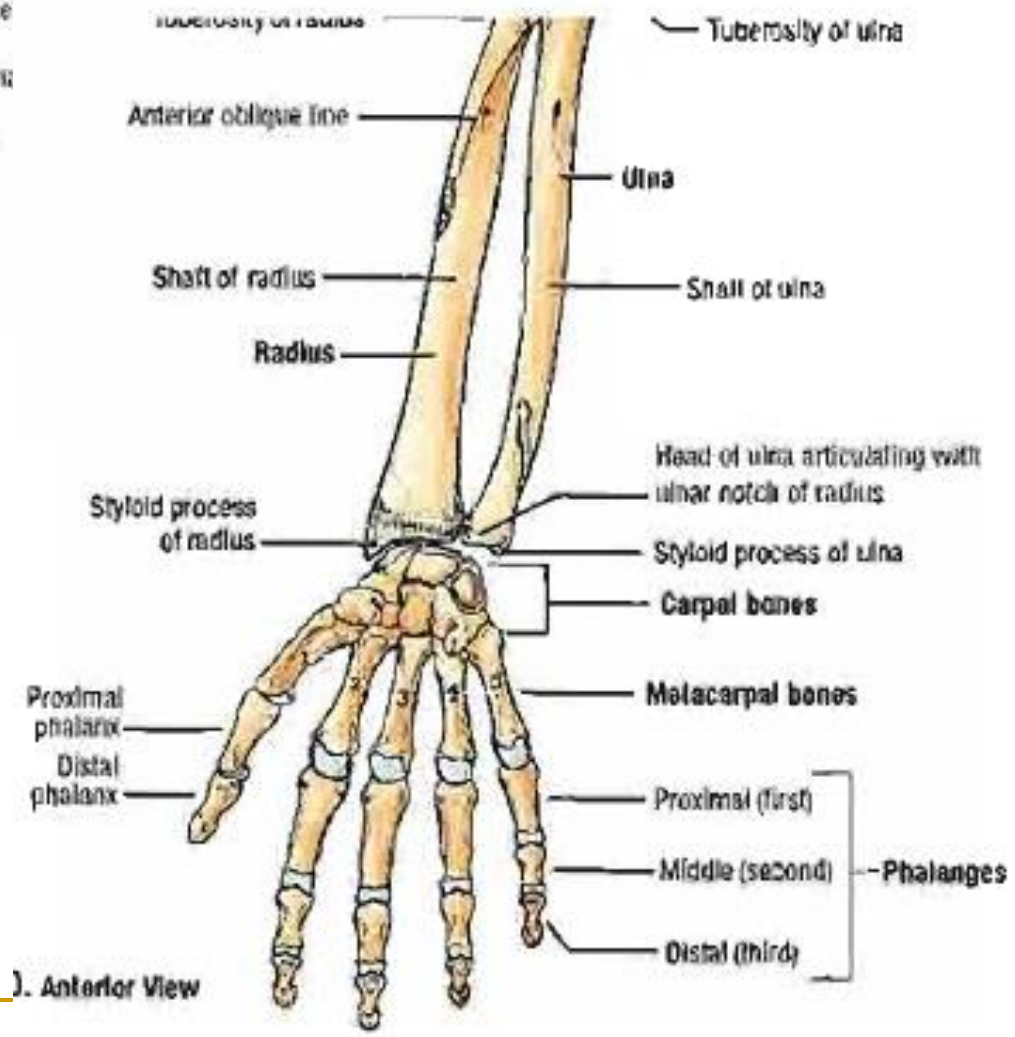
- The ulnar nerve enters the forearm posterior to the medial epicondyle, then descends between the flexor digitorum profundus and flexor carpi ulnaris and is joined by the ulnar artery.
- At the wrist the ulnar nerve and artery pass anterior to the flexor retinaculum and lateral to the pisiform to enter the palm.

- At the elbow, the ulnar nerve supplies the flexor carpi ulnaris and the medial half of the flexor digitorum profundus muscles; superior to the wrist, it gives off the dorsal (cutaneous) branch.

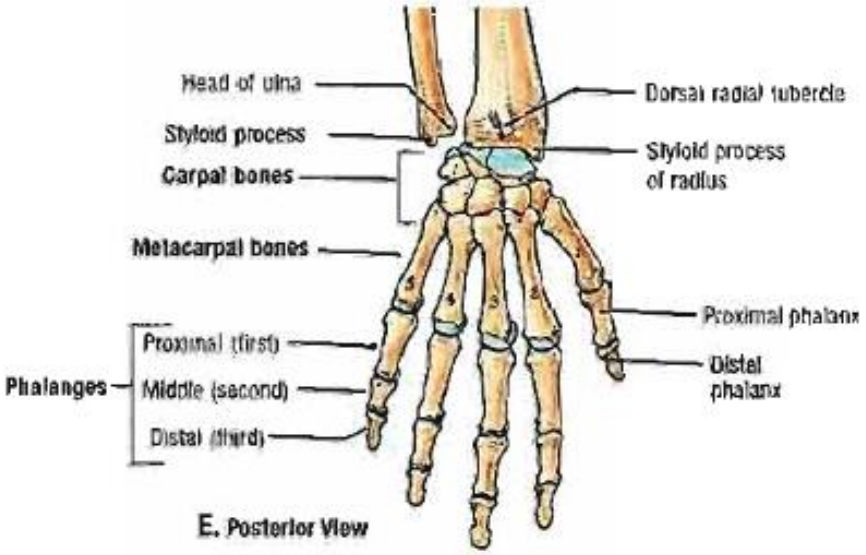

Ulna
Radius
Pronator quadratus
Anterior View


Triceps brachii (medial head)
Superficial branch of radial nerve (sensory)
Brachioradialis
Anconeus
Deep branch of radial nerve
Extensor carpi radialis longus
Extensor carpi radialis brevis
Posterior interosseous nerve
Supinator
Posterior compartment of forearm
Extensor carpi ulnaris
Abductor pollicis longus
Extensor digiti minimi
Extensor pollicis brevis
Extensor digitorum
Extensor pollicis longus
Extensor indicis

Trochlear notch
Olecranon
Radial notch
Head
Neck
Coronoid proce
Tuberosity
Tuberosity of ulna
Supinator fossa
Anterior oblique line
Ulna
Radius


Tuberosity of ulna
Anterior oblique line
Ulna
Shaft of radius
Shaft of ulna
Radius
Head of ulna articulating with ulnar notch of radius
Styloid process of radius
Styloid process of ulna
Carpal bones
Metacarpal bones
Proximal phalanx
Distal phalanx
Proximal (first)
Middle (second)
Distal (third)
Phalanges
). Anterior View

G. Posterior View



E. Posterior View

Medial cutaneous nerve of forearm

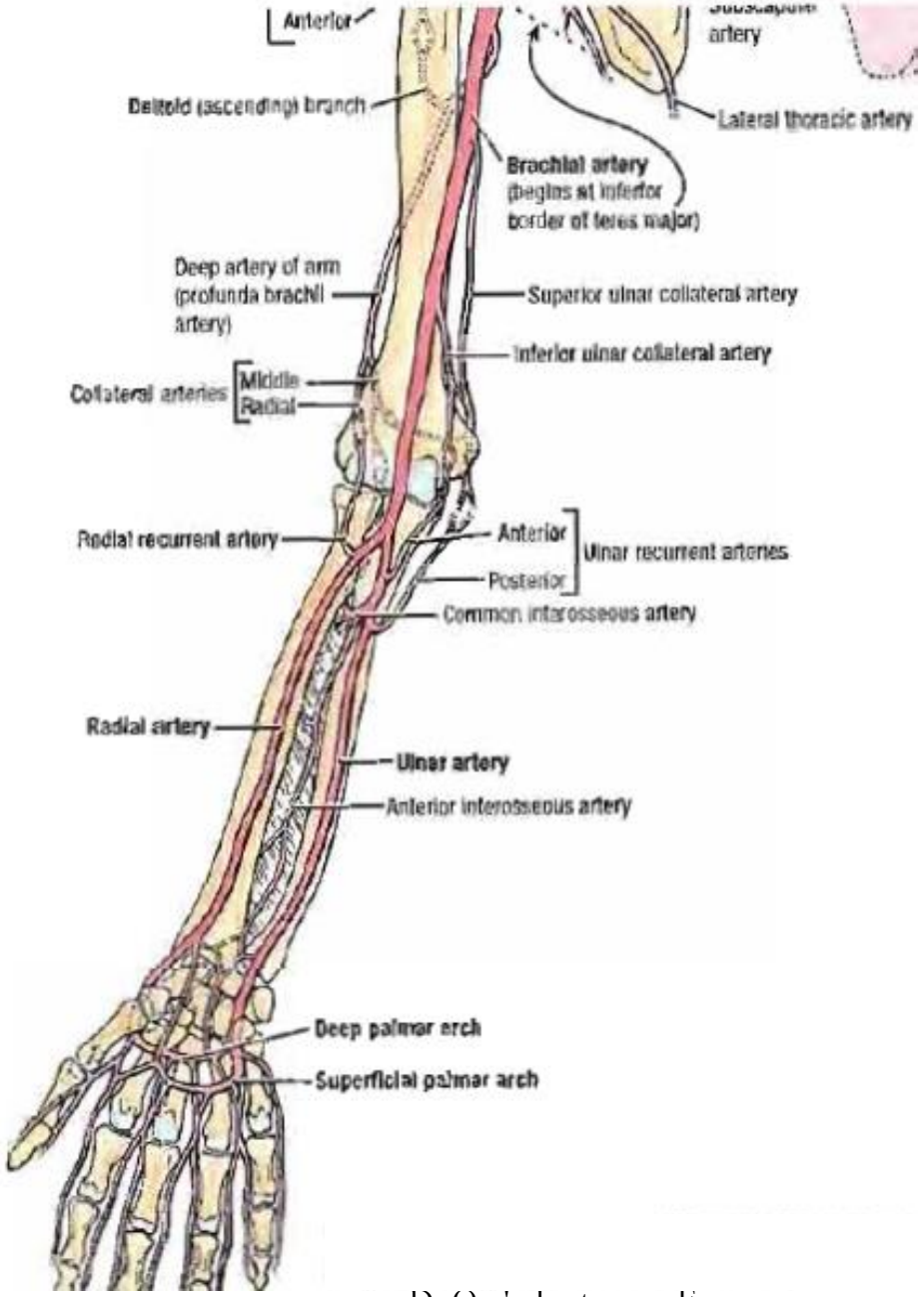
- C8, T1
- Pierces deep fascia with basilic vein in midarm; divides into anterior and posterior
- branches supplying skin over anterior and medial surfaces of forearm to wrist

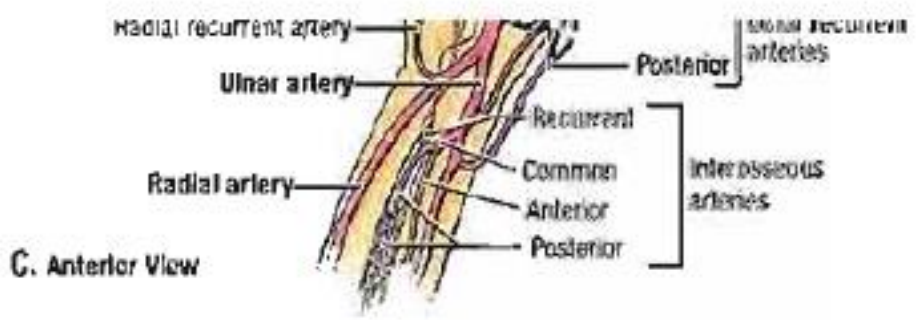
Median nerve

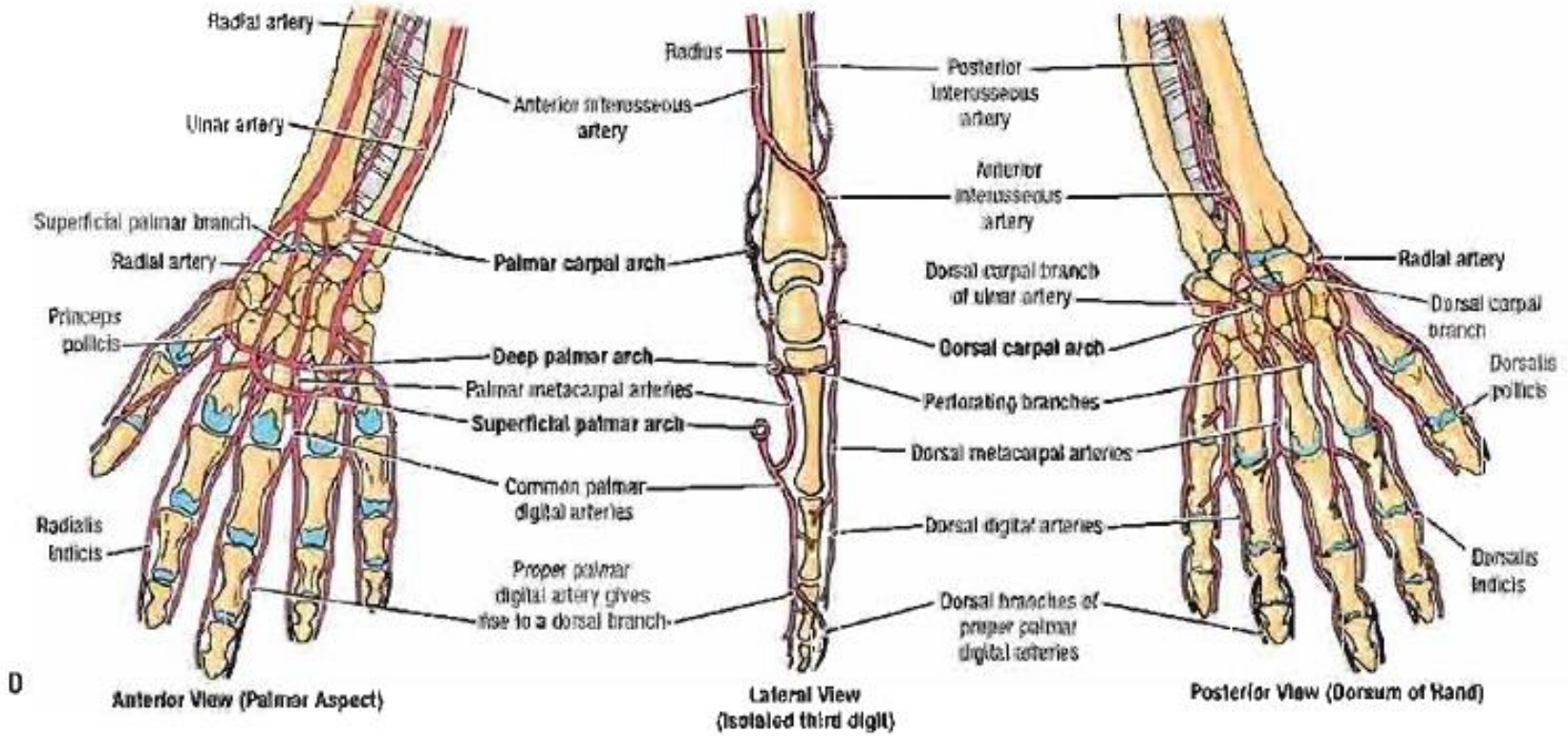
- C6, C7 (via lateral root);
- C8, T1 (via medial root)
- Lateral and medial cords of brachial plexus
- Courses with brachial artery in arm and deep to flexor digitorum superficialis in forearm; distal to origin of palmar cutaneous branch, traverses carpal tunnel to supply skin of palmar aspect of radial 3Â½ digits and adjacent palm, plus distal dorsal aspects of same, including nail beds

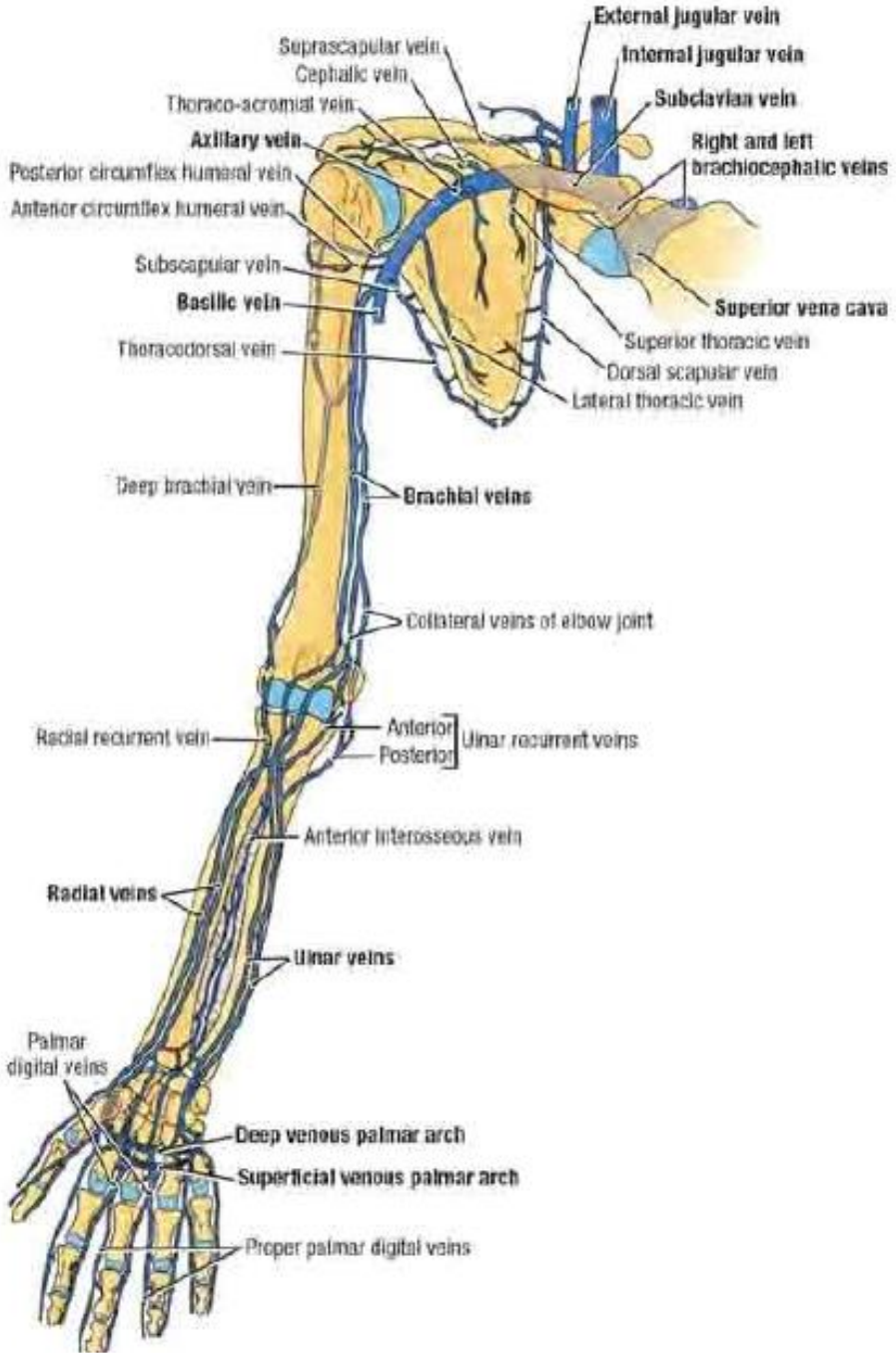
Ulnar nerve

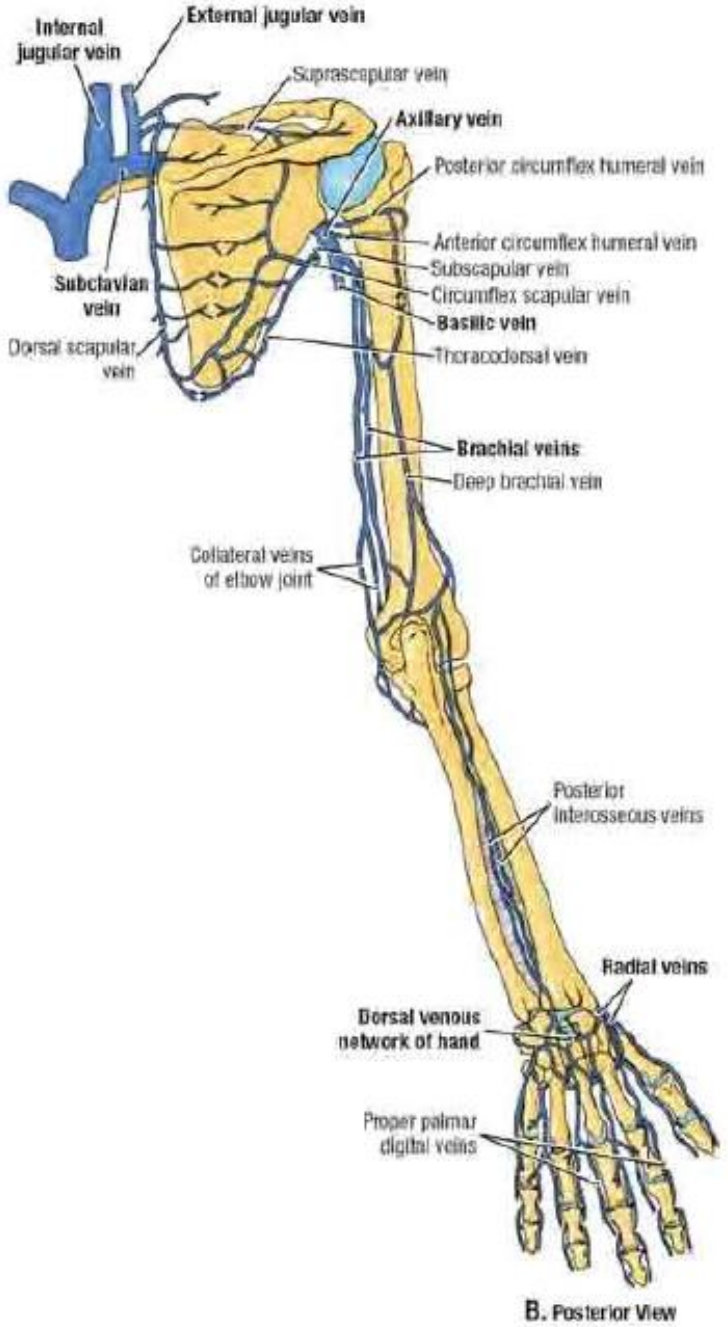
- (C7), C8,T1
- Medial cord of brachial plexus
- Courses with brachial, superior ulnar collateral, and ulnar arteries; supplies skin of palmar and dorsal aspects of medial (ulnar) 1½ digits and palm and dorsum of hand proximal to those digits

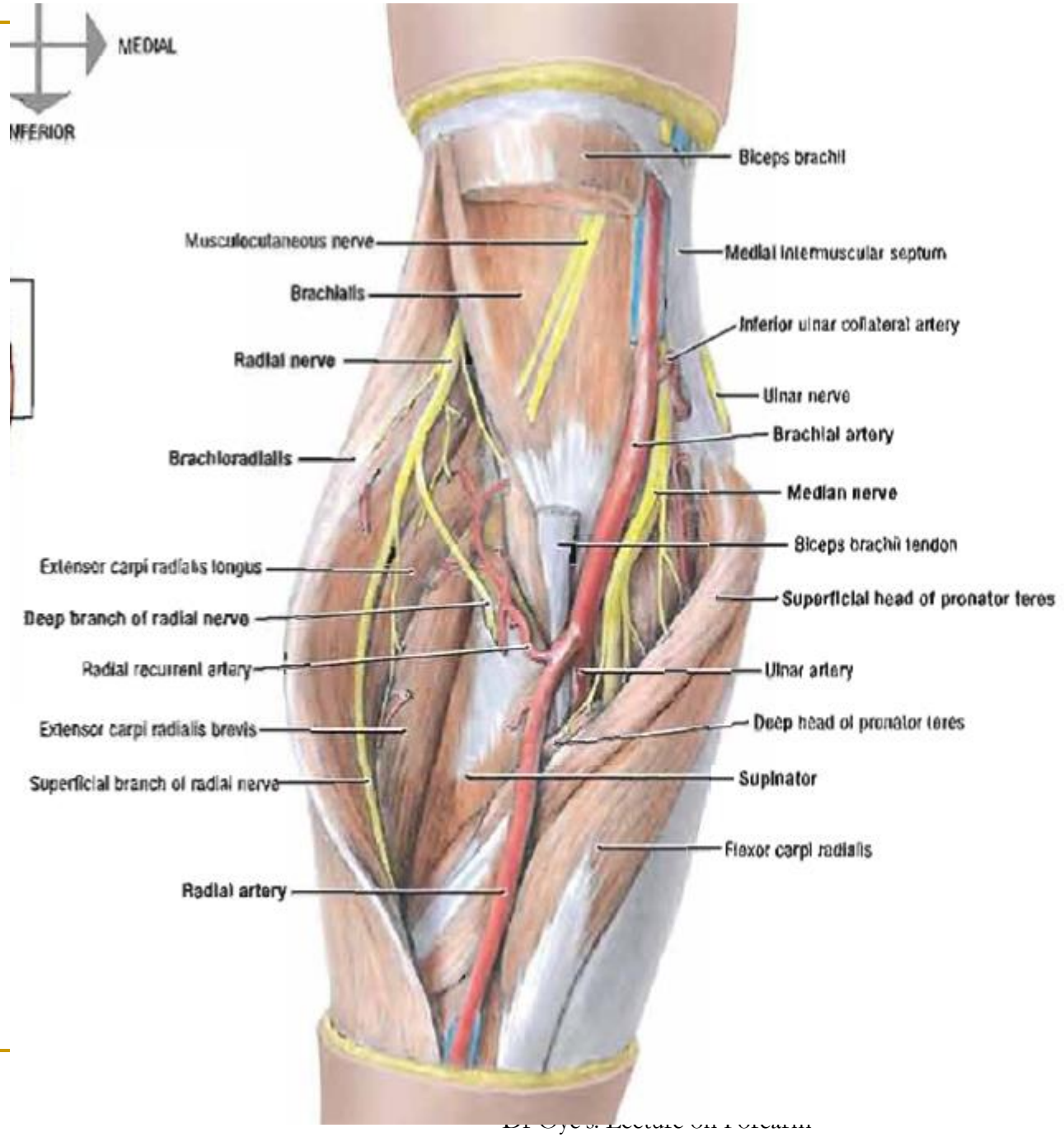

Anterior
artery
Deltoid (ascending) branch
Lateral thoracic artery
Brachial artery
(begins at inferior
border of teres major)
Deep artery of arm
(profunda brachii
artery)
Superior ulnar collateral artery
Inferior ulnar collateral artery
Collateral arteries
Middle
Radial
Radial recurrent artery
Anterior
Posterior
Ulnar recurrent arteries
Common interosseous artery
Radial artery
Ulnar artery
Anterior interosseous artery
Deep palmar arch
Superficial palmar arch

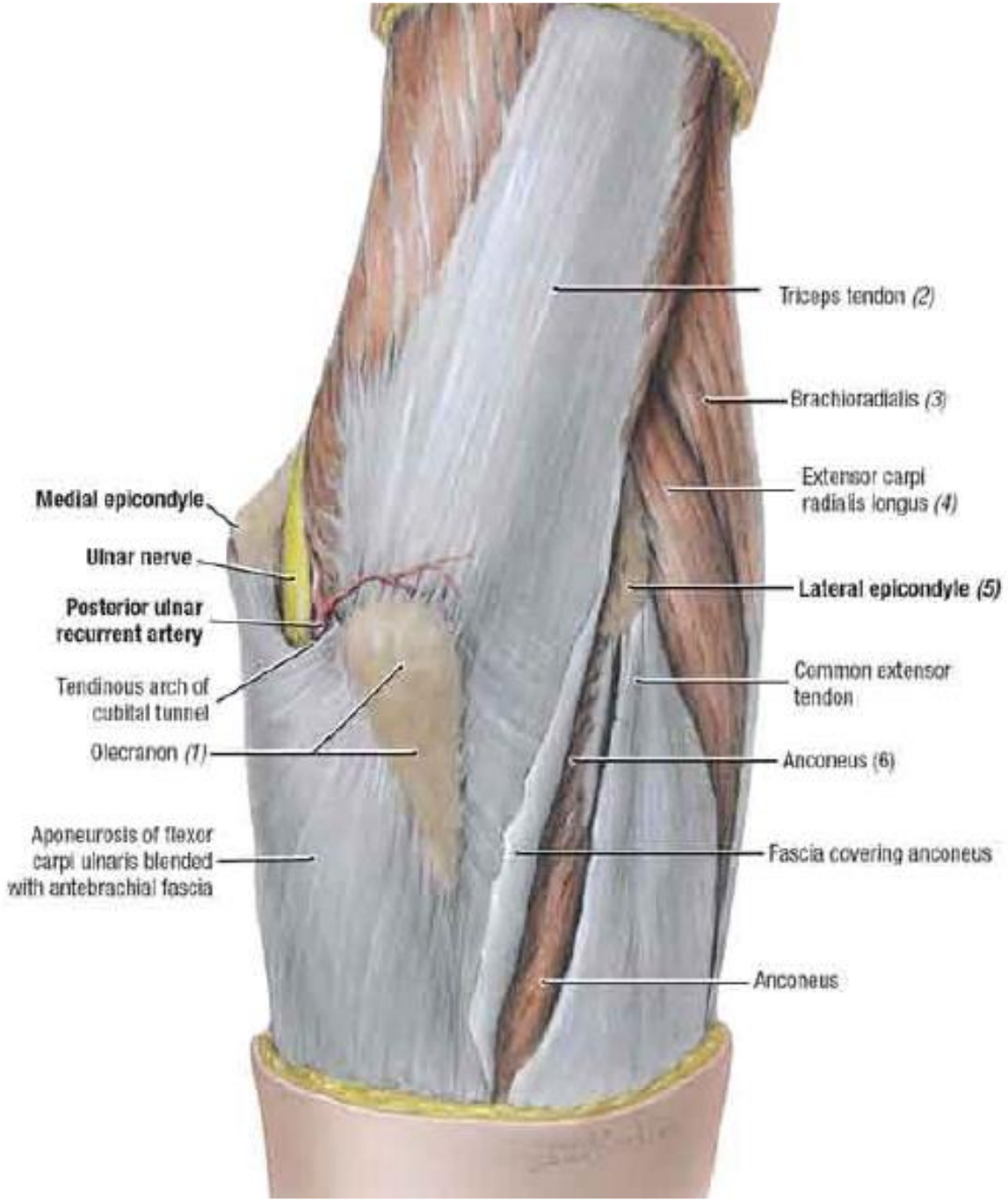

Radial recurrent artery
Ulnar artery
Radial artery
Posterior
arteries
Recurrent
Common
Anterior
Posterior
Interosseous
arteries
C. Anterior View

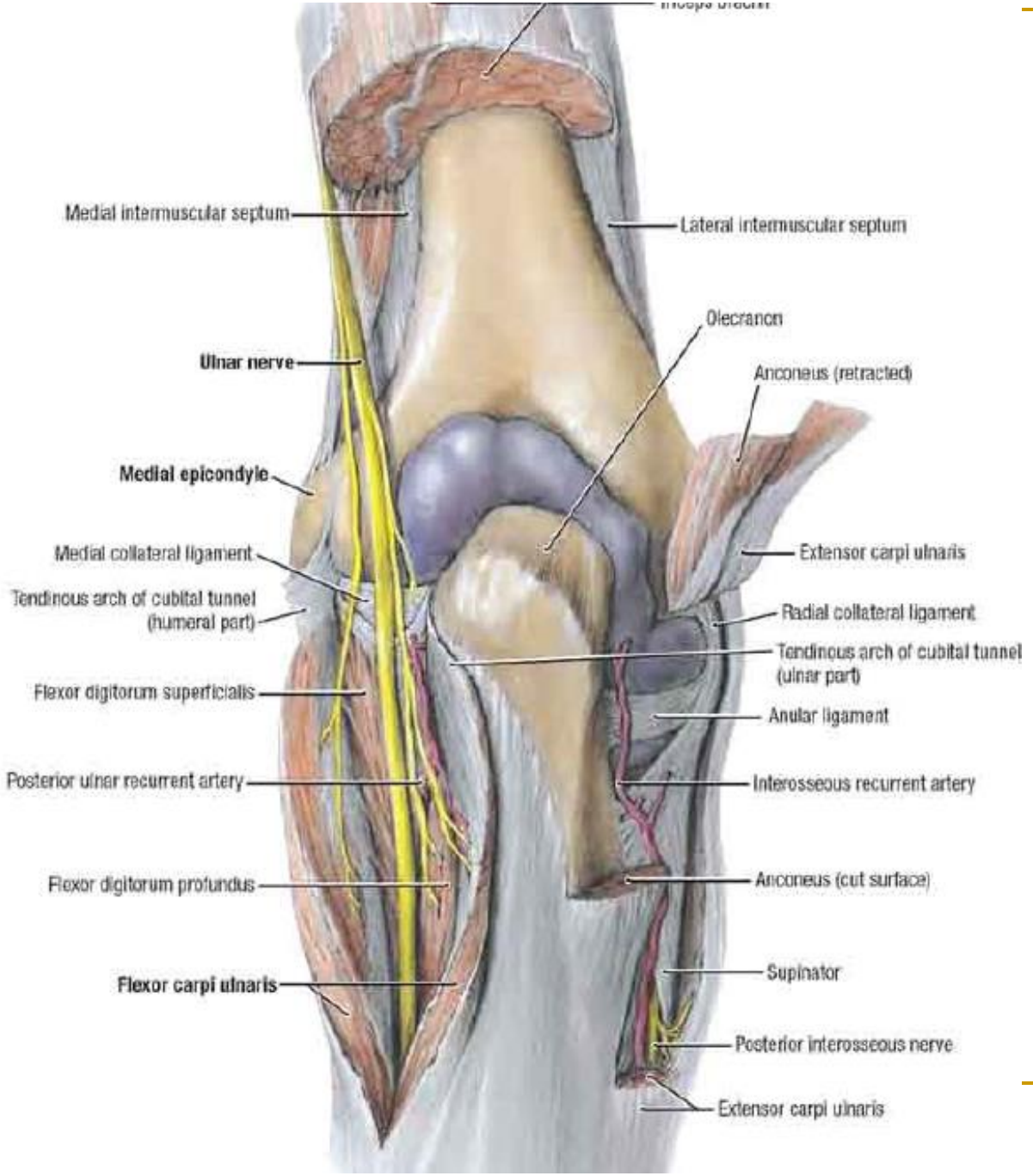

Radial artery
Ulnar artery
Superficial palmar branch
Radial artery
Princeps pollicis
Radialis indicis
Anterior interosseous artery
Palmar carpal arch
Deep palmar arch
Palmar metacarpal arteries
Superficial palmar arch
Common palmar digital arteries
Proper palmar digital artery gives rise to a dorsal branch
D
Anterior View (Palmar Aspect)
Radius
Lateral View (Isolated third digit)
Posterior interosseous artery
Anterior interosseous artery
Dorsal carpal branch of ulnar artery
Dorsal carpal arch
Perforating branches
Dorsal metacarpal arteries
Dorsal digital arteries
Dorsal branches of proper palmar digital arteries
Radial artery
Dorsal carpal branch
Dorsalis pollicis
Dorsalis indicis
Posterior View (Dorsum of Hand)


External jugular vein
Suprascapular vein
Internal jugular vein
Cephalic vein
Subclavian vein
Thoraco-acromial vein
Axillary vein
Right and left brachiocephalic veins
Posterior circumflex humeral vein
Anterior circumflex humeral vein
Subscapular vein
Basilic vein
Superior vena cava
Superior thoracic vein
Thoracodorsal vein
Dorsal scapular vein
Lateral thoracic vein
Deep brachial vein
Brachial veins
Collateral veins of elbow joint
Radial recurrent vein
Anterior
Posterior
Ulnar recurrent veins
Anterior interosseous vein
Radial veins
Ulnar veins
Palmar digital veins
Deep venous palmar arch
Superficial venous palmar arch
Proper palmar digital veins


Internal jugular vein
External jugular vein
Suprascapular vein
Axillary vein
Posterior circumflex humeral vein
Anterior circumflex humeral vein
Subscapular vein
Circumflex scapular vein
Basilic vein
Thoracodorsal vein
Subclavian vein
Dorsal scapular vein
Brachial veins
Deep brachial vein
Collateral veins of elbow joint
Posterior interosseous veins
Radial veins
Dorsal venous network of hand
Proper palmar digital veins
B. Posterior View


MEDIAL
INFERIOR
Biceps brachii
Musculocutaneous nerve
Medial intermuscular septum
Brachialis
Inferior ulnar collateral artery
Radial nerve
Ulnar nerve
Brachial artery
Brachioradialis
Median nerve
Biceps brachii tendon
Extensor carpi radialis longus
Superficial head of pronator teres
Deep branch of radial nerve
Radial recurrent artery
Ulnar artery
Extensor carpi radialis brevis
Deep head of pronator teres
Superficial branch of radial nerve
Supinator
Flexor carpi radialis
Radial artery


Triceps tendon (2)
Brachioradialis (3)
Extensor carpi radialis longus (4)
Medial epicondyle
Ulnar nerve
Posterior ulnar recurrent artery
Lateral epicondyle (5)
Tendinous arch of cubital tunnel
Common extensor tendon
Olecranon (1)
Anconeus (6)
Aponeurosis of flexor carpi ulnaris blended with antebrachial fascia
Fascia covering anconeus
Anconeus


Medial intermuscular septum
Lateral intermuscular septum
Olecranon
Ulnar nerve
Anconeus (retracted)
Medial epicondyle
Medial collateral ligament
Extensor carpi ulnaris
Tendinous arch of cubital tunnel (humeral part)
Radial collateral ligament
Tendinous arch of cubital tunnel (ulnar part)
Flexor digitorum superficialis
Anular ligament
Posterior ulnar recurrent artery
Interosseous recurrent artery
Flexor digitorum profundus
Anconeus (cut surface)
Supinator
Flexor carpi ulnaris
Posterior interosseous nerve
Extensor carpi ulnaris

Muscles acting on the elbow

Flexors

- biceps
- brachialis
- brachioradialis
- the forearm flexor muscles

Pronators

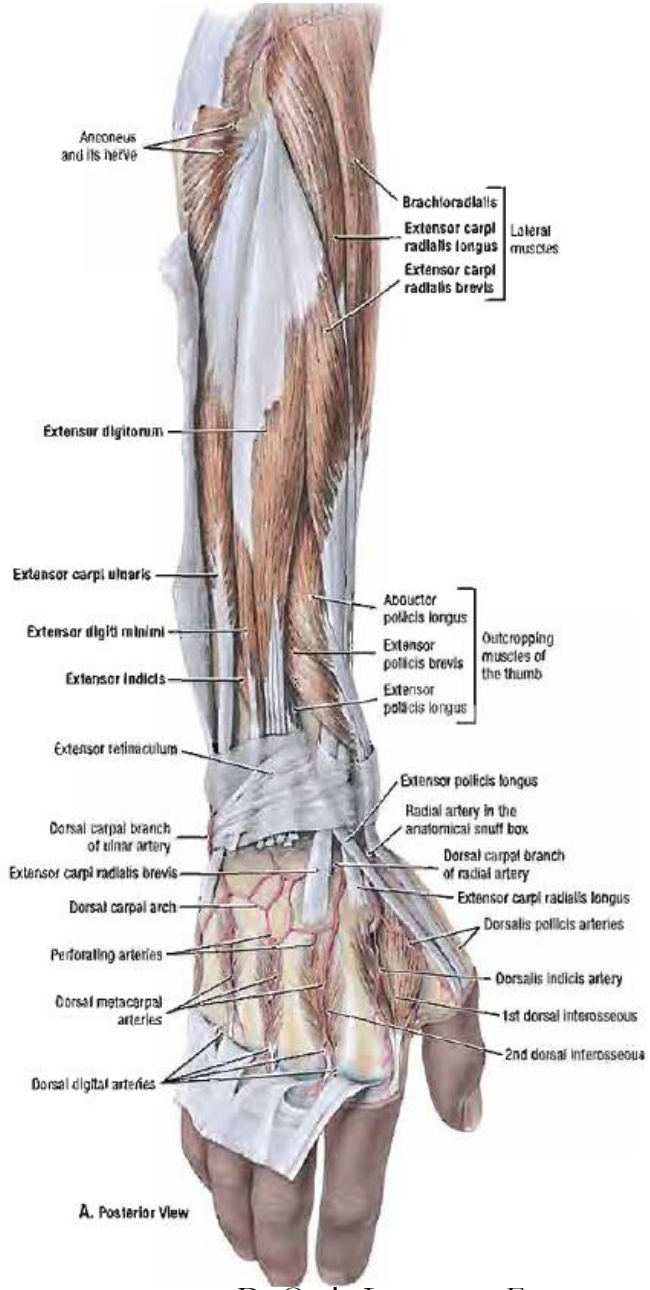
- pronator teres
- pronator quadratus
- flexor carpi radialis

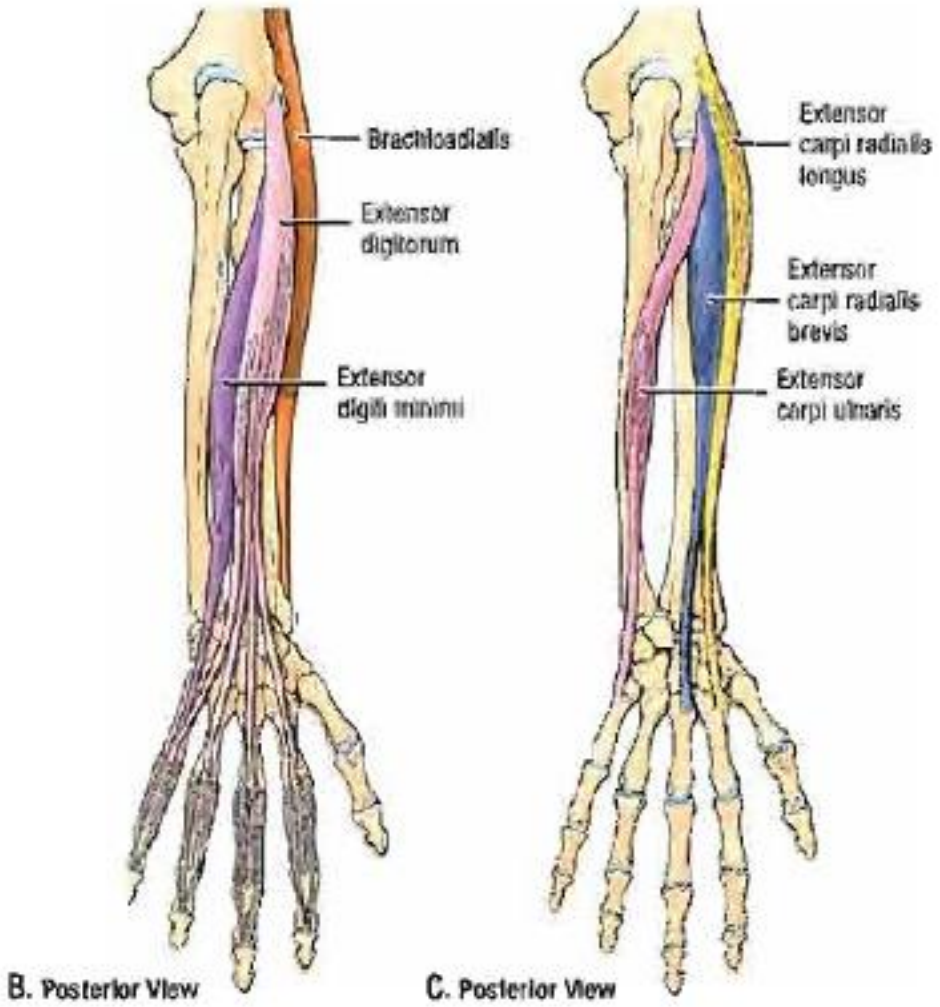
Extensors

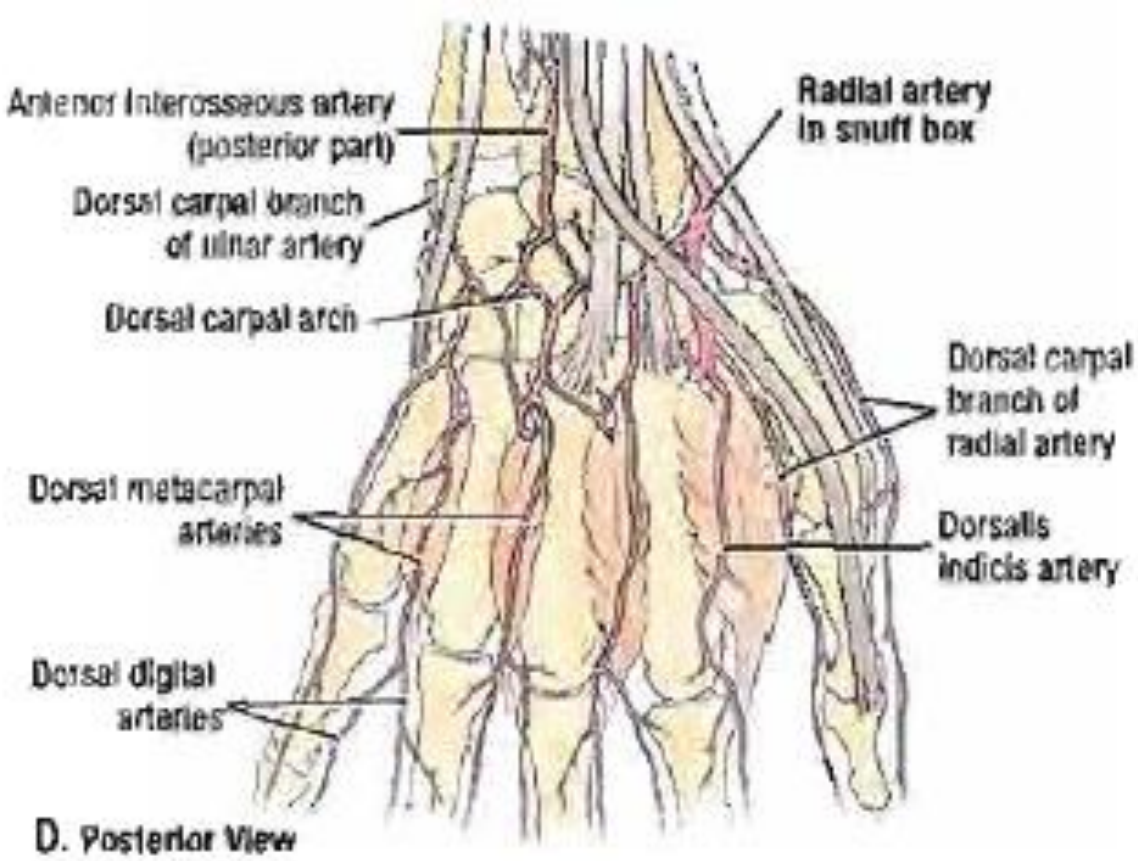
- triceps
- anconeus

Supinators

- biceps
- supinator
- extensor pollicis longus
- extensor pollicis brevis
- abductor pollicis longus


Anconeus and its nerve
Brachioradialis
Extensor carpi radialis longus
Extensor carpi radialis brevis
Lateral muscles
Extensor digitorum
Extensor carpi ulnaris
Extensor digiti minimi
Extensor indicis
Abductor pollicis longus
Extensor pollicis brevis
Extensor pollicis longus
Outcropping muscles of the thumb
Extensor retinaculum
Extensor pollicis longus
Radial artery in the anatomical snuff box
Dorsal carpal branch of ulnar artery
Dorsal carpal branch of radial artery
Extensor carpi radialis brevis
Extensor carpi radialis longus
Dorsal carpal arch
Dorsalis pollicis arteries
Perforating arteries
Dorsalis indicis artery
Dorsal metacarpal arteries
1st dorsal interosseous
2nd dorsal interosseous
Dorsal digital arteries
A. Posterior View


Brachioradialis
Extensor digitorum
Extensor digiti minimi
Extensor carpi radialis longus
Extensor carpi radialis brevis
Extensor carpi ulnaris
B. Posterior View
C. Posterior View

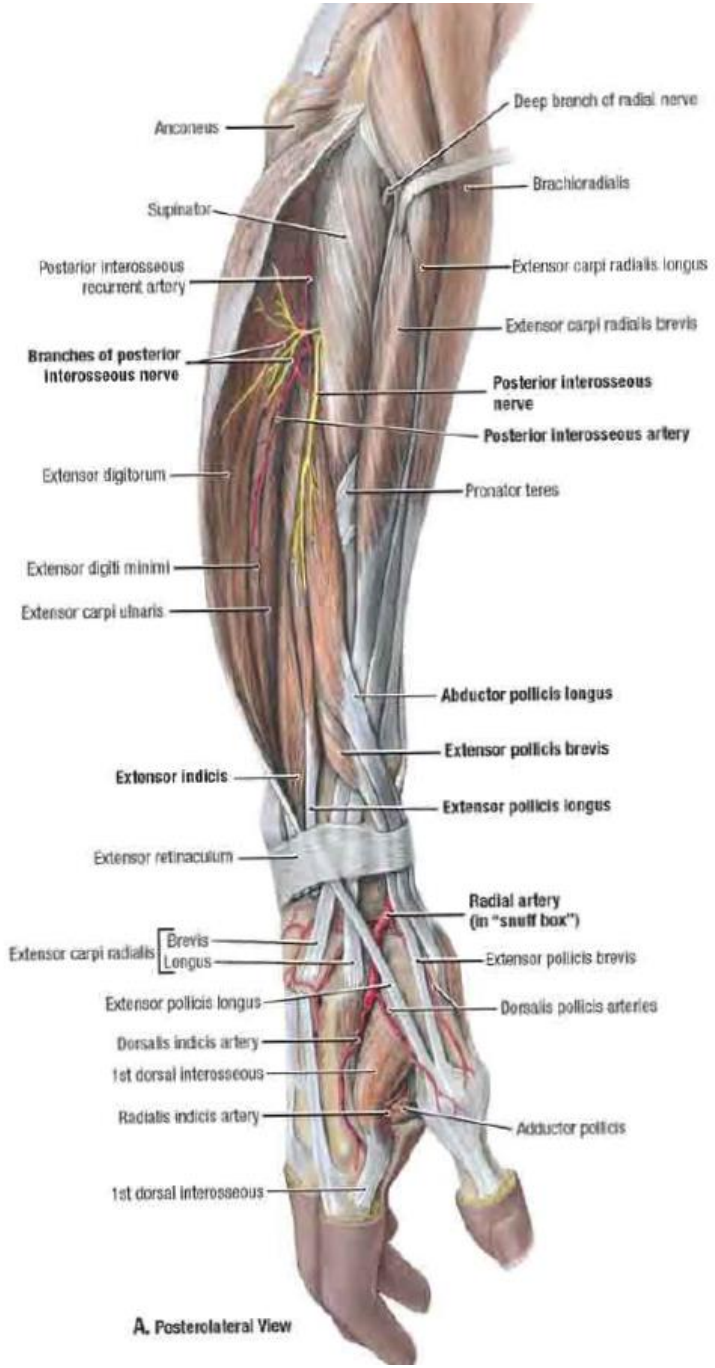

Anterior Interosseous artery (posterior part)
Dorsal carpal branch of ulnar artery
Dorsal carpal arch
Dorsal metacarpal arteries
Dorsal digital arteries
Radial artery in snuff box
Dorsal carpal branch of radial artery
Dorsalis indicis artery
D. Posterior View

- Three muscles of the thumb (abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus) emerge between the extensor carpi radialis brevis and the extensor digitorum.

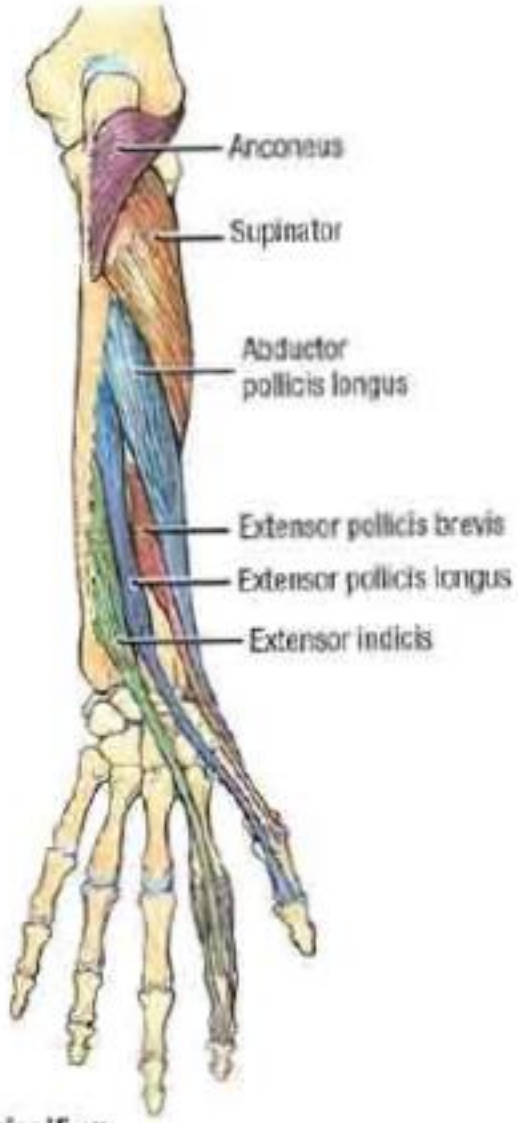
- The laterally retracted brachioradialis and extensor carpi radialis longus and brevis muscles and supinator muscles are **innervated by the deep branch of the radial nerve;**
- The other extensor muscles are supplied by the posterior interosseous nerve, which is a continuation of the deep branch of the radial nerve that pierced the supinator.

Clinical Relevance

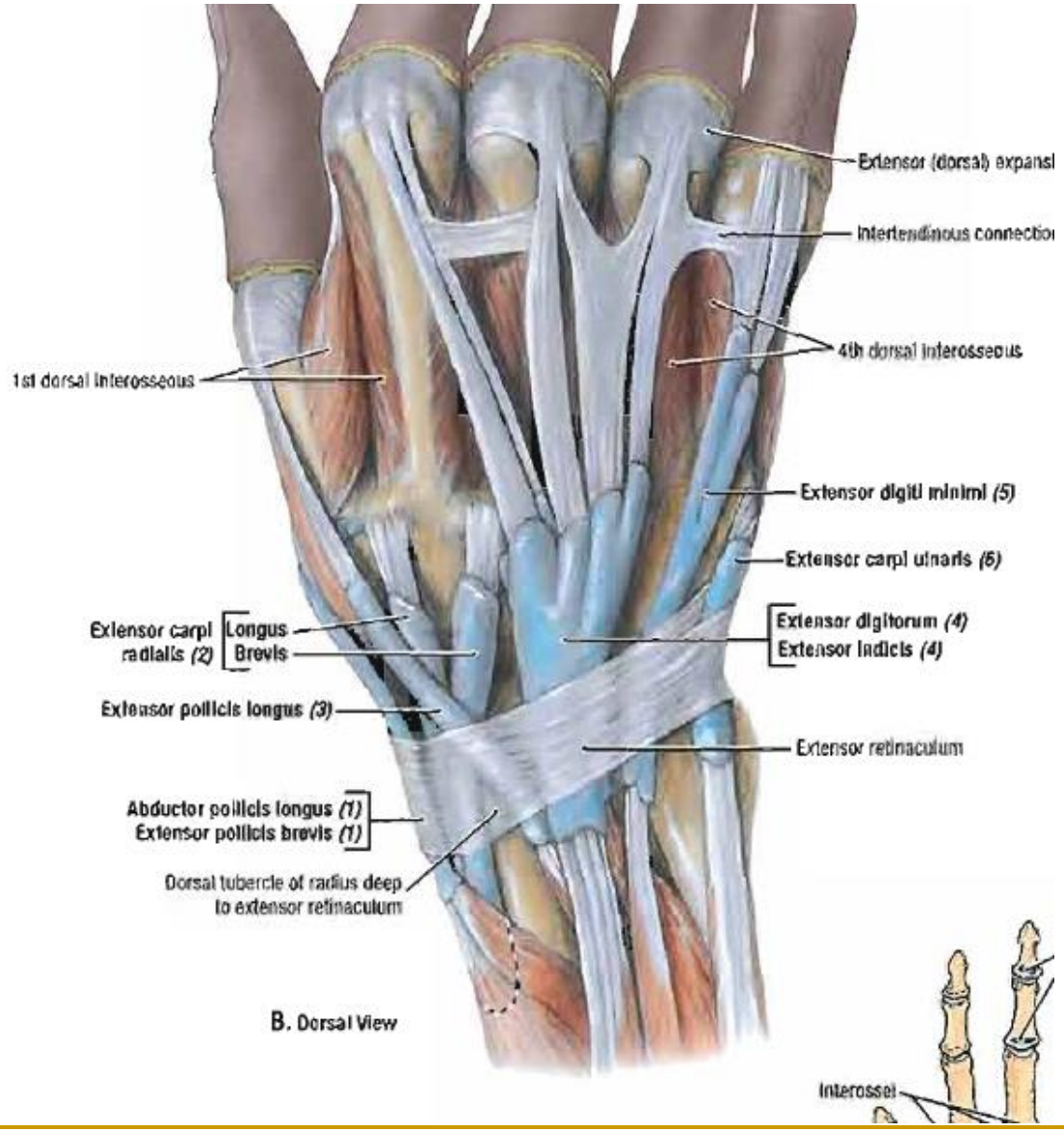
- **Severance of the deep branch of the radial nerve** results in **an inability to extend the thumb** and the *metacarpophalangeal joints of the other digits.*
- Loss of sensation does not occur because the deep branch is entirely muscular and articular in distribution.

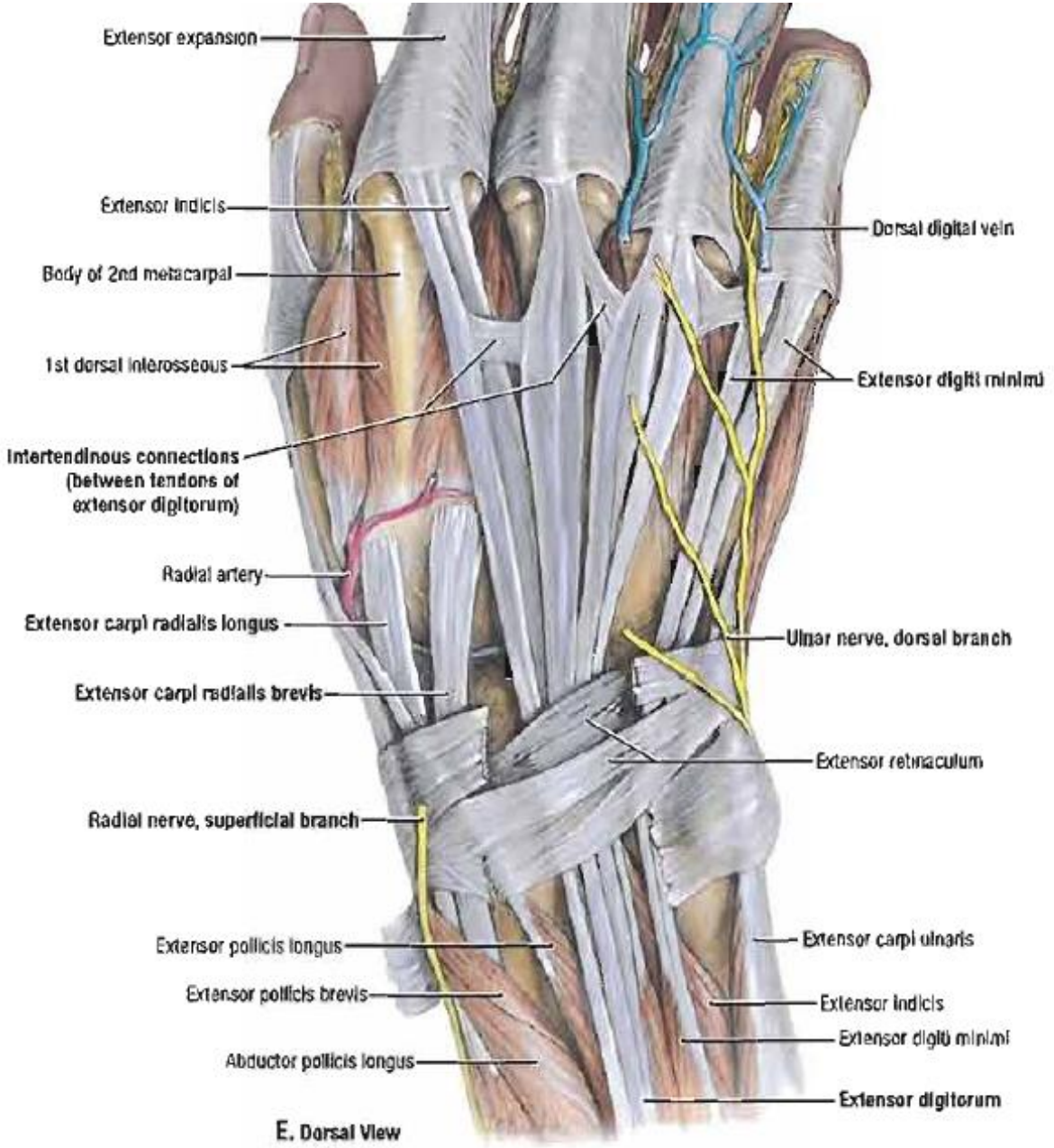


A. Posterolateral View



B. Posterior View

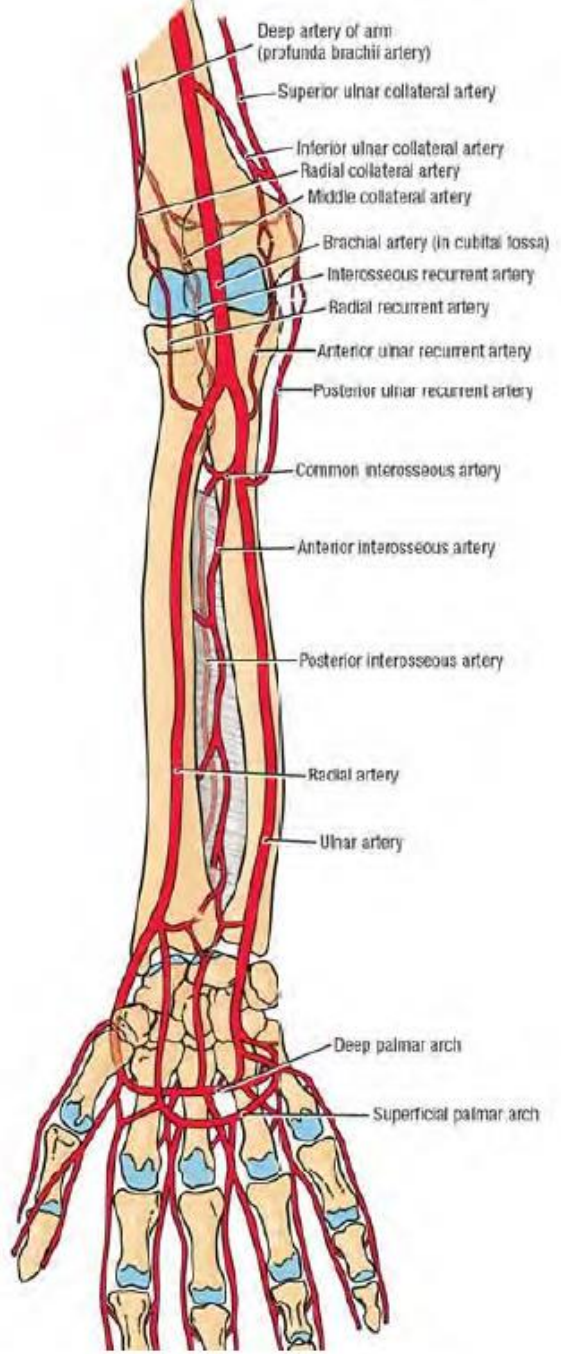

Extensor (dorsal) expansi
Intertendinous connectio
4th dorsal interosseous
1st dorsal interosseous
Extensor digiti minimi (5)
Extensor carpi ulnaris (6)
Extensor digitorum (4)
Extensor indicis (4)
Extensor carpi radialis (2) Longus Brevis
Extensor pollicis longus (3)
Extensor retinaculum
Abductor pollicis longus (1)
Extensor pollicis brevis (1)
Dorsal tubercle of radius deep to extensor retinaculum
B. Dorsal View
Interossei

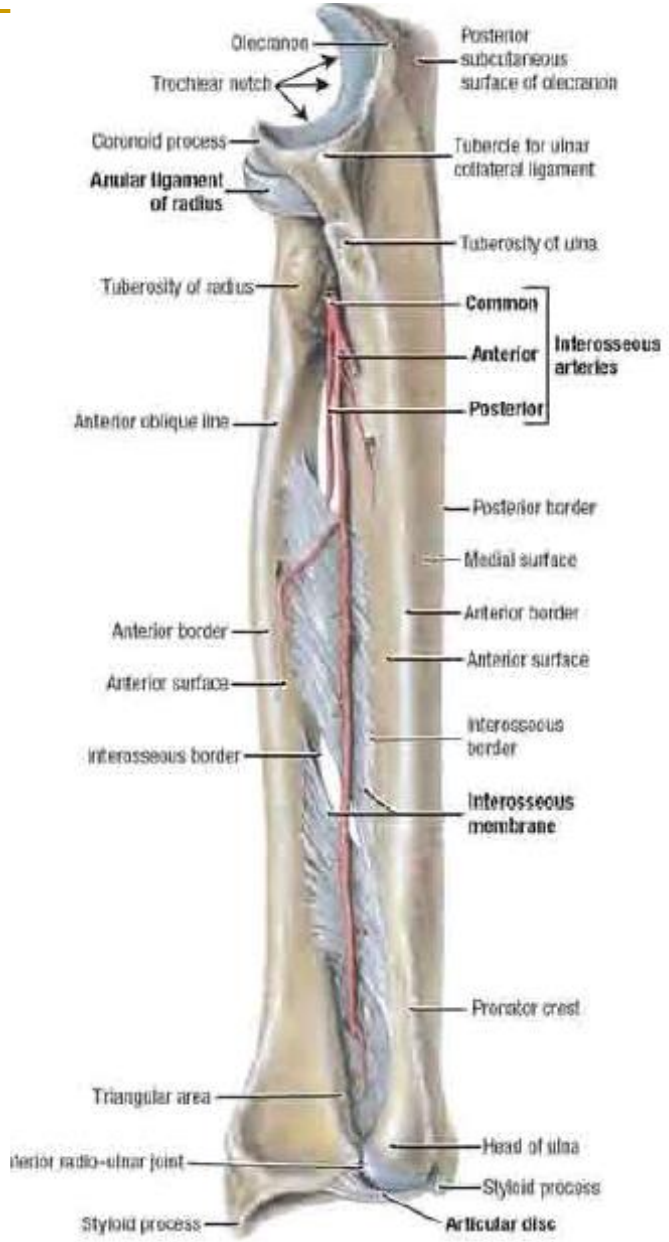


E. Dorsal View

A. Anterior View, Supination

B. Anterior View, Pronation

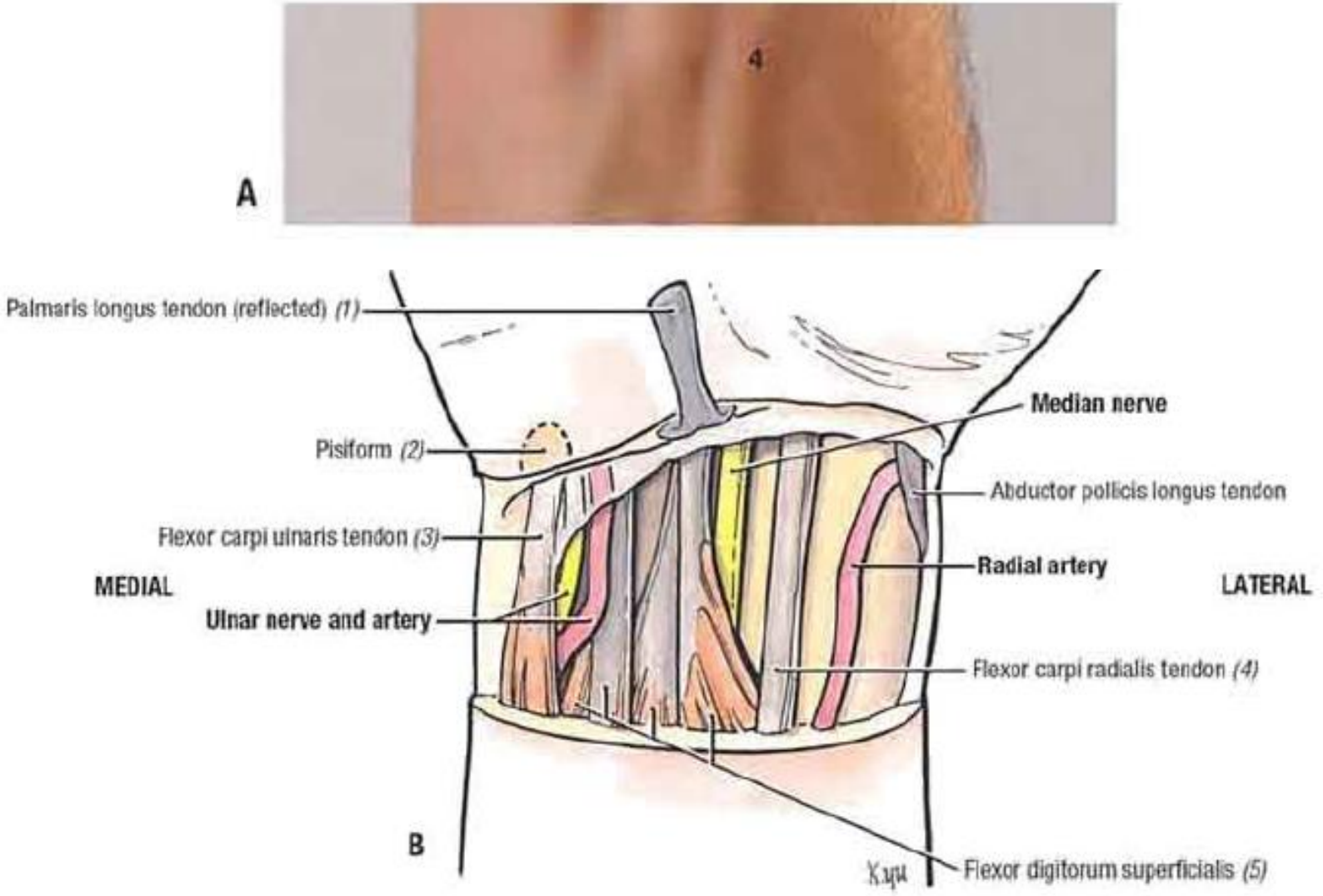

Deep artery of arm
(profunda brachii artery)
Superior ulnar collateral artery
Inferior ulnar collateral artery
Radial collateral artery
Middle collateral artery
Brachial artery (in cubital fossa)
Interosseous recurrent artery
Radial recurrent artery
Anterior ulnar recurrent artery
Posterior ulnar recurrent artery
Common interosseous artery
Anterior interosseous artery
Posterior interosseous artery
Radial artery
Ulnar artery
Deep palmar arch
Superficial palmar arch



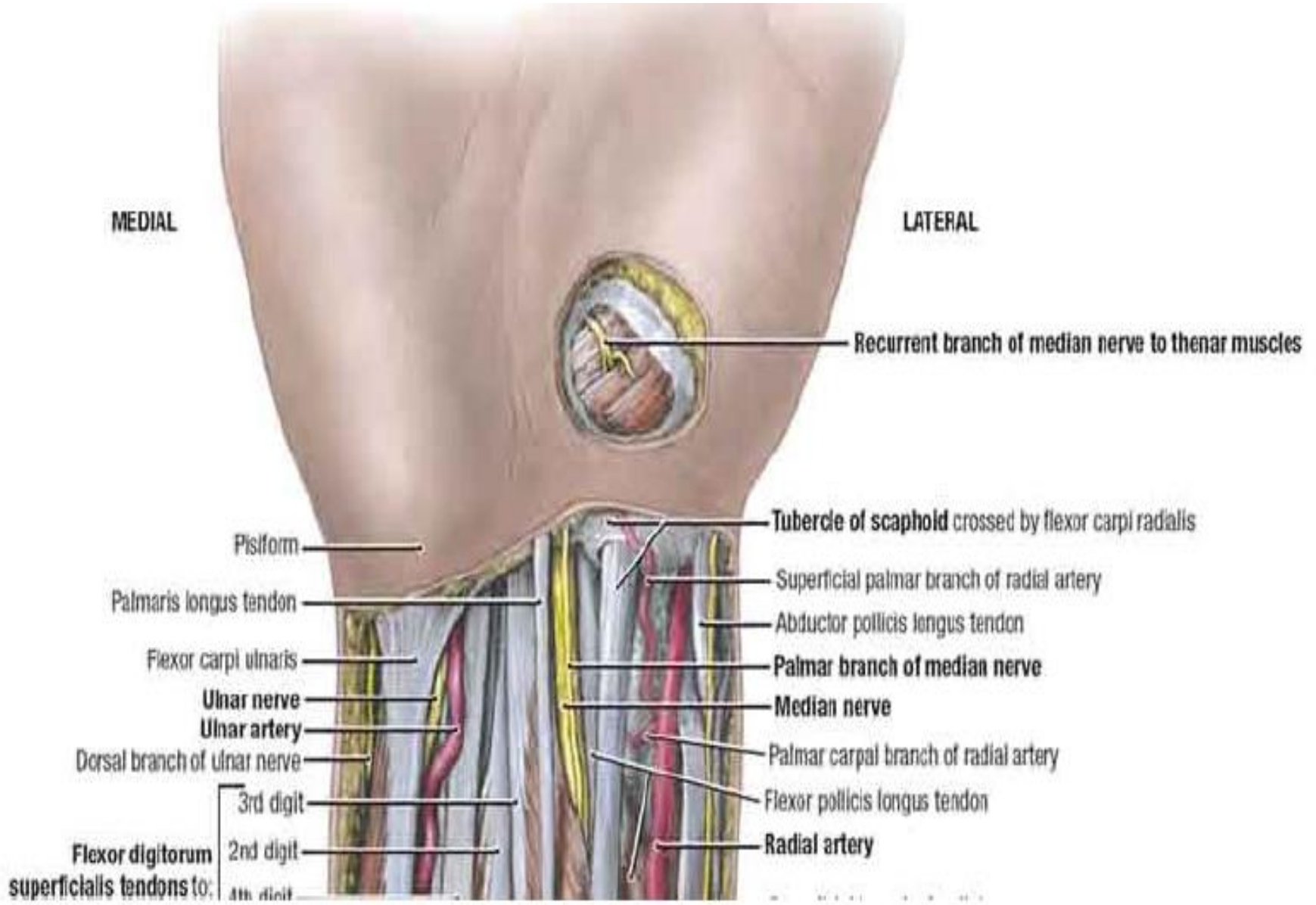
C. Anterior View

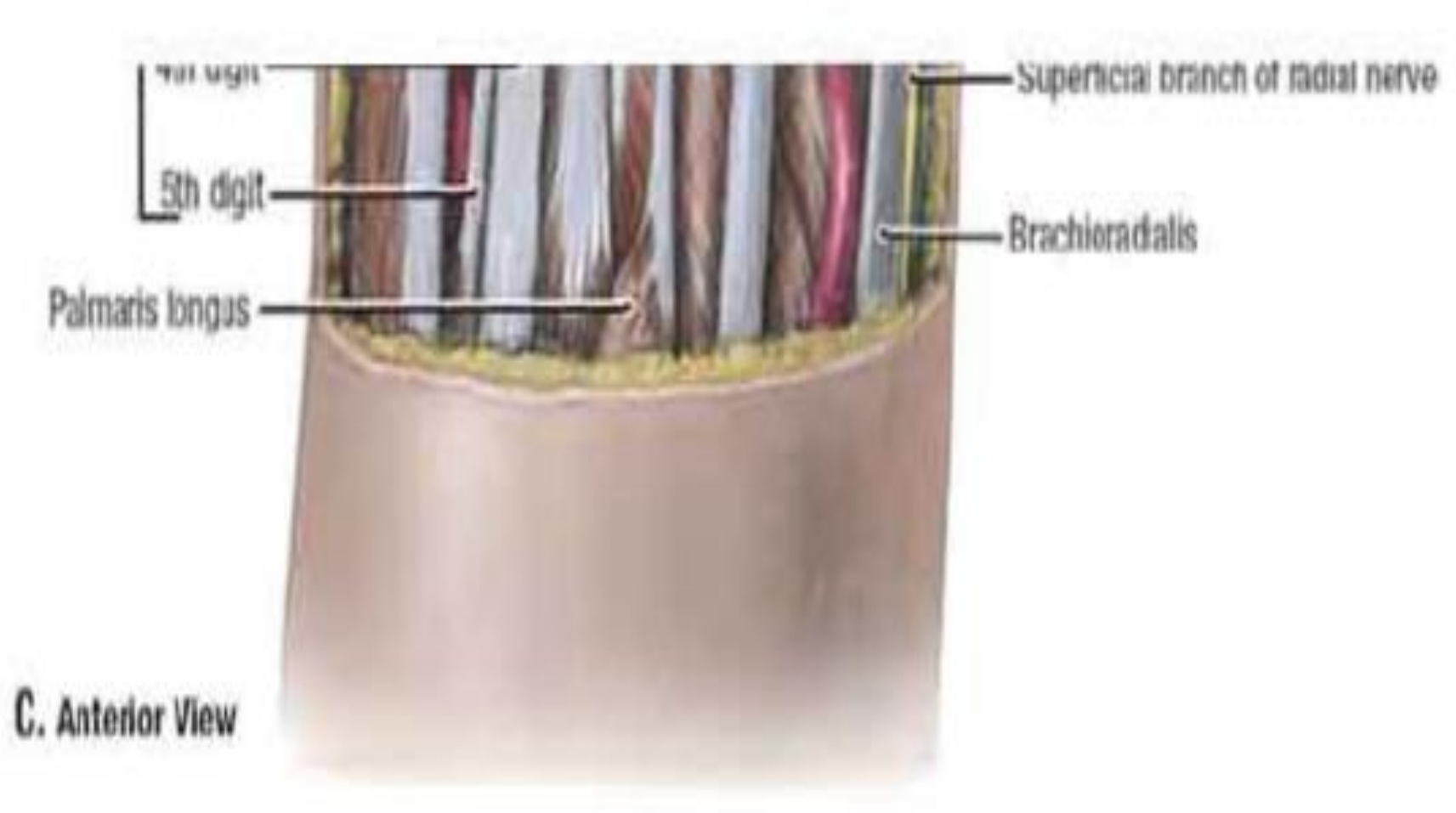
- The five tendons of the deep digital flexors (flexor pollicis longus and flexor digitorum profundus) lie side by side as they enter the carpal tunnel.
- The biceps brachii muscle attaches to the medial aspect of the radius; hence, it can supinate the forearm, whereas the **pronator teres muscle**, by attaching to the lateral surface, **can pronate the forearm.**

- The **deep branch of the radial nerve pierces and innervates the supinator muscle.**
- The **anterior interosseous nerve** and **artery** disappear between the flexor pollicis longus and flexor digitorum profundus muscles to lie on the interosseous membrane.



Anterior Views of Right Hand and Wrist


MEDIAL
LATERAL
Recurrent branch of median nerve to thenar muscles
Tubercle of scaphoid crossed by flexor carpi radialis
Pisiform
Superficial palmar branch of radial artery
Palmaris longus tendon
Abductor pollicis longus tendon
Flexor carpi ulnaris
Palmar branch of median nerve
Ulnar nerve
Median nerve
Ulnar artery
Palmar carpal branch of radial artery
Dorsal branch of ulnar nerve
Flexor pollicis longus tendon
3rd digit
Flexor digitorum superficialis tendons to:
2nd digit
Radial artery


5th digit
Palmaris longus
Superficial branch of radial nerve
Brachioradialis
C. Anterior View

- The **radial artery** passes deep to the tendon of the abductor pollicis longus muscle.
- The flexor digitorum superficialis tendons to the 3rd and 4th digits become anterior to those of the 2nd and 5th digits.
- The recurrent branch of the median nerve to the thenar muscles lies within a circle whose center is 2.5 to 4 cm distal to the tubercle of the scaphoid

Lesions of the median nerve usually occur in two places:

- The forearm and wrist. The most common site is where the nerve passes though the carpal tunnel.
- Lacerations of the wrist often cause **median nerve injury because this nerve is relatively close to the surface**.
- This results in paralysis of the **thenar muscles** and the **first two lumbricals**.
- Hence opposition of the thumb is not possible and fine control movements of the 2nd and 3rd digits are impaired. Sensation is also lost over the thumb and adjacent two and a half fingers.

- **Median nerve injury resulting from a perforating wound** in the elbow region results in loss of flexion of the proximal and distal interphalangeal joints of the 2nd and 3rd digits.
- The ability to flex the metacarpophalangeal joints of these digits is also affected because digital branches of the median nerve supply the 1st and 2nd lumbricals.
- The palmar cutaneous branch of the median nerve does not traverse the carpal tunnel.
- It supplies the skin of the central palm, which remains sensitive in carpal tunnel syndrome.

Radial artery

Origin:

- In cubital fossa, as smaller terminal division of brachial artery

Course/Distribution:

- Runs distally under brachioradialis, lateral to flexor carpi radialis, defining boundary between the flexor and extensor compartments and supplying the radial aspect of both.
- Gives rise to a superficial palmar branch near the radiocarpal joint; it then transverses the anatomical snuff box to pass between the heads of the 1st dorsal interosseous muscle joining the deep branch of the ulnar artery to form the deep palmar arch

Ulnar Artery

Origin:

- In cubital fossa, as larger terminal division of brachial artery

Course/Distribution:

- **Passes distally between 2nd and 3rd layers of forearm flexor muscles, supplying ulnar aspect of flexor compartment**; passes superficial to flexor retinaculum at wrist, ……..continuing as the superficial palmar arch (with superficial branch of radial) after its deep palmar branch joins the deep palmar arch

Radial recurrent artery

Origin:

- In cubital fossa, as 1st (lateral) branch of radial artery

Course/Distribution:

- Courses proximally, superficial to supinator, passing between brachioradialis and brachialis to anastomose with radial collateral artery

Anterior and posterior ulnar recurrent arteries

Origin:

- In and immediately distal to cubital fossa, as 1st and 2nd medial branches of ulnar artery

Course/Distribution:

- Course proximally to anastomose with the inferior and superior ulnar collateral arteries,
- respectively, forming collateral pathways anterior and posterior to the medial epicondyle of the humerus

Common interosseous artery

Origin:

- Immediately distal to the cubital fossa, as 1st lateral branch of ulnar artery

Course/Distribution:

- Terminates almost immediately, dividing into anterior and posterior interosseous arteries

Anterior and posterior interosseous arteries

Origin:

- Distal to radial tubercle, as terminal branches of common interosseous

Course/Distribution:

- Pass to opposite sides of interosseous membrane; anterior artery runs on interosseous membrane; posterior artery runs between superficial and deep layers of extensor muscles as primary artery of compartment

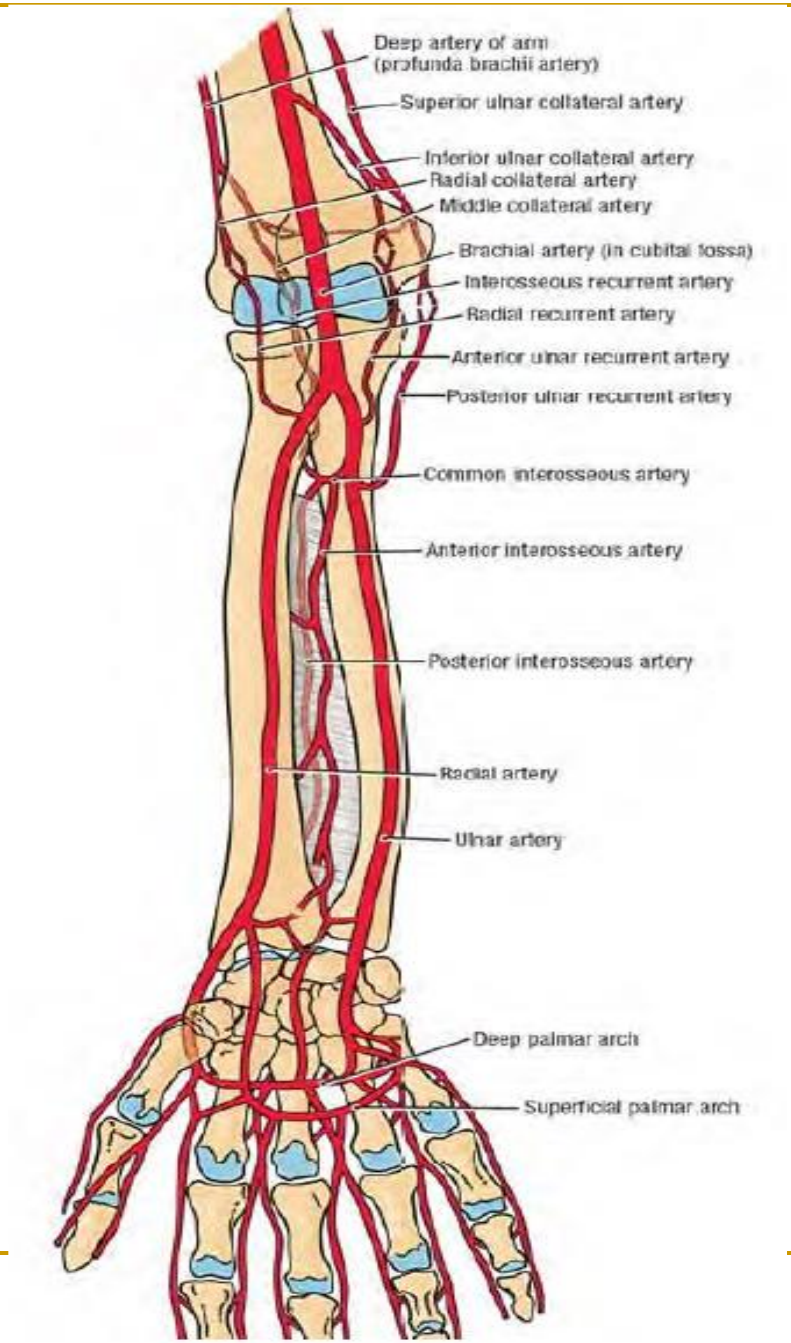
Interosseous recurrent artery

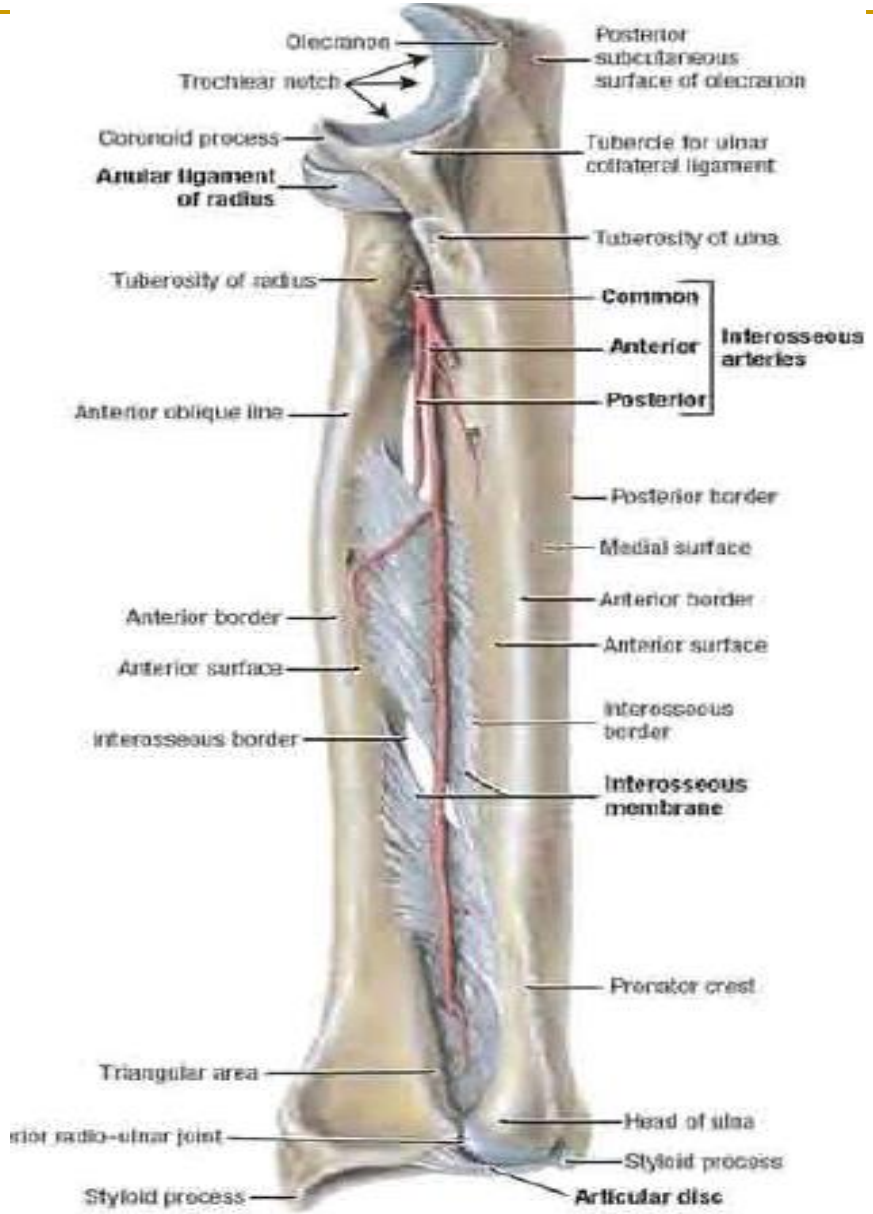
Origin:

- **Initial part of posterior interosseous artery**

Course/Distribution:

- **Courses proximally between lateral epicondyle and olecranon, deep to anconeus, to anastomose with middle collateral artery**


Deep artery of arm
(profunda brachii artery)
Superior ulnar collateral artery
Inferior ulnar collateral artery
Radial collateral artery
Middle collateral artery
Brachial artery (in cubital fossa)
Interosseous recurrent artery
Radial recurrent artery
Anterior ulnar recurrent artery
Posterior ulnar recurrent artery
Common interosseous artery
Anterior interosseous artery
Posterior interosseous artery
Radial artery
Ulnar artery
Deep palmar arch
Superficial palmar arch



C. Anterior View

WRIST JOINT

OYESHIKU

ARTICULAR SURFACES

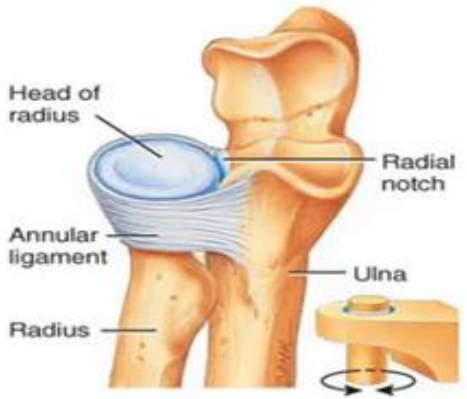
Elbow Joint

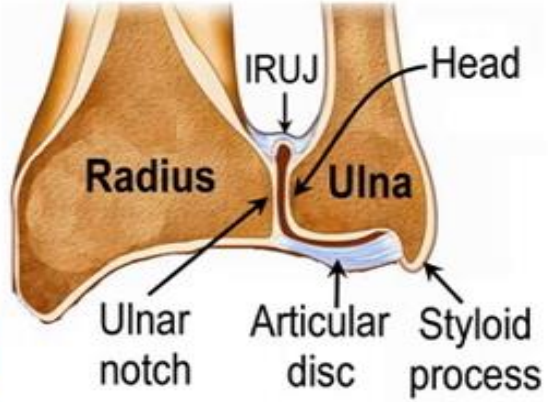
1. Humerus: trochlea & capitulum
2. Ulna: trochlear notch
3. Radius: head

Radio-Ulnar Joints

Superior RadioUlnar Joint:
1. Head of the radius
2. Radial notch of the ulna

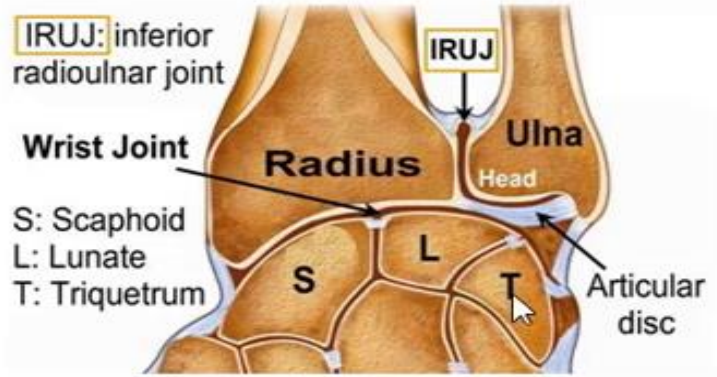
Inferior RadioUlnar Joint:
1. Head of the ulna
2. Ulnar notch of the radius





Wrist Joint

Superior:
lower end of radius + articular disc
Inferior:
Scaphoid + Lunate + Triquetrum



RADIOULNAR JOINTS

TYPE:

Synovial Joint, Pivot

ARTICULAR SURFACES:

Superior RadioUlnar Joint:

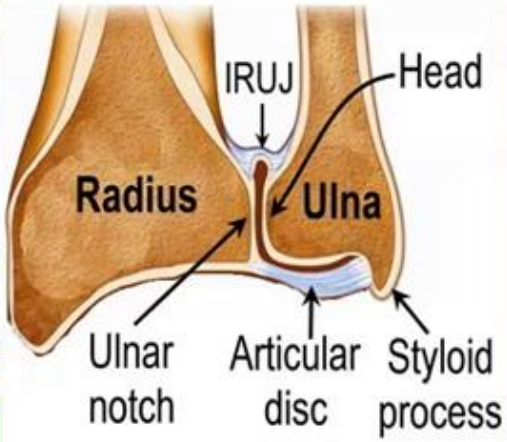
1. Head of the radius
2. Radial notch of the ulna

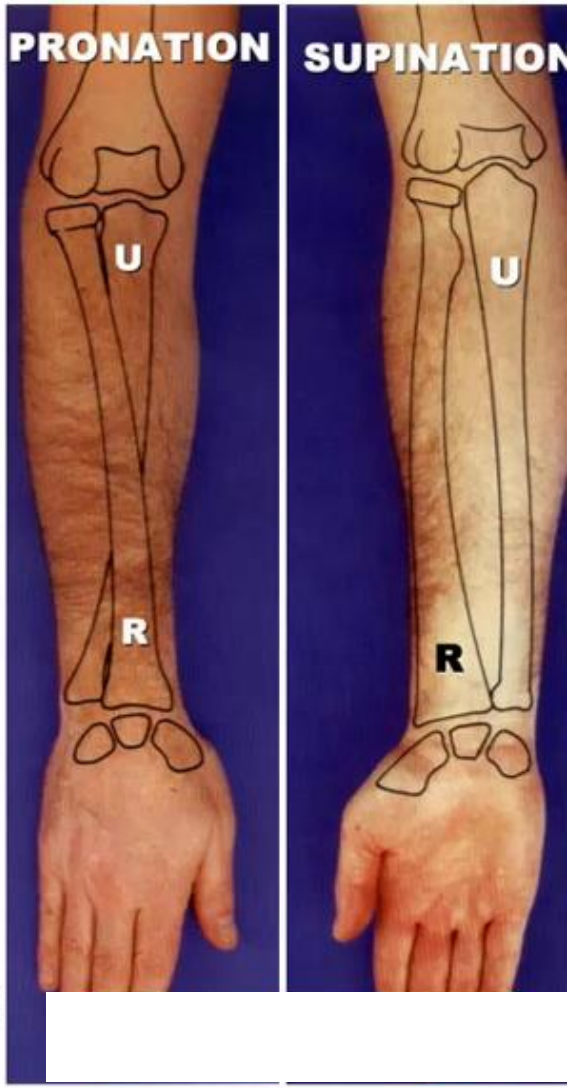
Inferior RadioUlnar Joint:

1. Head of the ulna
2. Ulnar notch of the radius

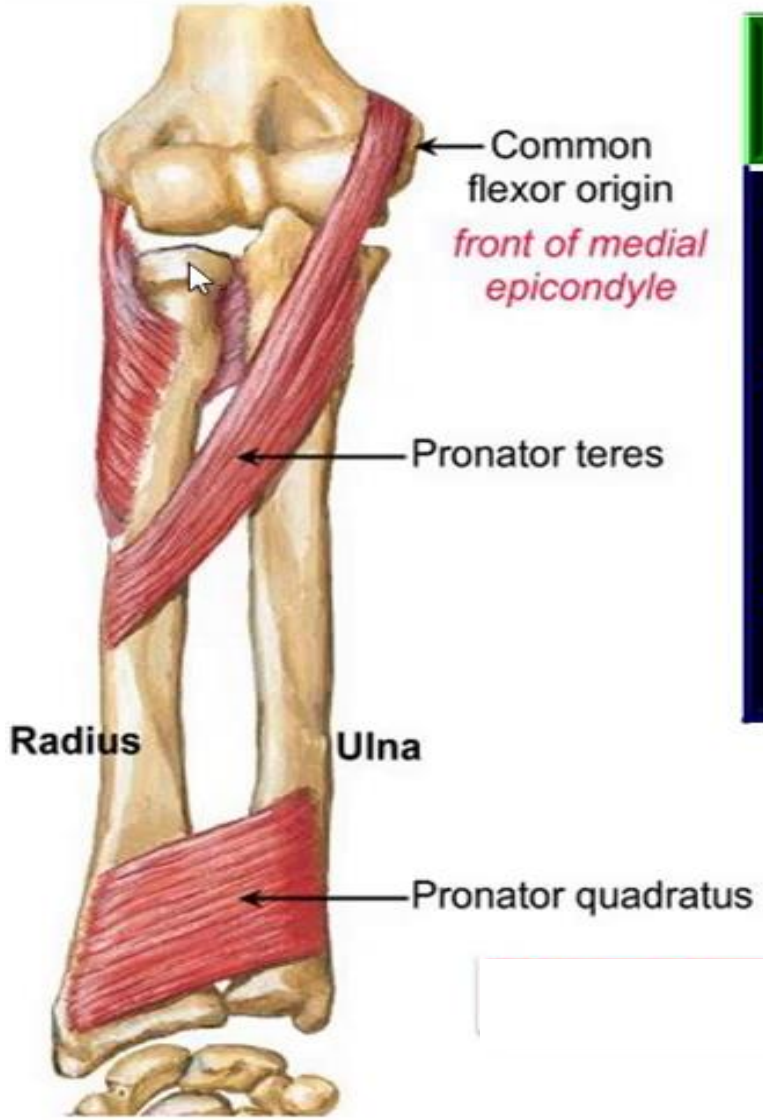
Ligament:

Annular ligament





SUPINATION & PRONATION



Supination

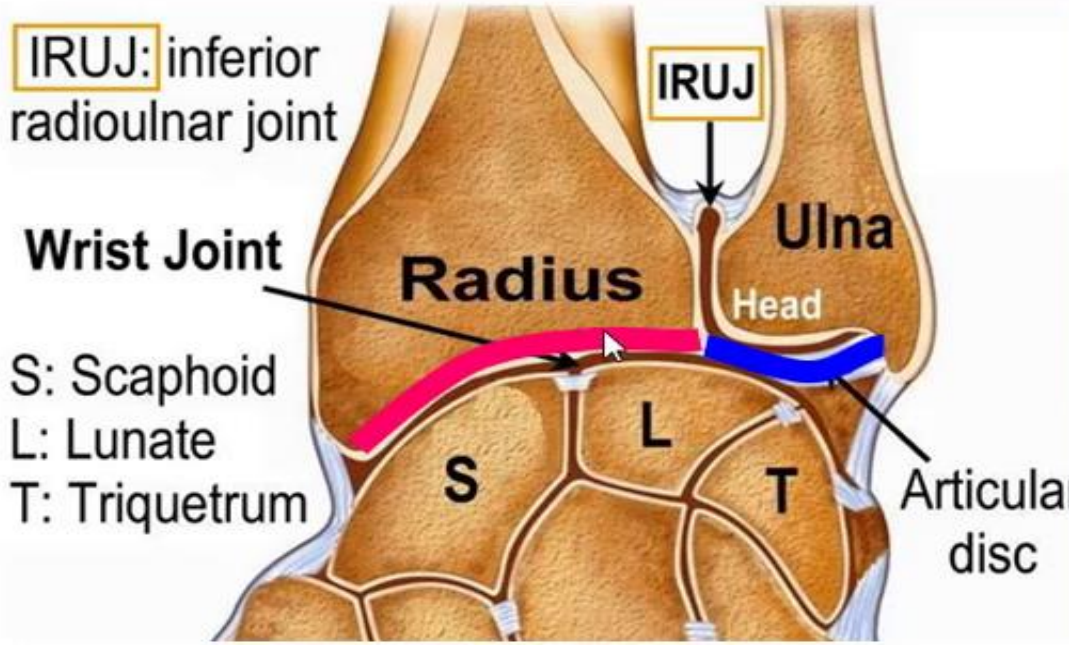
1. **Biceps:** when elbow is flexed
2. **Supinator:** when elbow is extended
3. **Brachioradialis:** initiates supination

Pronation

1. **Pronator teres**
2. **Pronator quadratus**
3. **Brachioradialis:** initiates pronation

Supination

WRIST JOINT



TYPE: Synovial Joint, Ellipsoid

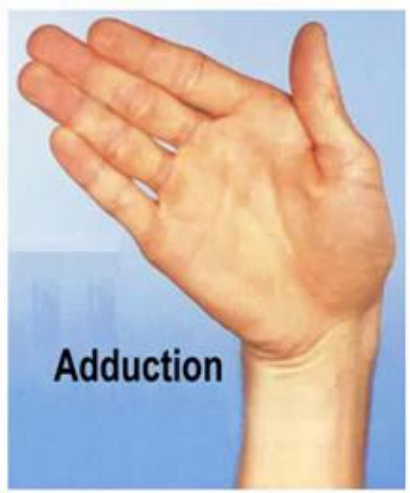
ARTICULAR SURFACES:

Superior Articular Surfaces: Lower end of radius + articular disc

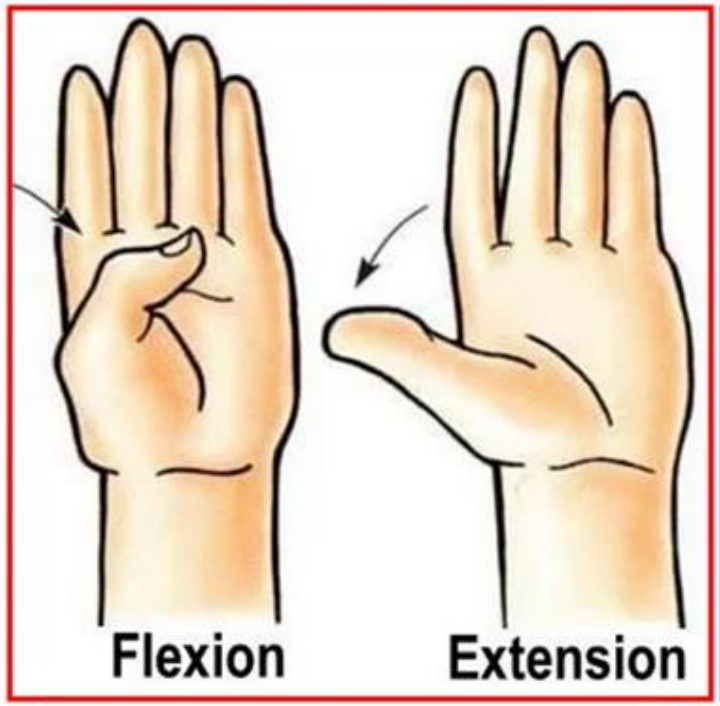
Inferior Articular Surfaces: Scaphoid + Lunate + Triquetrum

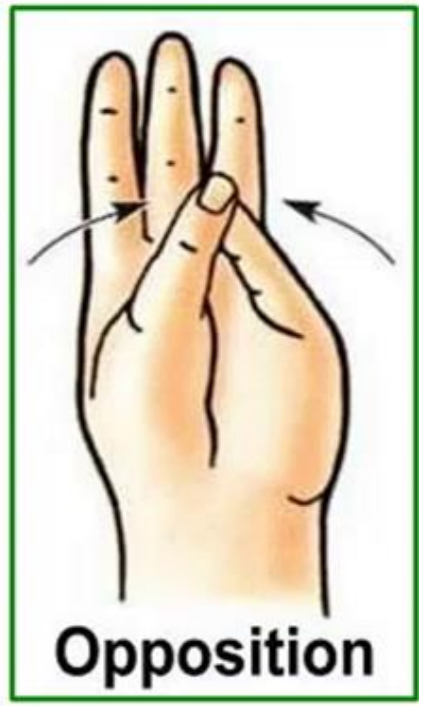
Movements of wrist joint

1 Flexion	**Main Flexors:** 1. Flexor carpi ulnaris 2. Flexor carpi radialis 3. Palmaris longus	**2 Extension**	**Main Extensors:** 1. Ext carpi ulnaris 2. Ext carpi radialis longus 3. Ext carpi radialis brevis
3 Adduction	1. Flexor carpi ulnaris 2. Extens carpi ulnaris	**4 Abduction**	1. Flexor carpi radialis 2. Ext carpi radialis longus 3. Ext carpi radialis brevis



Movements of the Thumb





Flexion: Flexor pollicis longus and brevis

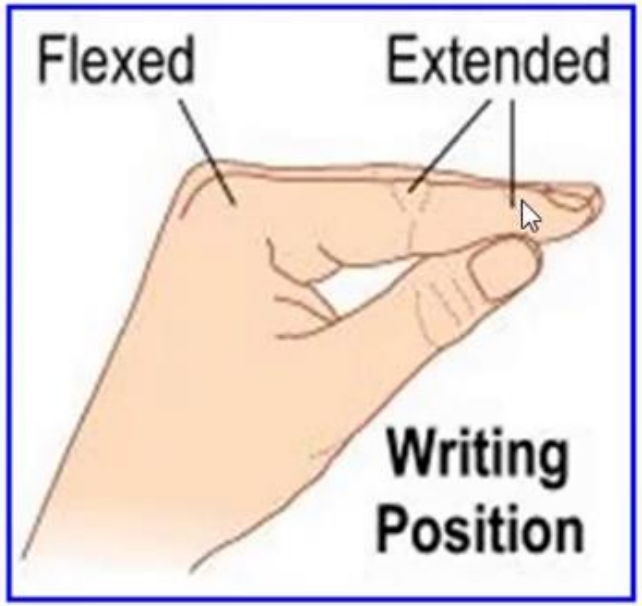
Extension: Extensor pollicis longus and brevis

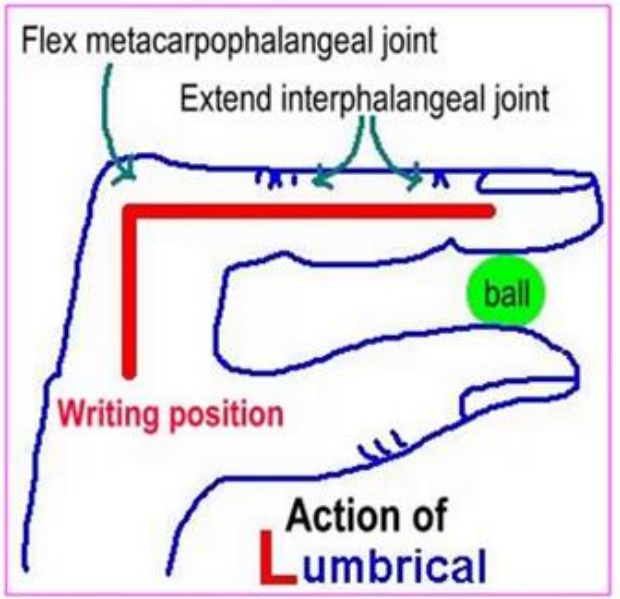
Adduction: Adductor pollicis

Abduction: Abductor pollicis longus &brevis

Opponens pollicis

Writing Position





Flexion of the metacarpophalangeal joint

Extension of the interphalangeal joint

By interossei & lumbrical muscles

Thank you